

Chapter 9 Custom Power Devices for Distribution Power Quality: DSTATCOM, DVR, UPQC and Advanced Compensators

Learning Objectives

By the end of this chapter, you should be able to:

- distinguish custom power devices from the transmission FACTS controllers of Chapters 6 to 8, and explain why the same converter principles serve a different purpose at distribution voltage;
- classify voltage-quality and current-quality problems, read a voltage-tolerance curve and compute THD, TDD, displacement and true power factor and voltage unbalance;
- explain the operating principle, classification, topologies, control algorithms and design of a distribution static compensator (DSTATCOM) for power-factor correction, voltage regulation, load balancing, harmonic elimination and neutral-current compensation;
- explain the components, energy-source options, compensation strategies, control, protection and design of a dynamic voltage restorer (DVR), and state the exact condition under which a DVR can restore a sag without real power;
- derive the voltage, current and VA ratings of a unified power quality conditioner (UPQC) in its UPQC-P, UPQC-Q and UPQC-S forms, for right-shunt and left-shunt connections, and distinguish an operating duty from an installed rating;
- size DC-link storage and capacitance, interfacing inductors, ripple filters, injection transformers and semiconductor ratings;
- derive the instantaneous pq theory and the synchronous-reference-frame method used to generate compensator references;
- analyse shunt, series and hybrid active filters from the harmonic-source model and the network impedance, and size a hybrid filter;
- apply IEEE Std 519-2022 current and voltage criteria at a point of common coupling, including the statistical assessment, and size a compensator for compliance;
- select a custom power device from an audited disturbance profile using rating, energy, protection and life-cycle cost arguments rather than device capability alone.

9.1 From Transmission FACTS to Distribution Custom Power

Chapters 6 to 8 used voltage-source converters to control power flow, voltage and stability on transmission networks. When the same converter technology is applied inside a distribution system, at a customer bus or a feeder, the goal changes. The converter now protects a load from a bad supply, or protects the supply from a bad load. Hingorani called this family custom power [1]. Three members dominate practice and give this chapter its structure: the distribution static compensator (DSTATCOM), the dynamic voltage restorer (DVR) and the unified power quality conditioner (UPQC). Active power filters (APFs), solid-state transfer switches and fault-current limiters belong to the same family.

The three main devices differ by how they connect (Figure 9.1). A DSTATCOM is connected in shunt and injects current, so it naturally corrects problems that originate in the load current: poor power factor, unbalance, neutral current and harmonics. A DVR is connected in series through an injection transformer and inserts voltage, so it corrects problems in the supply voltage: sags, swells, unbalance, notches and voltage harmonics. A UPQC places a series converter and a shunt converter on a common DC link and can do both jobs, subject to the ratings of each converter and the energy available on the shared link.

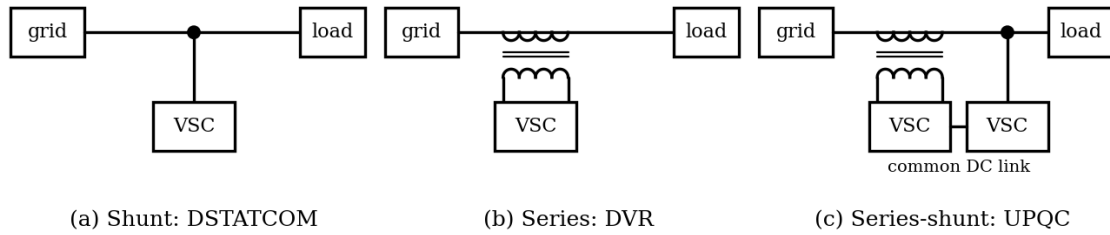


Figure 9.1. Connection families of custom power devices: (a) shunt, (b) series and (c) series-shunt.

Typical installations range from a few kVA at 230 V to several MVA at medium voltage. Neither voltage nor size defines the family; the purpose and control objective do. The series-voltage, shunt-current and DC-energy relationships of Chapters 6 to 8 remain valid, but distribution adds its own constraints: neutral conductors and zero-sequence current, single-phase and nonlinear loads, injection-transformer saturation, fault bypass, very short disturbance-detection times and the tolerance of the specific equipment being protected. A DVR controls protected-load voltage, while an SSSC mainly controls a line's effective series compensation; shared converter mathematics does not mean identical duties.

Figure 9.2 summarizes how the three devices are classified in the literature [2], [3], [4]: by converter type (current-source or voltage-source), by supply system (single-phase two-wire, three-phase three-wire and three-phase four-wire), by topology, by the source of DC energy and, for the UPQC, by shunt location and by how the series converter shares real and reactive power.

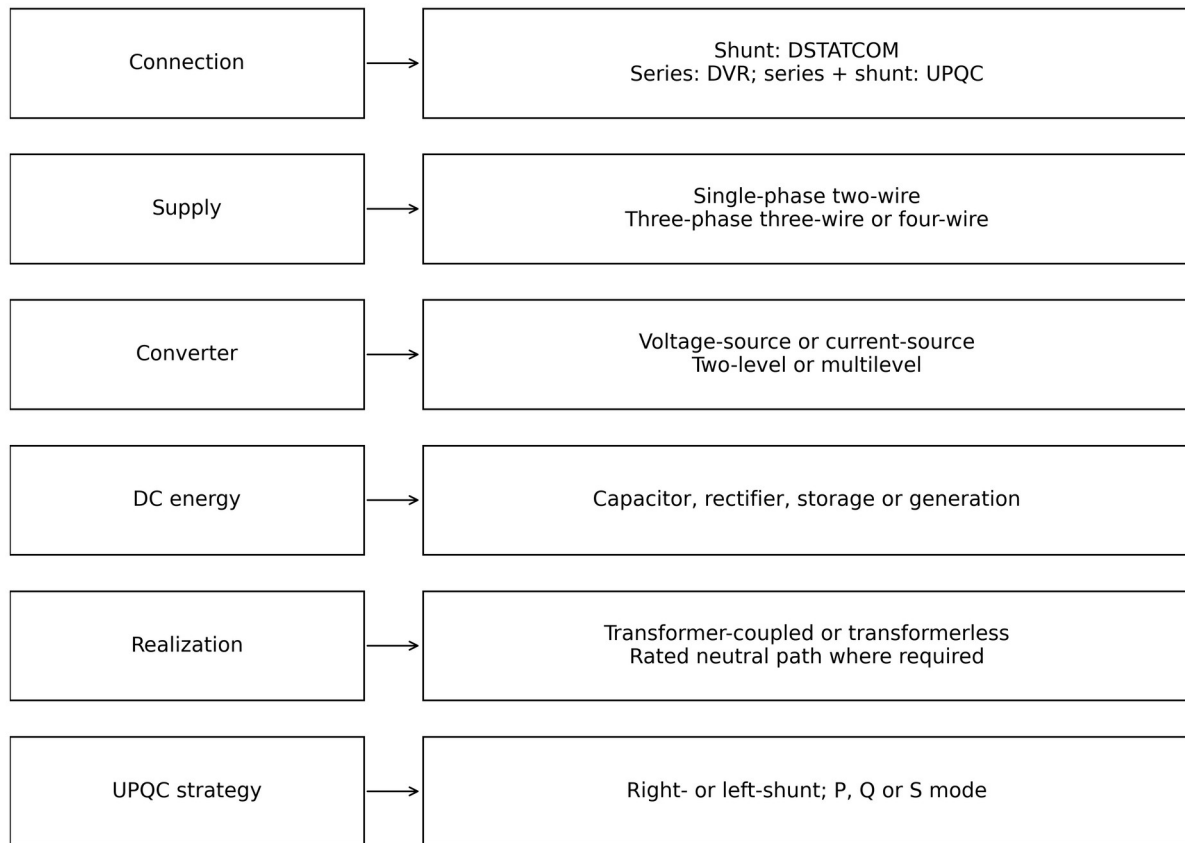


Figure 9.2. Classification of DSTATCOM, DVR and UPQC.

Voltage-source converters are widely used for custom-power conditioning because their DC capacitors, controllable AC voltage and modular voltage levels suit these duties. A current-source converter uses DC inductive energy storage and requires a compatible commutation and AC-filter arrangement. Efficiency, cost and size depend on semiconductor technology, operating point, cooling and topology; neither

converter family is universally cheaper or more efficient. Unless stated otherwise, this chapter uses PWM voltage-source converters.

Table 9.1. Transmission FACTS and distribution custom power compared.

Attribute	Transmission FACTS	Distribution custom power
Purpose	Network power flow, voltage and stability	Voltage and current quality at a load, bus or feeder
Typical scale	Tens to hundreds of MVA	kVA to a few MVA; voltage alone is not the definition
Typical disturbance	Contingencies, oscillations, transfer limits	Sags, swells, harmonics, unbalance, flicker, poor power factor
Response	Cycles to seconds for most functions	Often sub-cycle for sag correction and current shaping
Special constraints	Insulation level, protection coordination, SSR	Neutral current, single-phase loads, transformer saturation, bypass, equipment tolerance

9.1.1 A physical view of custom power compensation

The natural entry point is to ask where the disturbance originates and which electrical quantity must be synthesized. If the load current is undesirable while the supply voltage is acceptable, the compensator should inject current in shunt. If the supply voltage is undesirable while the load current must continue, it should inject voltage in series. If both are unacceptable, the two actions can be coordinated through a common DC link. A shunt converter cannot directly create a missing series voltage, although its current can move the PCC voltage through the feeder impedance; a series converter cannot directly cancel a nonlinear load current at the source, although its voltage can shape that current indirectly. Network impedance, converter rating, available DC energy and control bandwidth therefore decide what can actually be corrected (Table 9.2).

Table 9.2. Compensation location and its principal physical limit.

Location	Controlled quantity	Primary disturbance	Typical device	Principal physical limit
Shunt	Injected current	Reactive, harmonic, unbalanced or neutral load current	DSTATCO M / shunt APF	Converter current and PCC voltage
Series	Injected voltage	Sag, swell, unbalance, voltage harmonics	DVR / series APF	Injection voltage, line current and stored energy
Series + shunt	Voltage and current together	Simultaneous supply and load problems	UPQC	Both converter ratings and common DC-link balance
Multi-feeder	Coordinated series voltages	Sag support across several feeders	IDVR / MC-UPQC	Cross-feeder real-power balance and converter ratings

9.1.2 From ideal compensator to practical converter

Each device is developed in four layers. The first is the ideal compensation law obtained from Kirchhoff's laws and phasors. The second is the converter realization: VSC topology, transformer or coupling inductor, DC link and filters. The third is the control system that estimates the disturbance and forces the converter to reproduce the required voltage or current. The fourth is the practical envelope set by semiconductor current, modulation voltage, stored energy, thermal limits, protection and the upstream

network. This layering guards against a common error in custom-power studies, in which a controller computes a mathematically perfect reference that the converter cannot produce. The ideal relationships must therefore be checked against the practical operating envelope before they are used for equipment design.

Conventions used throughout. Phasors and unqualified AC voltages/currents are RMS; waveform amplitudes and amplitude-invariant Clarke/Park variables are peak quantities. $V_{LL} = \sqrt{3} V_{ph}$ for balanced voltages. Compensating current is positive from the converter into the bus: $i_s = i_L - i_c$. Series injection is a voltage rise: $V_L = V_s + V_{inj}$. Converter AC export is positive, giving $\dot{W}_{dc} = -(P_{se} + P_{sh} + P_{loss})$. The load displacement angle is measured from the actual load voltage, whereas unity source power factor is referred to the supply voltage. Transformer ratio $n = V_{converterwinding} / V_{linewinding}$ gives $I_{converter} = I_{line} / n$ and $L_{line} = L_{converter} / n^2$. In Section 9.7 only, I_L denotes the fixed maximum-demand RMS current, rather than instantaneous load current. An operating VA duty is distinct from the installed equal-leg, thermal and fault ratings.

9.2 Power-Quality Problems in Distribution Systems

9.2.1 Voltage-quality and current-quality problems

It is useful to sort distribution power-quality problems by where they originate. A voltage-quality problem is already present in the supply voltage before the protected load sees it: a sag caused by a remote fault, a swell on the healthy phases during a single-line-to-ground fault on an ungrounded or impedance-grounded system, voltage unbalance, notching caused by a neighbour's converter, flicker or voltage harmonics. A current-quality problem is created by the load and pushed back into the network: reactive current, unbalanced phase currents, excessive neutral current and harmonic current. Current-quality problems become voltage-quality problems for other customers once the current flows through the network impedance (Table 9.3).

Table 9.3. Voltage-quality and current-quality problems and the natural compensator.

Voltage-quality problems (supply side)	Current-quality problems (load side)
Sag, swell, interruption	Poor displacement power factor (reactive current)
Voltage unbalance	Unbalanced phase currents
Notching, flicker, fluctuation	Excessive neutral current (triplen harmonics, single-phase loads)
Voltage harmonics, transients	Harmonic currents, load fluctuation
Natural compensator: series (DVR)	Natural compensator: shunt (DSTATCOM, shunt APF)

9.2.2 Disturbance categories

Figure 9.3 shows eight commonly encountered disturbances; it is not an exhaustive classification. Under IEEE Std 1159-2019 [5], a short-duration voltage sag has an RMS magnitude of 0.1 to 0.9 pu lasting from half a cycle to one minute. A swell starts at about 1.1 pu, and its typical upper magnitude depends on duration. An interruption has an RMS voltage below 0.1 pu; a sustained interruption lasts longer than one minute. Harmonics are integer multiples of the fundamental, usually generated by nonlinear loads. Notches are short, periodic voltage dips produced when a line-commutated converter momentarily shorts two phases during commutation. Voltage fluctuations at a few hertz produce visible lamp flicker. Transients may be impulsive (lightning) or oscillatory (capacitor switching). Unbalance, interharmonics, DC offset, noise and frequency variation also matter.

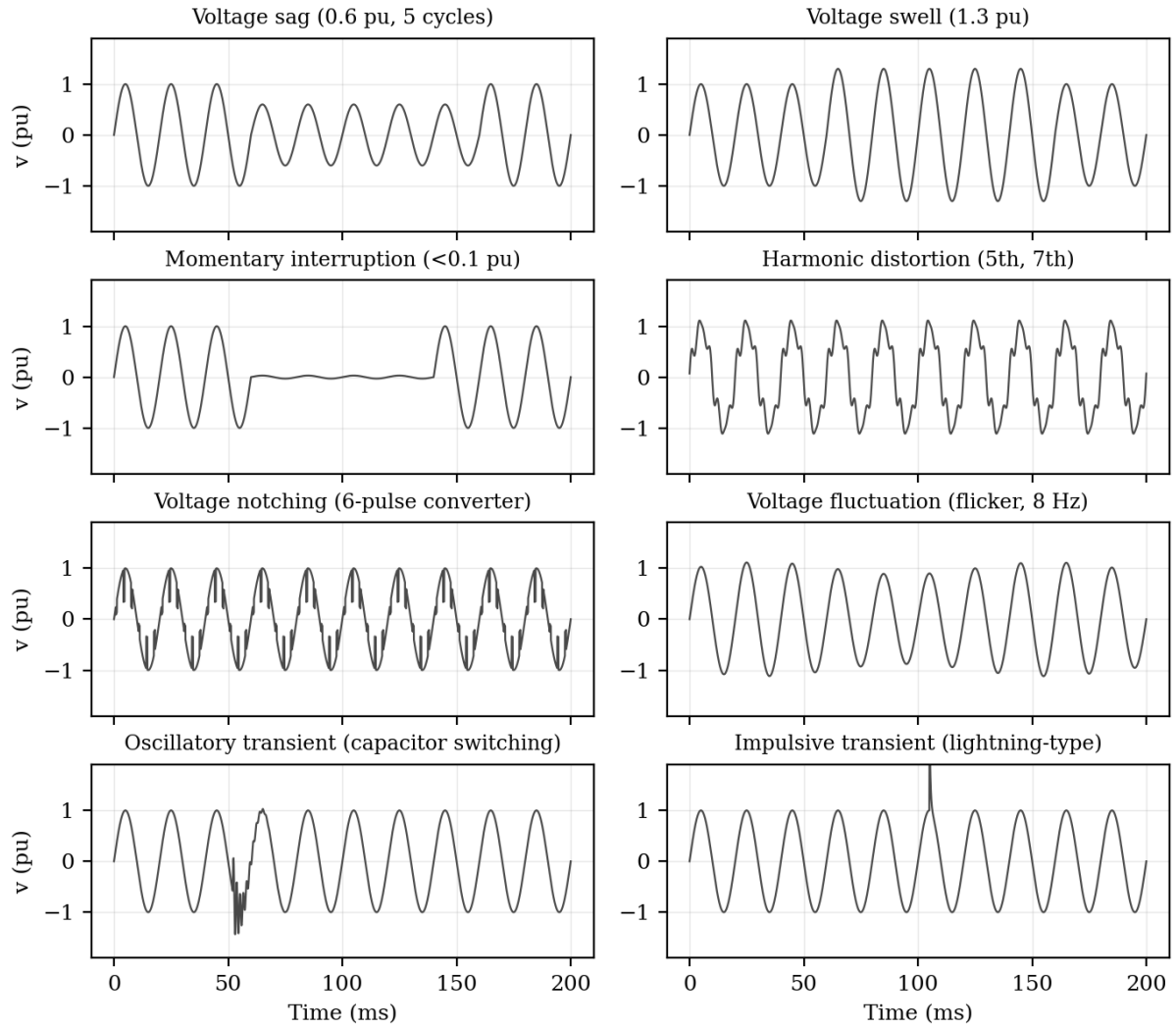


Figure 9.3. Illustrative power-quality disturbances (simulated, 50 Hz, per unit).

Table 9.4. Selected categories and typical characteristics, after IEEE Std 1159-2019.

Category	Typical duration	Typical magnitude
Instantaneous sag / swell	0.5 to 30 cycles	0.1 to 0.9 pu / 1.1 to 1.8 pu
Momentary sag / swell	30 cycles to 3 s	0.1 to 0.9 pu / 1.1 to 1.4 pu
Temporary sag / swell	3 s to 1 min	0.1 to 0.9 pu / 1.1 to 1.2 pu
Momentary / temporary interruption	0.5 cycle to 3 s / 3 s to 1 min	below 0.1 pu
Sustained interruption	longer than 1 min	0 pu
Undervoltage / overvoltage	longer than 1 min	0.8 to 0.9 pu / 1.1 to 1.2 pu
Voltage imbalance	steady state	0.5 to 5 % (negative sequence)
Harmonics	steady state	0 to 20 %
Voltage fluctuation (flicker)	intermittent	0.1 to 7 %, below 25 Hz

9.2.3 Reading a voltage-tolerance curve

Equipment sensitivity to sags and swells is commonly summarized by a tolerance curve such as the one published by the Information Technology Industry Council (ITI), historically the CBEMA curve [6]. Figure 9.4 reproduces only its general shape. The ITI curve is a specific reference for 120 V, 60 Hz information-technology equipment, and its application note should be consulted for exact coordinates. For a real installation, use the protected equipment's own ride-through specification at its actual supply voltage and frequency.

The curve plots acceptable voltage against disturbance duration. A brief, deep sag may be tolerated, while the same depth held longer causes a trip. Duration, and not only depth, is therefore a first-class design parameter throughout the chapter: it sets the DVR's stored energy and the response time its controller must achieve.

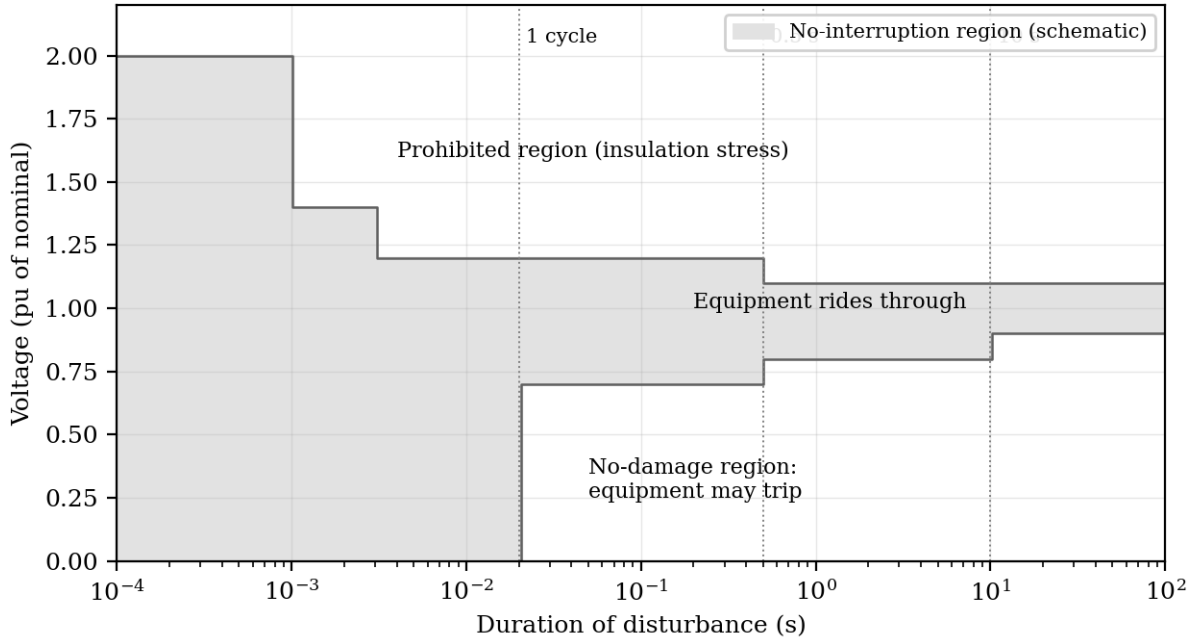


Figure 9.4. Schematic equipment voltage-tolerance regions (shape only, not a compliance curve).

9.2.4 Power-quality indices

Several indices recur in this chapter. For a periodic quantity with RMS harmonic components X_h , total harmonic distortion is

$$THD = \frac{\sqrt{\sum_{h \geq 2} X_h^2}}{X_1} \times 100\% \quad (9.1)$$

where X_1 is the RMS fundamental. Total demand distortion (TDD) has the same numerator but uses the maximum demand load current as denominator (Section 9.7). For a load with RMS fundamental current I_1 , displacement angle ϕ_1 and total RMS current I , the true power factor separates into a displacement factor and a distortion factor:

$$PF = \frac{P}{VI} = \frac{I_1}{I} \cos \phi_1 = \frac{\cos \phi_1}{\sqrt{1 + THD_I^2}} \quad (9.2)$$

with THD_I as a fraction, assuming sinusoidal voltage. A plain capacitor bank targets fundamental displacement current; it does not actively cancel a prescribed harmonic spectrum and may alter distortion

through network resonance. A suitably rated DSTATCOM can compensate both reactive fundamental and harmonic current. Voltage unbalance is expressed by the voltage unbalance factor,

$$VUF = \frac{|V_n|}{|V_p|} \times 100\% \quad (9.3)$$

where V_p and V_n are the positive- and negative-sequence components of Chapter 2.

9.2.5 Matching problems to devices

A DVR corrects load-voltage sags and swells within its voltage, current, energy and control limits. A DSTATCOM or shunt APF compensates load-generated reactive, unbalanced, neutral and harmonic currents, and through the feeder impedance can also regulate the bus voltage. A UPQC combines these services. A UPQC can replenish the series converter's real power from the grid during a sag only while adequate source voltage remains; with the source absent, it needs usable stored energy and a topology that can safely supply an isolated load (Section 9.8.2). Flicker mitigation needs enough bandwidth and current or voltage margin. Very fast impulsive transients usually need surge protection, not a converter (Table 9.5).

Table 9.5. Power-quality problems and candidate devices.

Dominant problem	Candidate device and required checks
Low power factor, reactive demand	DSTATCOM in UPF mode; capacitor bank if the load is steady and linear
Feeder voltage regulation	DSTATCOM in ZVR mode; voltage effect depends on feeder impedance
Unbalanced load, neutral current	Three-phase four-wire DSTATCOM (four-leg, split-capacitor or transformer-assisted)
Load-current harmonics	DSTATCOM / shunt APF; hybrid filter after a resonance study
Voltage sag or swell	DVR with enough injection voltage, current, energy and phase-transition capability
Sags on several feeders	Interline DVR, if joint operating points and shared-link balance are feasible
Supply-voltage harmonics, notches	DVR or series APF designed for harmonic voltage injection
Both voltage and current problems	UPQC, subject to both converter duties, source capacity and DC balance
Sustained interruption	UPS, standby source or island-capable storage scheme with isolation

9.3 Distribution Static Compensator (DSTATCOM)

The DSTATCOM is a shunt-connected VSC whose job is to make the current drawn from the supply look like the current of an ideal load: sinusoidal, balanced and, if required, in phase with the voltage. It is the distribution counterpart of the STATCOM of Chapter 6 [7], but its duty is wider. A transmission STATCOM mainly exchanges positive-sequence fundamental reactive power; a DSTATCOM is commonly asked to cancel harmonics, balance single-phase loads and remove neutral current as well. Commercial units range from a few hundred kVA to several MVA at medium voltage, often coordinating switched capacitor or reactor banks.

9.3.1 Operating principle

Figure 9.5 shows the single-line arrangement. The VSC is connected at the point of common coupling (PCC) through an interfacing inductor L_r , and a small R_f - C_f ripple filter at the PCC absorbs switching-frequency current. The DC side has only a capacitor; no external energy source is needed because, ideally, the DSTATCOM exchanges no average real power. Voltage and current sensors feed a DSP or rapid-prototyping controller that generates the gate pulses.

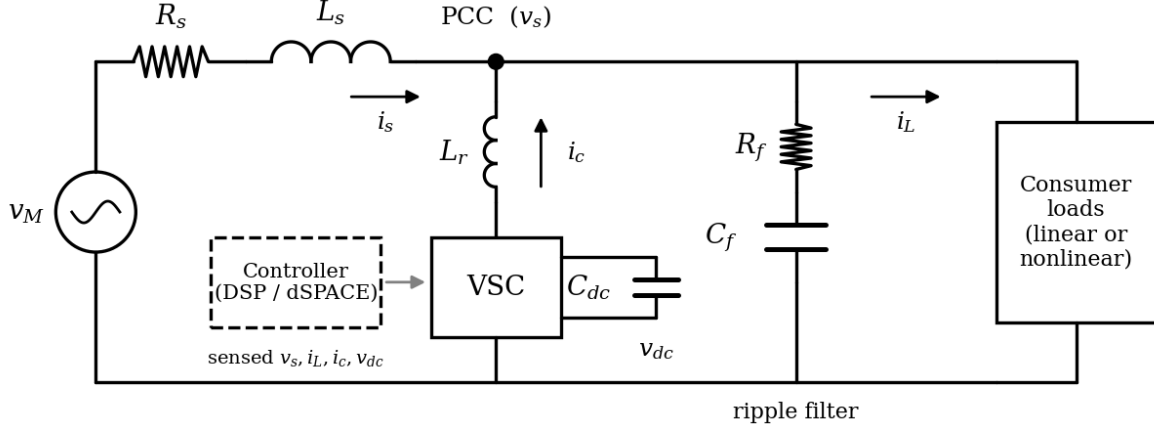


Figure 9.5. DSTATCOM single-line diagram with interfacing inductor, ripple filter and DC-link capacitor.

With the compensator current i_c defined positive from the VSC into the PCC, Kirchhoff's current law gives

$$i_s = i_L - i_c \quad (9.4)$$

so whatever part of the load current the DSTATCOM supplies locally no longer flows from the source. Decompose the load current of phase k into four parts:

$$i_{Lk} = i_{pk} + i_{qk} + i_{uk} + i_{hk} \quad (9.5)$$

where i_{pk} is the fundamental positive-sequence current in phase with the voltage (it carries the load's average real power), i_{qk} the fundamental positive-sequence current in quadrature, i_{uk} the negative- and zero-sequence fundamental components and i_{hk} the harmonics. In full compensation the source reference is $i_{sk}^* = i_{pk} + i_{loss,k}$, where i_{loss} is a small in-phase current that supplies converter losses and holds the DC-link voltage, and the DSTATCOM must inject $i_{ck}^* = i_{qk} + i_{uk} + i_{hk} - i_{loss,k}$. Which components remain in the source current is a design choice: in unity-power-factor operation only i_p remains, while in zero-voltage-regulation operation a deliberate fundamental reactive component is retained so that the feeder drop yields the required PCC magnitude.

Treating the VSC fundamental output as a voltage $V_c \angle \delta$ behind a reactance X_r , with PCC voltage $V \angle 0$, the power delivered to the PCC per phase is

$$P_c = \frac{V V_c \sin \delta}{X_r}, Q_c = \frac{V(V_c \cos \delta - V)}{X_r} \quad (9.6)$$

When $V_c \cos \delta > V$ the converter exports reactive power; when $V_c \cos \delta < V$ it absorbs reactive power. With the export convention of Equation (9.6), a small **negative** δ draws real power to charge the capacitor and cover losses. Current-controlled PWM implements the resulting voltage and current relationship.

The same hardware supports several objectives, often simultaneously:

- *Power-factor correction or unity power factor (UPF) mode.* Source current is forced into phase with the PCC voltage, so the source supplies only real power. The PCC voltage settles wherever the feeder impedance puts it.

- *Zero voltage regulation (ZVR) mode.* The DSTATCOM injects enough reactive current to hold the PCC voltage magnitude at its reference. The source current then contains a reactive component, so the source power factor is no longer unity.
- *Load balancing.* Negative-sequence and, in four-wire systems, zero-sequence load currents are supplied locally, so the source sees a balanced set.
- *Harmonic elimination.* Load harmonic currents are supplied locally; this is shunt active filtering.
- *Neutral-current compensation.* In a four-wire system the zero-sequence and triplen currents return through the compensator's own neutral path, so the source neutral carries almost nothing.

Figure 9.6 compares the three cases that matter most for fundamental quantities. Without compensation, the lagging load current produces a drop across the feeder and the load voltage falls. In UPF mode the source current is aligned with the PCC voltage; the drop $jX_s I_s$ is now in quadrature with V_L , so the magnitude falls less but still falls. In ZVR mode the source current is advanced until $|V_L| = |V_s|$.

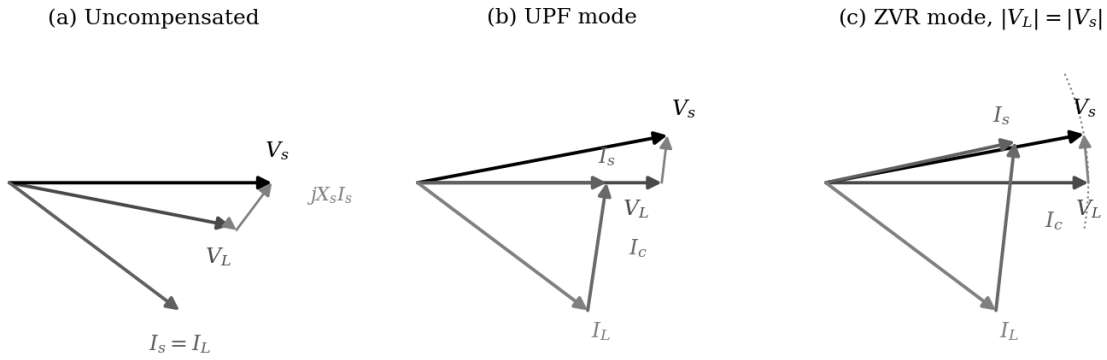


Figure 9.6. Phasor diagrams: (a) uncompensated lagging load, (b) DSTATCOM in UPF mode, (c) DSTATCOM in ZVR mode.

Reactive current needed for ZVR. Take the PCC voltage V_L as reference and the source EMF magnitude $|V_s|$ as given. With per-phase load real power P and a lossless compensator, the source current is $I_s = I_p + jI_q$ with $I_p = P/V_L$. KVL across the feeder impedance $Z_s = R_s + jX_s$ gives $V_s = V_L + Z_s I_s$, and requiring $|V_s|$ to equal the specified source magnitude leads to a quadratic in I_q :

$$(R_s^2 + X_s^2)I_q^2 + 2(BR_s - AX_s)I_q + A^2 + B^2 - |V_s|^2 = 0, A = V_L + R_s I_p, B = X_s I_p \quad (9.7)$$

Check the discriminant of Equation (9.7) first: a negative value means the specified steady-state target has no real solution under these assumptions. When two roots exist, the smaller-current feasible branch is normally selected, subject to converter current, voltage and stability limits; its sign depends on the target and network. For the voltage-raising case of Example 9.4 it gives leading source current. Then obtain $I_c = I_L - I_s$ phase by phase. A stiff bus needs large reactive current for appreciable voltage movement. Retaining R_s matters on low-voltage feeders with appreciable R/X.

Compared with an SVC whose maximum capacitive current decreases with voltage, a DSTATCOM can maintain commanded current over a wider low-voltage range, subject to current limit, usable DC voltage, control stability and energy balance. It can command harmonic and unbalanced current in a suitable topology. Its coupling filter and closed-loop impedance can still resonate with the network, so replacing a capacitor bank does not eliminate the need for an impedance and damping study.

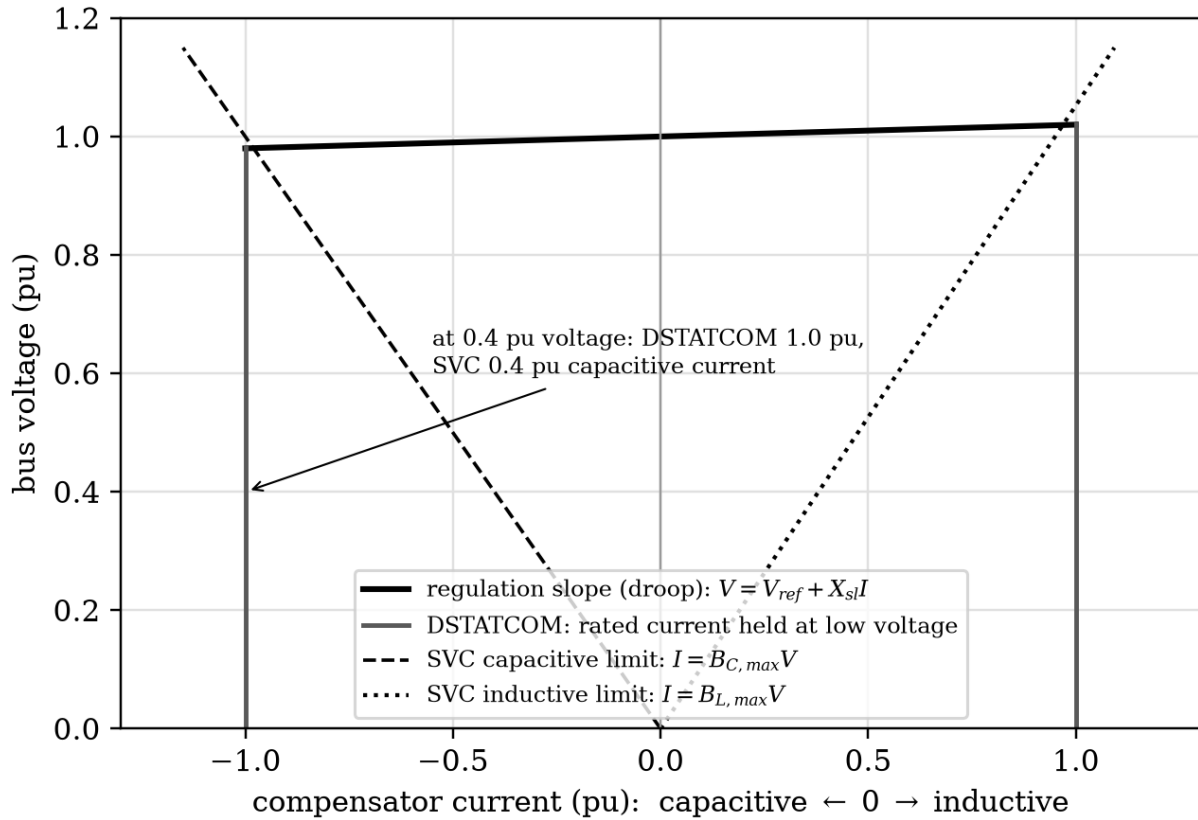


Figure 9.7. V-I characteristic of a DSTATCOM compared with an SVC (schematic). Along the droop line the bus voltage rises as the compensator moves from capacitive to inductive current.

9.3.2 Classification and topologies

Converter type. A current-fed DSTATCOM uses a CSC with a DC inductor; a voltage-fed DSTATCOM uses a VSC with a DC capacitor. For the reasons given in Section 9.1, VSCs dominate.

Three-phase three-wire topologies. Figure 9.8 shows the four common arrangements. The three-leg VSC (a) uses six switches and one DC capacitor and is the standard choice. The two-leg VSC with split capacitors (b) saves two switches by tying the third phase to the capacitor midpoint, at the cost of a higher DC voltage and capacitor-voltage balancing. Three single-phase H-bridges (c) each need an isolation transformer but give independent phase control and redundancy. An isolating star/delta transformer (d) matches a low-voltage converter to a medium-voltage bus and provides galvanic isolation.

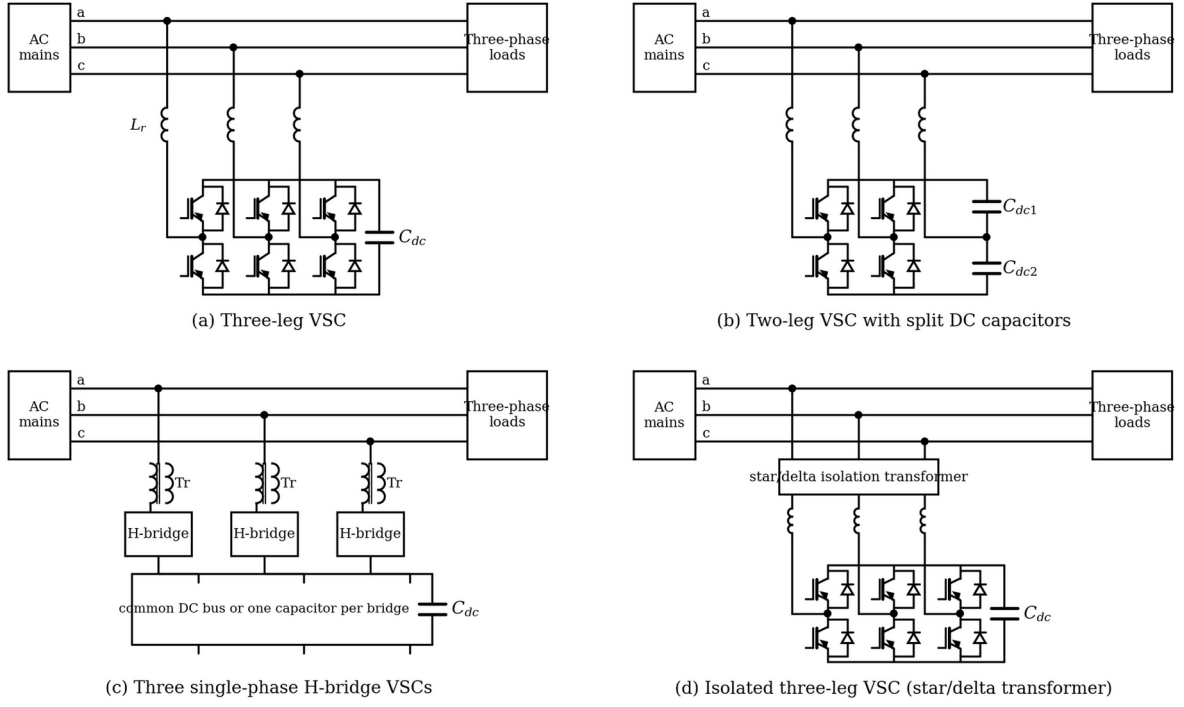


Figure 9.8. Three-phase three-wire DSTATCOM topologies.

Three-phase four-wire topologies. Four-wire distribution feeds single-phase loads between line and neutral, so zero-sequence current flows and triplen harmonics add in the neutral. A six-switch three-leg bridge cannot synthesize zero-sequence current, so a neutral path must be added (Figure 9.9). The four-leg VSC (a) adds a fourth leg with its own inductor L_{rn} and controls the neutral current directly. The split-capacitor three-leg VSC (b) returns the neutral current through the capacitor midpoint; it needs no extra switches but large, balanced capacitors because the neutral current flows through them. Transformer-assisted schemes (c) and (d) let a three-leg VSC handle positive and negative sequence while a passive transformer provides a low-impedance zero-sequence path. Zig-zag, star/delta, T-connected, star/hexagon, star/polygon and Scott-connected transformers have all been used [8]. Three single-phase H-bridges with isolation transformers are also used. Table 9.6 summarizes the trade-offs.

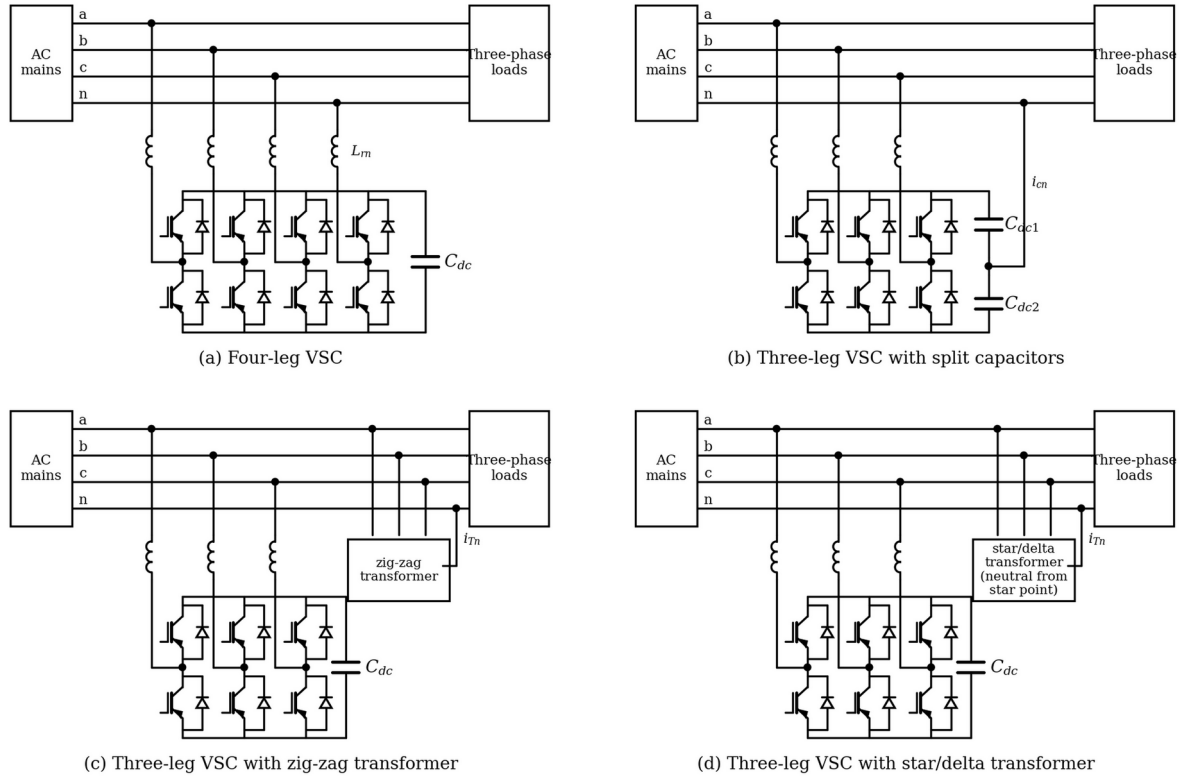
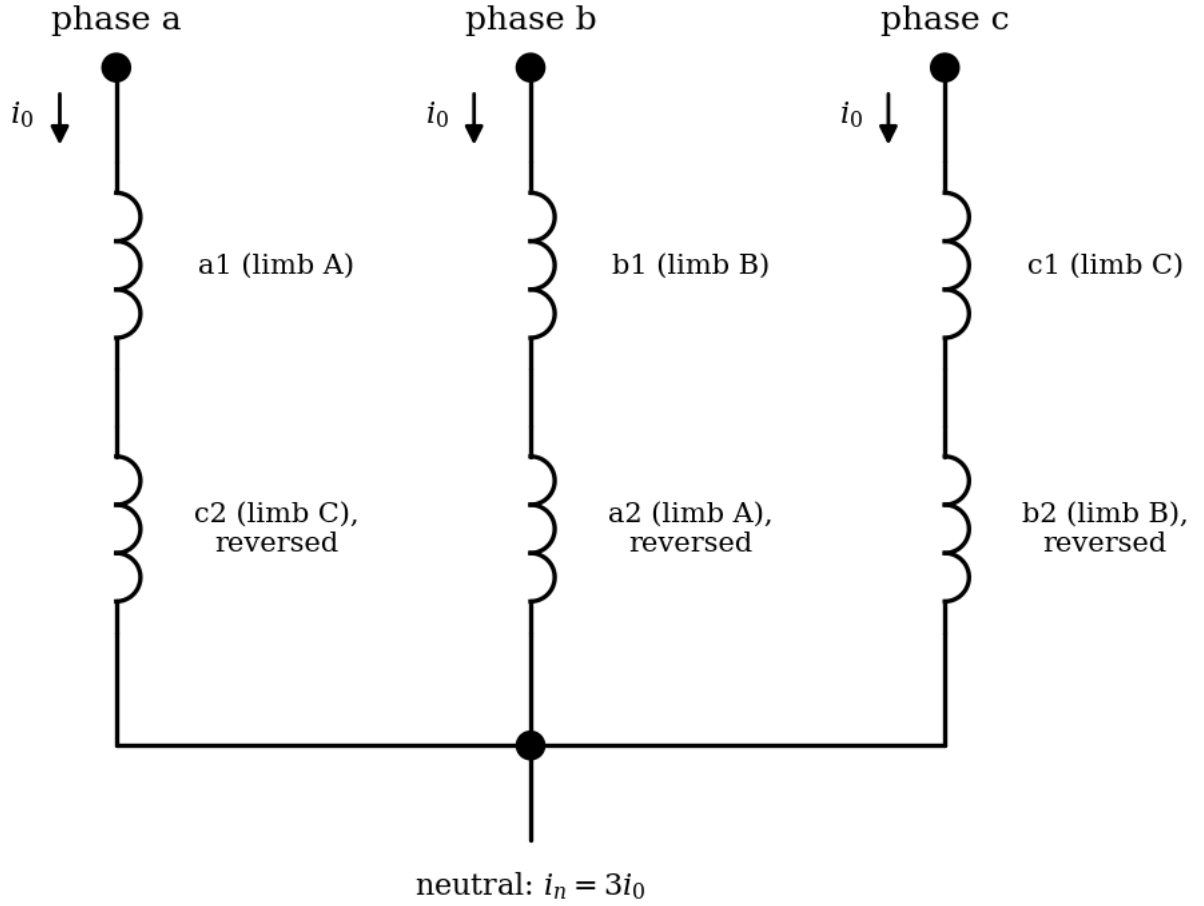


Figure 9.9. Three-phase four-wire DSTATCOM topologies.

Table 9.6. Four-wire DSTATCOM topologies compared.

Topology	Zero-sequence capability	Main advantage	Main design concern
Three-leg VSC	No direct neutral-current path	Lowest switch count for three-wire systems	Cannot compensate feeder neutral current
Four-leg VSC	Full controlled path	Independent neutral-current control	Extra leg, three-dimensional modulation and current limit
Split DC link	Neutral tied to midpoint	No fourth leg	Capacitor-voltage imbalance and ripple
Zig-zag assisted	Passive zero-sequence path	Reduces converter neutral duty	Transformer size, losses and harmonic heating

The zig-zag transformer works because each limb carries two windings belonging to different phases, connected in opposition (Figure 9.10). Zero-sequence currents are equal in all three phases, so the two mmfs on each limb cancel and the zero-sequence current sees only leakage impedance. Positive- and negative-sequence currents do not cancel and see the full magnetizing impedance, so the transformer draws almost nothing from the balanced system. Connected at the load, it diverts most of the neutral current away from the source. Its effectiveness depends on its zero-sequence impedance relative to the source neutral path, so it is usually combined with active control rather than relied on alone.



Limb A carries a1 (current i_a) and a2 (current i_b , reversed): net mmf $\propto i_a - i_b$.
 Zero-sequence currents ($i_a = i_b = i_c = i_0$) give zero mmf on every limb,
 so i_0 meets only leakage impedance and returns through the neutral.

Figure 9.10. Zig-zag transformer as a passive zero-sequence (neutral-current) path.

9.3.3 Control algorithms

A DSTATCOM controller has three layers. The reference generator extracts from measured voltages and currents the source-current reference $\hat{i}_s^c$ (indirect current control) or the compensator-current reference $\hat{i}_c^c = \hat{i}_L - \hat{i}_s^c$ (direct current control). A DC-link voltage regulator, usually a PI controller, adds the small active current $\hat{i}_{loss}$ that keeps v_{dc} at its reference. A current controller, hysteresis or PWM, makes the VSC track the reference. The current loop must be substantially faster than the outer voltage loops: an outer DC-voltage loop that is too fast tries to correct switching-scale and twice-frequency energy pulsations and injects distortion into the reference, while a slow current loop cannot reproduce harmonic commands. Bandwidth separation is therefore a physical requirement, not a tuning preference.

Many reference-generation algorithms have been published [2], [9]. In the time domain they include unit-template (PI-based) control, the power balance theory, the $I \cos \phi$ algorithm, current synchronous detection, instantaneous reactive power (pq) theory, synchronous reference frame (SRF or dq) theory, instantaneous symmetrical component theory, single-phase pq and dq theories, Adaline (LMS) estimators and enhanced-PLL methods. In the frequency domain they include Fourier, DFT and recursive-DFT extraction, Kalman filtering and wavelet methods. The principal methods are developed below; the others are summarized in Table 9.7.

Unit-template (PI-controller-based) control

This is the simplest scheme (Figure 9.11). The amplitude of the PCC voltage is estimated from the three sampled phase voltages,

$$V_t = \sqrt{2/3 (v_{sa}^2 + v_{sb}^2 + v_{sc}^2)} \quad (9.8)$$

and in-phase unit templates are formed as $u_a = v_{sa}/V_t$, $u_b = v_{sb}/V_t$, $u_c = v_{sc}/V_t$. For balanced sinusoidal voltages these are unit-amplitude sinusoids locked to the supply. Quadrature unit templates, each leading its in-phase template by 90° , follow algebraically:

$$w_a = \frac{-u_b + u_c}{\sqrt{3}}, w_b = \frac{\sqrt{3}u_a}{2} + \frac{u_b - u_c}{2\sqrt{3}}, w_c = -\frac{\sqrt{3}u_a}{2} + \frac{u_b - u_c}{2\sqrt{3}} \quad (9.9)$$

A PI controller acting on the DC-voltage error gives the amplitude I_{sp}^* of the in-phase source current. It settles automatically at the load's real-power current plus losses: if the source supplies too little, the capacitor makes up the difference, v_{dc} falls and the PI output rises. In ZVR mode a second PI acting on $V_t^* - V_t$ gives the quadrature amplitude I_{sq}^* . The source-current references are

$$i_{sk}^* = I_{sp}^* u_k + I_{sq}^* w_k, k = a, b, c \quad (9.10)$$

With $I_{sq}^* = 0$ the scheme gives UPF for balanced sinusoidal templates. Its DC loop trades ripple rejection against settling time. Raw distorted or unbalanced voltages contaminate the templates; a suitable fundamental positive-sequence extractor or PLL is needed, with its own delay and sequence-rejection limitations.

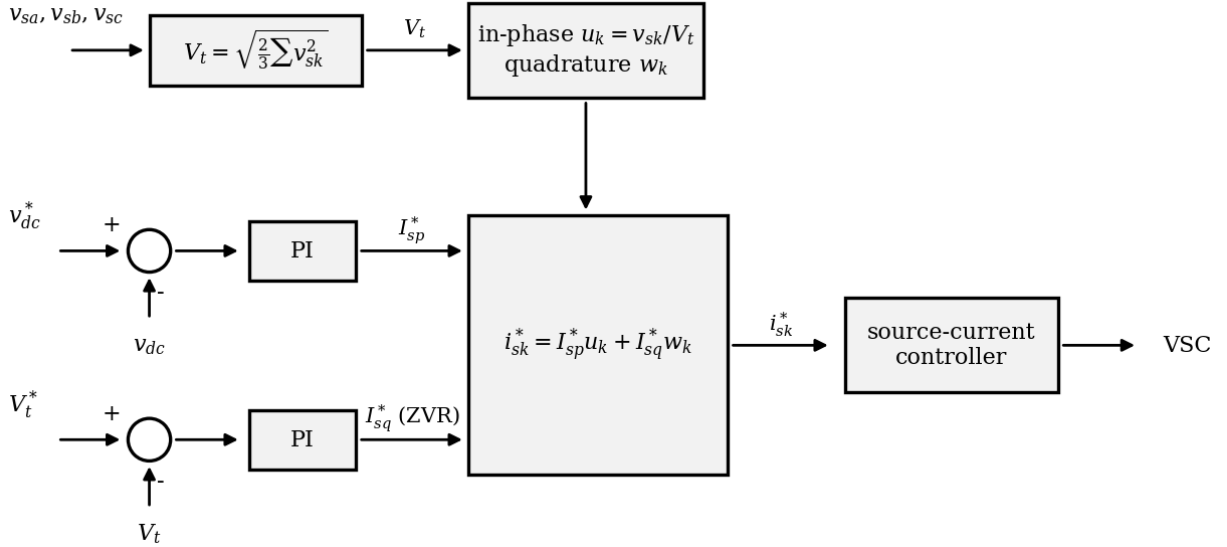


Figure 9.11. Unit-template (PI-controller-based) reference generation for UPF and ZVR modes.

Synchronous reference frame (SRF, dq) theory

The SRF method transforms the load currents into a frame rotating with the positive-sequence fundamental voltage, where that component appears as DC [10]. With θ from a PLL locked to the PCC voltage, the amplitude-invariant Park transformation is

$$\begin{bmatrix} i_{Ld} \\ i_{Lq} \\ i_{L0} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) \\ -\sin \theta & -\sin\left(\theta - \frac{2\pi}{3}\right) & -\sin\left(\theta + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} i_{La} \\ i_{Lb} \\ i_{Lc} \end{bmatrix} \quad (9.11)$$

With voltage $v_a = V_m \cos \theta$, a fundamental positive-sequence current $I_m \cos(\theta - \phi)$ maps to $i_{Ld} = I_m \cos \phi$ and $i_{Lq} = -I_m \sin \phi$, both constant: a lagging load has negative i_{Lq} in this convention. Harmonics and negative sequence appear as ripple; for a six-pulse rectifier on a 50 Hz supply the 5th and 7th harmonics both map to 300 Hz in the dq frame. A low-pass filter, or a moving average over one sixth of a cycle for balanced six-pulse loads, extracts the DC part $\dot{i}_{Ld}$. The references are then (Figure 9.12)

$$\dot{i}_d^* = \dot{i}_{Ld} + i_{loss}, \dot{i}_q^* = \begin{cases} 0, & \text{UPF mode} \\ i_{qr}, & \text{ZVR mode} \end{cases} \quad (9.12)$$

where i_{loss} is the output of the DC-voltage PI and i_{qr} that of an AC-voltage PI. The inverse transformation $\dot{i}_{sa}^* = \dot{i}_d^* \cos \theta - \dot{i}_q^* \sin \theta$ (and likewise for b and c with $\theta \mp 2\pi/3$) returns balanced, sinusoidal source-current references. A positive $\dot{i}_q^*$ is a leading source current, which raises the PCC voltage through the feeder inductance. Because the reference is built from the PLL angle rather than from the measured voltage waveform, SRF control gives clean sinusoidal source currents even when the PCC voltage is distorted. Its accuracy depends on the PLL and on the low-pass filter, which trades dynamic response against ripple rejection.

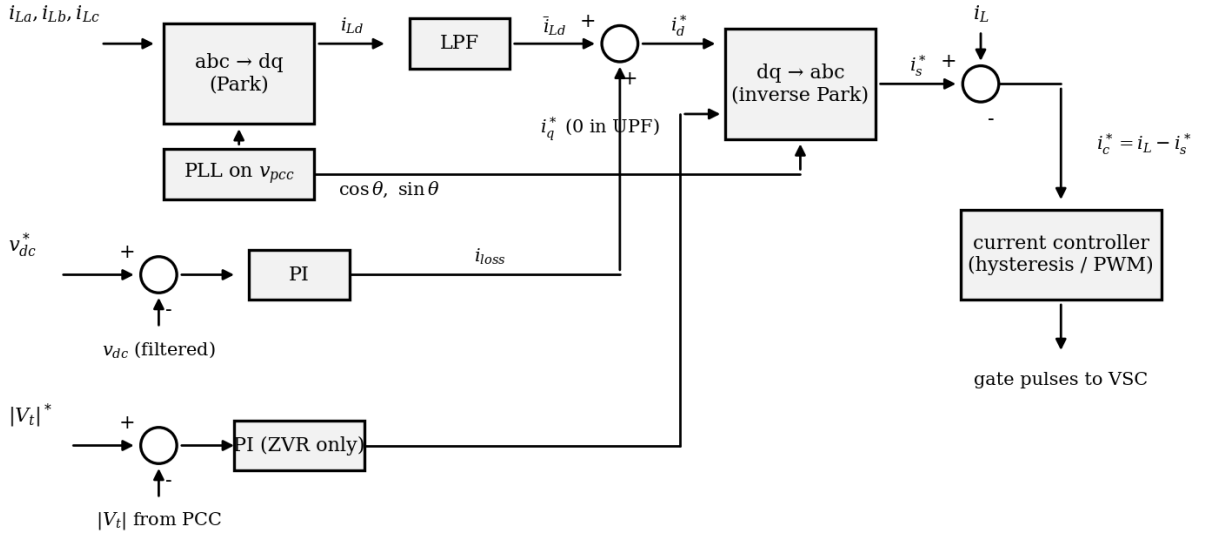


Figure 9.12. SRF (dq) control of a three-leg DSTATCOM.

Instantaneous reactive power (pq) theory

Akagi, Kanazawa and Nabae's instantaneous power theory [11], [12] works in the stationary $\alpha\beta$ frame. The first step applies the Clarke transformation to voltages and currents (Figure 9.13):

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (9.13)$$

The $2/3$ factor is amplitude-invariant: balanced currents of peak I_m trace an $\alpha\beta$ circle of radius I_m . With zero sequence absent, true instantaneous three-phase power is $P_{3\phi} = \frac{3}{2}p$; a power-invariant transform uses a different scaling.

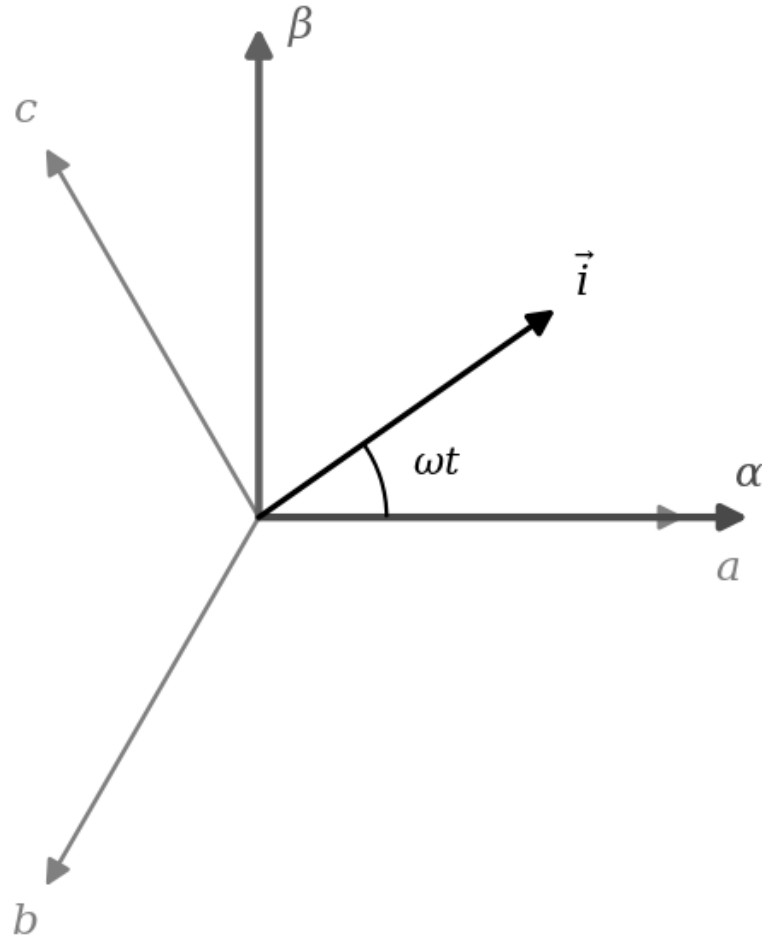


Figure 9.13. Amplitude-invariant Clarke axes.

The instantaneous real and imaginary powers are

$$p = v_\alpha i_\alpha + v_\beta i_\beta, q = v_\alpha i_\beta - v_\beta i_\alpha \quad (9.14)$$

With the positive- β axis and cosine phase sequence used here, a lagging balanced load has $q < 0$ and its conventional reactive power is $Q = -\frac{3}{2}\dot{q}$; this sign must be carried into the inverse and into the controller. For balanced sinusoidal voltages, a linear balanced load gives constant p and q ; harmonics and unbalance in the current produce oscillating parts. Distorted or unbalanced voltage also produces oscillating power, so practical controllers often build the current reference from the extracted positive-sequence fundamental voltage.

Split load powers into average and oscillating parts: $p_L = \dot{p}_L + \tilde{p}_L$ and $q_L = \dot{q}_L + \tilde{q}_L$. For harmonic-only compensation choose $\dot{p}_C = \tilde{p}_L - p_{loss}$ and $\dot{q}_C = \tilde{q}_L$; for harmonic and fundamental-reactive compensation choose $\dot{q}_C = q_L$. Because converter current is positive **into the PCC**, losses require negative average

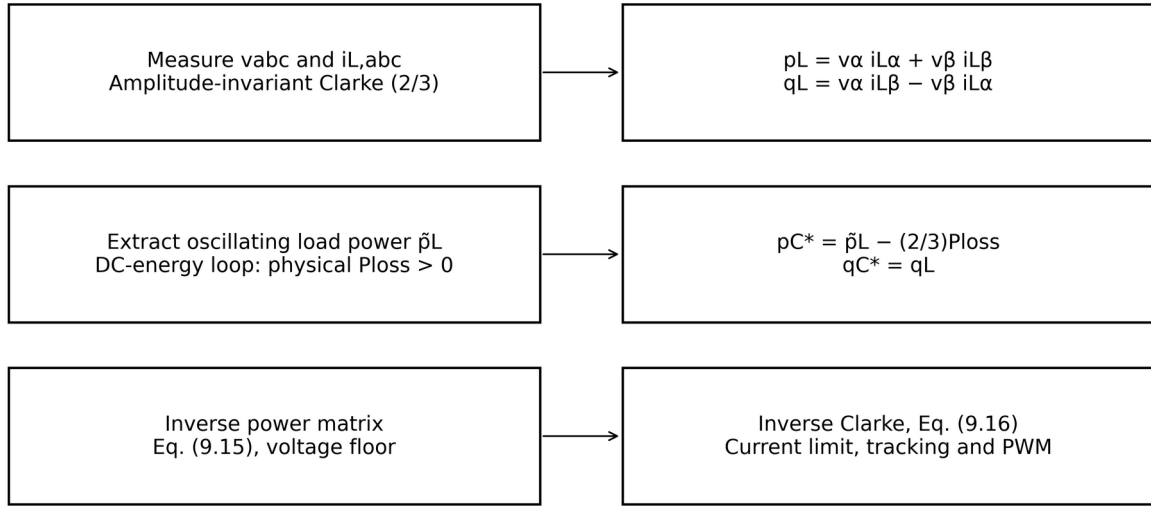
converter export: the loss term is **subtracted**. With amplitude-invariant Clarke variables, $p_{loss} = (2/3)P_{loss,3\phi}$. Inverting Equation (9.14) gives the current references of Equation (9.15).

$$i_{\alpha,C} = \frac{v_{\alpha} p_C - v_{\beta} q_C}{v_{\alpha}^2 + v_{\beta}^2}, i_{\beta,C} = \frac{v_{\beta} p_C + v_{\alpha} q_C}{v_{\alpha}^2 + v_{\beta}^2} \quad (9.15)$$

and the phase references follow from the inverse amplitude-invariant transformation,

$$\begin{bmatrix} i_{C,a} \\ i_{C,b} \\ i_{C,c} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 \end{bmatrix} \begin{bmatrix} i_{\alpha,C} \\ i_{\beta,C} \\ i_{0,C} \end{bmatrix} \quad (9.16)$$

with $i_{0,C} = 0$ in a three-wire installation. The denominator in Equation (9.15) must be kept away from zero during deep sags; a practical controller saturates the current, gives priority to the most important duty and has a declared fallback. Figure 9.14 shows the complete signal chain.



Current positive from converter into PCC; source current = load – injection

Figure 9.14. pq-based compensating-current reference.

Four-wire extension. A four-wire load can have a nonzero phase-current sum. Extend the transformation with

$$i_0 = \frac{1}{3}(i_a + i_b + i_c), v_0 = \frac{1}{3}(v_a + v_b + v_c) \quad (9.17)$$

For neutral current defined as returning from the load, $i_n = i_a + i_b + i_c = 3i_0$, and the complete real-power identity is

$$P_{3\phi} = \frac{3}{2}(v_{\alpha} i_{\alpha} + v_{\beta} i_{\beta}) + 3v_0 i_0 \quad (9.18)$$

A four-wire DSTATCOM commands $i_{0,C} = i_{0,L}$ so that the source carries no zero sequence. When $v_0 = 0$ this term must be controlled directly as a current; it cannot be reconstructed by dividing a zero-sequence power by v_0 , because nonzero i_0 can exist with zero p_0 . A three-wire bridge without a neutral path cannot deliver zero-sequence current at all.

Other algorithms in brief

Power balance control averages total load real power and uses the chosen voltage templates to obtain the active source-current reference. In $I \cos \phi$ control, first isolate the fundamental load current. With $v = V_m \cos \theta$ and $i_{L1} = I_p \cos \theta + I_Q \sin \theta$, the sample at the positive voltage peak ($\theta = 0$) gives I_p , while the positive sine-template peak ($\theta = \pi/2$, a voltage zero crossing) gives I_Q . Equivalently, sample at the zero crossings of the complementary orthogonal template, with the correct sign. Average the phase coefficients and use in-phase/quadrature templates. Sampling raw distorted current would contaminate these coefficients. Current synchronous detection also depends on template extraction. The balanced-voltage form of instantaneous symmetrical-component reference generation is

$$i_{sa}^* = \frac{v_{sa} + \beta(v_{sb} - v_{sc})}{\sum_k v_{sk}^2} \dot{P}_L, \beta = \frac{\tan \phi_p}{\sqrt{3}} \quad (9.19)$$

where b and c follow cyclically. In Equation (9.19), use balanced sinusoidal positive-sequence voltage templates, and take P_L as total average three-phase load real power (add converter losses and the DC-regulator correction once when required). For these templates the denominator is $3V_m^2/2$; positive ϕ_p gives a lagging source-current reference, and $\beta = 0$ gives UPF. Substituting raw unbalanced or distorted voltages does not generally yield balanced sinusoidal currents. An Adaline example estimates the active fundamental coefficient by adapting a weight W_{pa} multiplying the template:

$$W_{pa}(k+1) = W_{pa}(k) + \eta [i_{La}(k) - W_{pa}(k)u_a(k)]u_a(k) \quad (9.20)$$

In Equation (9.20), η is the adaptation step size. It must be chosen for template normalization, sampling rate, convergence and noise rejection; no single numerical range suits every design. One weight illustrates active-coefficient estimation, while simultaneous quadrature/harmonic identification needs additional orthogonal basis weights. Single-phase pq/dq methods synthesize an orthogonal signal using a quarter-cycle delay at nominal frequency or a frequency-adaptive quadrature generator; the delay is not an exact broadband Hilbert transform. PR and selective-harmonic controllers provide high gain around specified frequencies. Repetitive control uses periodic memory and needs delay, frequency-variation and stability checks.

Table 9.7. Comparison of DSTATCOM reference-generation and regulation methods (qualitative; after [9]).

Method	Main idea	Strengths	Limitations
Unit template / PI	DC-link PI sets in-phase amplitude	Very simple, few sensors	Slow (DC-loop limited); sensitive to distorted voltage
SRF (dq)	Park transform, LPF on i_{Ld}	Clean sinusoidal reference with PLL; widely used	Needs PLL; LPF delay; sequence extraction for unbalance
pq (IRPT)	Instantaneous p, q in $\alpha \beta$	Fast; clear harmonic and reactive separation	Needs ideal voltage or positive-sequence extraction
Power balance, $I \cos \phi$, CSD	Average active/reactive load power or current	Simple averaging	Response tied to averaging window
ISC theory	Symmetrical components, forced balanced current	Direct balancing, chosen pf	Sensitive to voltage distortion

Method	Main idea	Strengths	Limitations
Adaline (LMS)	Adaptive weight estimation	Fast tracking, no PLL	Step-size trade-off
DFT / recursive DFT	Fourier extraction per cycle	Selective harmonic control	One-cycle window; computation
PR / repetitive	Resonant or periodic internal model	Zero steady-state error at selected harmonics	Tuning burden; delay and robustness

Current control, limiting and anti-windup

A hysteresis controller switches when current leaves a band around its reference. It provides fast direct feedback without an explicit load model, but its switching frequency depends on the band, coupling inductance, DC voltage, grid voltage and reference slope. A fixed-frequency PWM regulator uses PI control in a suitable rotating frame or resonant control for sinusoidal stationary-frame references. Deadbeat, predictive and sliding-mode methods are also used. Verify tracking gain and phase error at every compensated harmonic, with sampling/PWM delay and stability margin included. Selective resonant gain can improve chosen harmonics without requiring a uniformly wide broadband response [35]. The DC link must also provide the voltage needed for $L_r di_c/dt$, resistance, grid voltage and control margin.

During a severe load step or a depressed PCC voltage, the requested current can exceed the converter rating, and the reference must be limited before modulation. A useful priority is semiconductor safety first, DC-link stability second and the chosen power-quality objective third, although the exact order is application dependent. Anti-windup is needed in every PI loop because a saturated converter cannot realize the controller output. Harmonic cancellation, power-factor correction and voltage regulation compete for the same current, so a controller that claims all three must state how the current limit is shared when their references conflict.

9.3.4 Design of a DSTATCOM

The design of a three-leg VSC-based DSTATCOM covers the DC-bus voltage and capacitance, the interfacing inductor, the ripple filter and the switch ratings [2].

DC-bus voltage. With sinusoidal PWM, the peak phase voltage a three-leg VSC can produce is $m V_{dc}/2$. The DC voltage must therefore exceed twice the peak phase voltage of the PCC:

$$V_{dc} = \frac{2\sqrt{2}V_{LL}}{\sqrt{3}m} \quad (9.21)$$

Equation (9.21) is the zero-drop SPWM lower bound, not a complete DC-voltage design. At 415 V it is 677.7 V for $m=1$. Check the **minimum** voltage in the permitted DC window against the converter-terminal voltage, including coupling-inductor drop: $V_{dc,min} \geq 2\sqrt{2}|V_{PCC} + j\omega L_r I_c|/m_{max}$. Harmonic tracking additionally needs voltage for $L_r di_c/dt$, resistance, dead time and control margin. A 700 V bus permitted to fall by 5% reaches 665 V, below even the zero-drop bound. Space-vector modulation has a different utilization limit and must be identified explicitly when used.

DC-bus capacitance. The physical requirement is a specified real-energy deficit ΔE_{real} , obtained by integrating the DC-power imbalance over the controller recovery interval. Then $C_{dc} \geq 2\Delta E_{real}/(V_{dc,max}^2 - V_{dc,min}^2)$. Equation (9.22) is a preliminary energy-budget rule: it assumes that a declared fraction of apparent-power duty acts as real-power imbalance for time t . It is not a claim that reactive volt-amperes are continuously taken from the capacitor.

$$\frac{1}{2}C_{dc}(V_{dc}^2 - V_{dc1}^2) = k3V_{ph}(aI)t \quad (9.22)$$

Here $k3V_{ph}aIt$ is the chosen buffer-energy budget, with overload factor a and duty fraction k . It must be checked against the transient power and minimum modulation voltage. In a split-capacitor bridge, neutral-current **peak amplitude** I_n flowing at the midpoint creates ripple on each equal capacitor. With an approximately stiff total DC voltage and sinusoidal neutral current, the midpoint current divides equally and each capacitor has $\Delta v_{pp} = I_n/(\omega C)$, giving Equation (9.23). RMS neutral-current data require multiplication by $\sqrt{2}$ before using that equation. Triplen harmonics require their individual frequency-dependent ripple contributions.

$$C \geq \frac{\hat{I}_n}{\omega \Delta v_{pp}} \quad (9.23)$$

A four-leg converter returns neutral current through its fourth leg rather than directly into a split-capacitor midpoint. Its DC link still buffers switching ripple and the power pulsation produced by single-phase or unbalanced loading.

Interfacing inductance. The inductor limits the switching-frequency ripple in i_c . For a three-leg VSC with switching frequency f_s and permitted peak-to-peak ripple Δi_{cr} ,

$$L_r = \frac{\sqrt{3}mV_{dc}}{12af_s\Delta i_{cr}} \quad (9.24)$$

Choose a permitted peak-to-peak ripple and state its current base. Equation (9.24) is the preliminary three-leg ripple rule adopted here, with overload factor a ; actual ripple also depends on PWM sequence, duty ratio and operating voltage. Confirm it with the selected modulation and switching model. Larger L_r reduces ripple but requires more voltage to track a given current slope. For a single-phase H-bridge the preliminary rule used here is $L_r = mV_{dc}/(4af_s\Delta i_{cr})$; bipolar and unipolar PWM have different ripple waveforms and must be checked explicitly.

Ripple filter. A series R_f - C_f branch at the PCC presents a low impedance at the switching frequency and a high impedance at 50 Hz. Its corner frequency is placed near half the switching frequency:

$$f_r = \frac{1}{2\pi R_f C_f} = \frac{f_s}{2} \quad (9.25)$$

This is an RC corner frequency, not a resonance. With $R_f = 5\Omega$ and $f_s = 10$ kHz, $C_f = 6.366\mu\text{F}$ gives a 5 kHz corner. Selecting $10\mu\text{F}$ moves it to 3.183 kHz; the new value must be checked for fundamental reactive current, resistor heating, switching attenuation and network interaction.

Semiconductor ratings. As an initial screen, this chapter uses $V_{sw} = 1.1V_{dc,max}$ and a 25% current margin in Equation (9.26). These are declared screening allowances, not safe ratings in themselves. Select the device voltage class from the maximum DC voltage plus measured or calculated commutation overshoot, and check repetitive blocking, short-circuit withstand, junction temperature, switching loss and cooling. The RMS compensating current I_c enters the sinusoidal peak estimate; distorted currents require their actual worst-case peak and crest factor.

$$I_{sw} = 1.25(\Delta i_{cr} + \sqrt{2}I_c) \quad (9.26)$$

Adding the full peak-to-peak ripple, as here, is slightly conservative; some texts add only half of it. A design should state its convention.

9.3.5 Analytical and reduced-model performance illustrations

The following illustrations are explicitly defined reduced models, not switched-converter or experimental validation. Use $V_{ph}=415/\sqrt{3}$ V, $V_m=\sqrt{2}V_{ph}$, 50 Hz and phase angles $\theta_k=\omega t+[0, -2\pi/3, +2\pi/3]_k$. The load is $i_{Lk}=A\cos(\theta_k-\cos^{-1}0.8)+A\sum_h c_h\cos(h\theta_k)$, with $h=[5, 7, 11, 13, 17]$ and $c_h=[0.10, 0.07, 0.04, 0.02, 0.02](0.131/\sqrt{0.0173})$. The fundamental peak A steps from 100 to 150 A at 0.25 s. Source current is $i_{sk}=B\cos\theta_k+0.08A\sum_h c_h\cos(h\theta_k)$; this specified residual represents incomplete tracking rather than a calculated PWM response.

The source-amplitude model is $\dot{F}=(0.8A-F)/0.004$, $\dot{B}^i=F+50(W_0-W)/(1.5V_m)$, $\dot{B}=(B^i-B)/0.0005$, and $\dot{W}=1.5V_m(B-0.8A)$, initialized with $F=B=80$ A and $W=W_0=1/2(0.02)(700)^2$ J. Forward Euler uses 50 μ s steps. The plotted DC voltage is reconstructed from the **instantaneous** AC exchange: $V_{dc}=\sqrt{2[W_0-\int\sum_k v_k(i_{Lk}-i_{sk})dt]}/0.02$, using trapezoidal integration. Losses, modulation saturation, PWM, PLL dynamics and protection are omitted. The 700 V bus here is an energy illustration, not a hardware headroom recommendation.

Figures 9.15 to 9.17 show the prescribed harmonic cancellation and the energy-buffer response. The source fundamental is aligned with the PCC voltage. The load THD is 13.1%; reducing harmonic amperes to 8% of their original values gives source THD 1.31%, since its fundamental amplitude is 0.8 of the load fundamental in steady state. A finite source-amplitude response produces a DC-voltage dip and subsequent recovery. These results illustrate reference generation and energy balance; they do not establish semiconductor or filter performance.

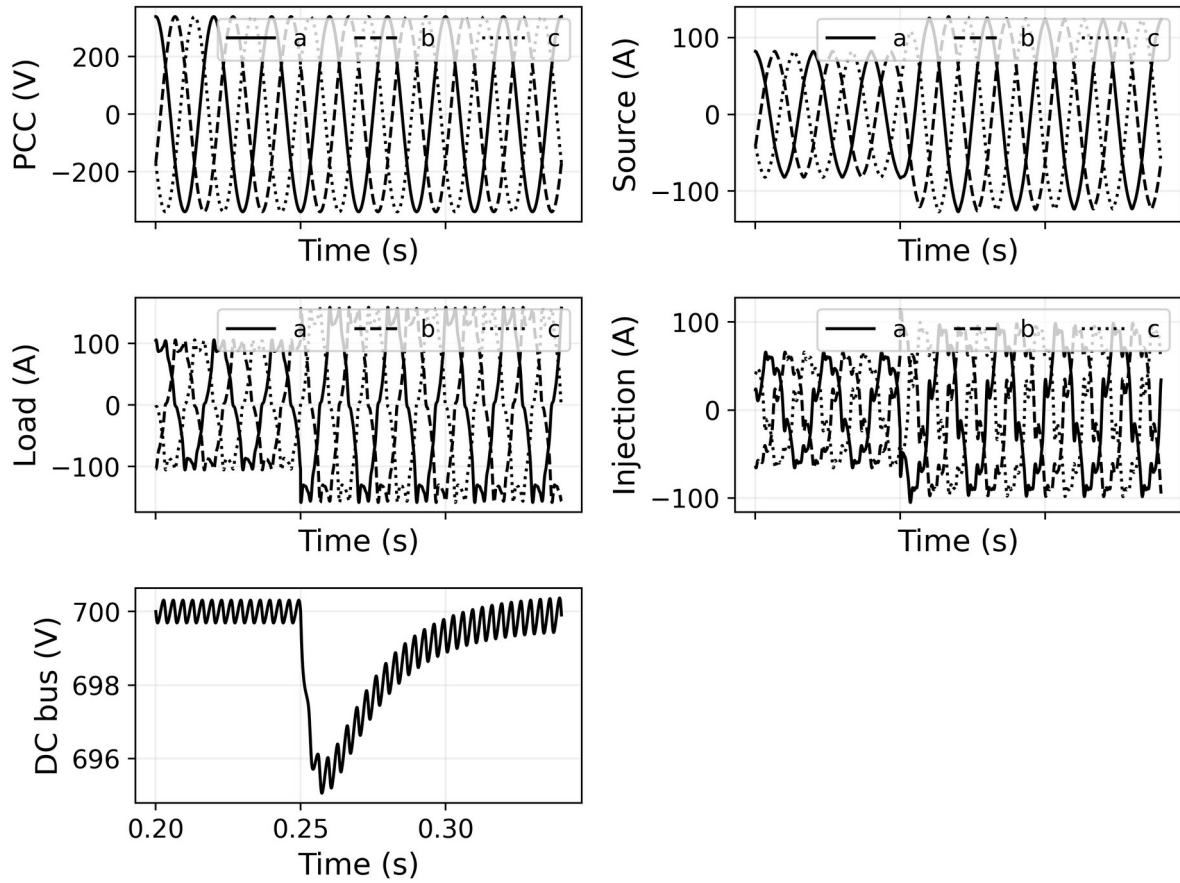


Figure 9.15. Defined reduced DSTATCOM model: PCC voltage, source current, load current, injected current and reconstructed DC voltage. Load fundamental peak changes from 100 to 150 A at 0.25 s; model and numerical method are specified in Section 9.3.5.

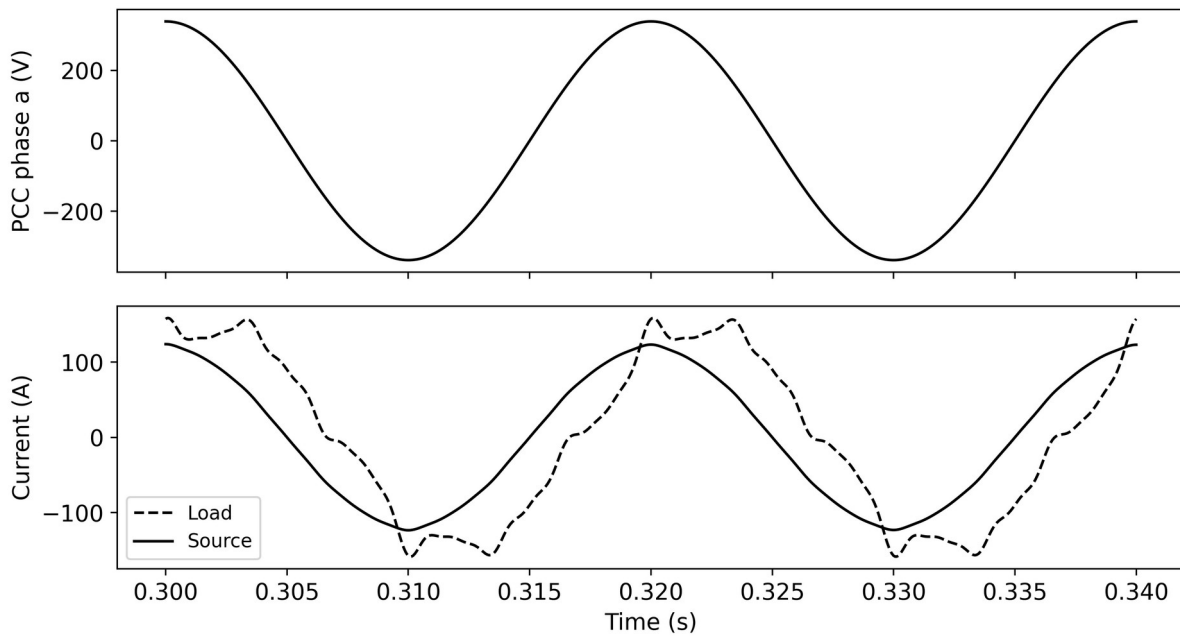


Figure 9.16. Phase-a detail in UPF mode: source current in phase with PCC voltage; load current distorted.

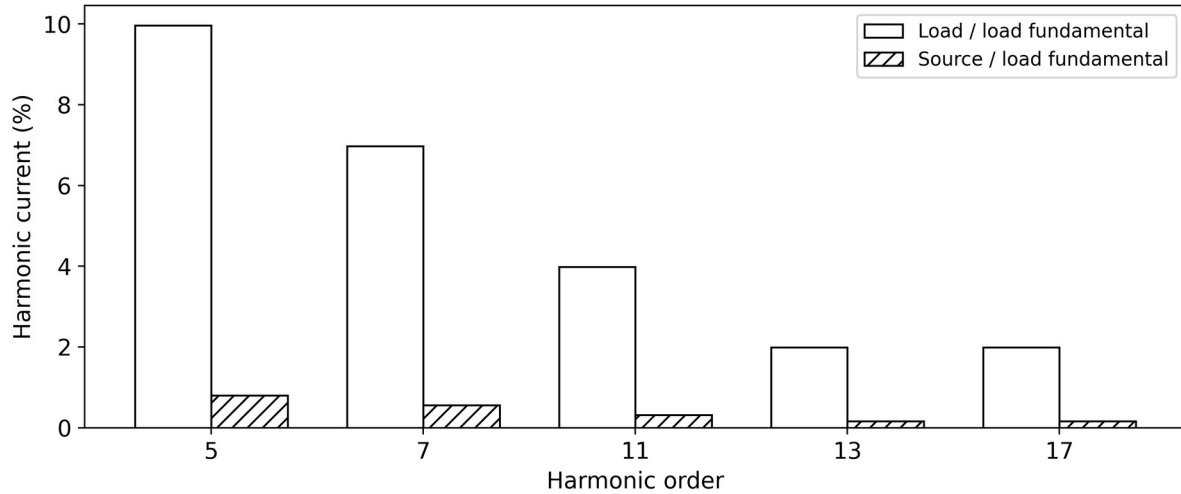


Figure 9.17. Prescribed harmonic spectra after the load step, normalized to the load fundamental. Load THD is 13.1%; the source fundamental is 80% of the load fundamental, so source THD is 1.31%.

Figure 9.18 is a separate fundamental-phasor envelope comparison, with a 415 V, 50 Hz source, $R_s = 0.02 \Omega$, $L_s = 2$ mH and constant load real power stepping from 30 to 45 kW at 0.25 s. For UPF solve $|V_L + (R_s + j\omega L_s)P/(3V_L)| = 415/\sqrt{3}$ on the higher-voltage branch. For ZVR set $V_L = 415/\sqrt{3}$ and use the smaller feasible root of Equation (9.7). The UPF voltage and both source-current peak envelopes interpolate between their steady solutions with $1 - e^{-(t-0.25)/0.006}$ after the step; ZVR voltage is the imposed ideal regulated target. This is a declared 6 ms envelope, not a derived closed-loop transient. ZVR requires extra reactive current and sufficient converter voltage/current capacity.

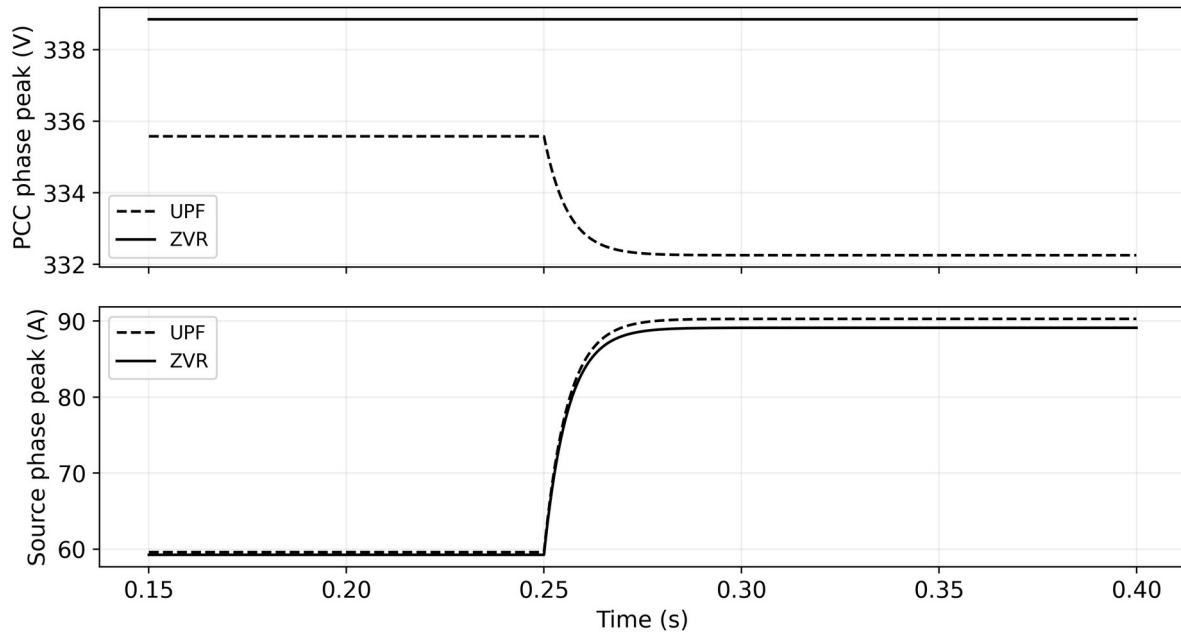


Figure 9.18. Fundamental-phasor envelope illustration of UPF versus ZVR on a weak feeder; phase voltage and source current are peak amplitudes. The 6 ms envelope is specified, and the ZVR target is ideal.

Figure 9.19 illustrates ideal four-wire current references. At 415 V, phase a has a 10 kW resistive load; phase b has a 5 kW, 0.8 lagging-pf load removed at 0.2 s; phase c draws $25 \cos \theta_c + 10 \cos 3\theta_c + 6 \cos 5\theta_c$ A. The balanced source-current peak is $P_L / (1.5 V_m)$, with $P_L = 15\,000 + 25 V_{ph} / \sqrt{2}$ W before the change and $10\,000 + 25 V_{ph} / \sqrt{2}$ W afterward. Set $i_c = i_L - i_s$. The load neutral return is $\sum i_{Lk}$ and the compensator supplies it locally. Instantaneous reference changes and perfect tracking are assumed; there is no claim of a simulated neutral-leg transient or DC-link response.

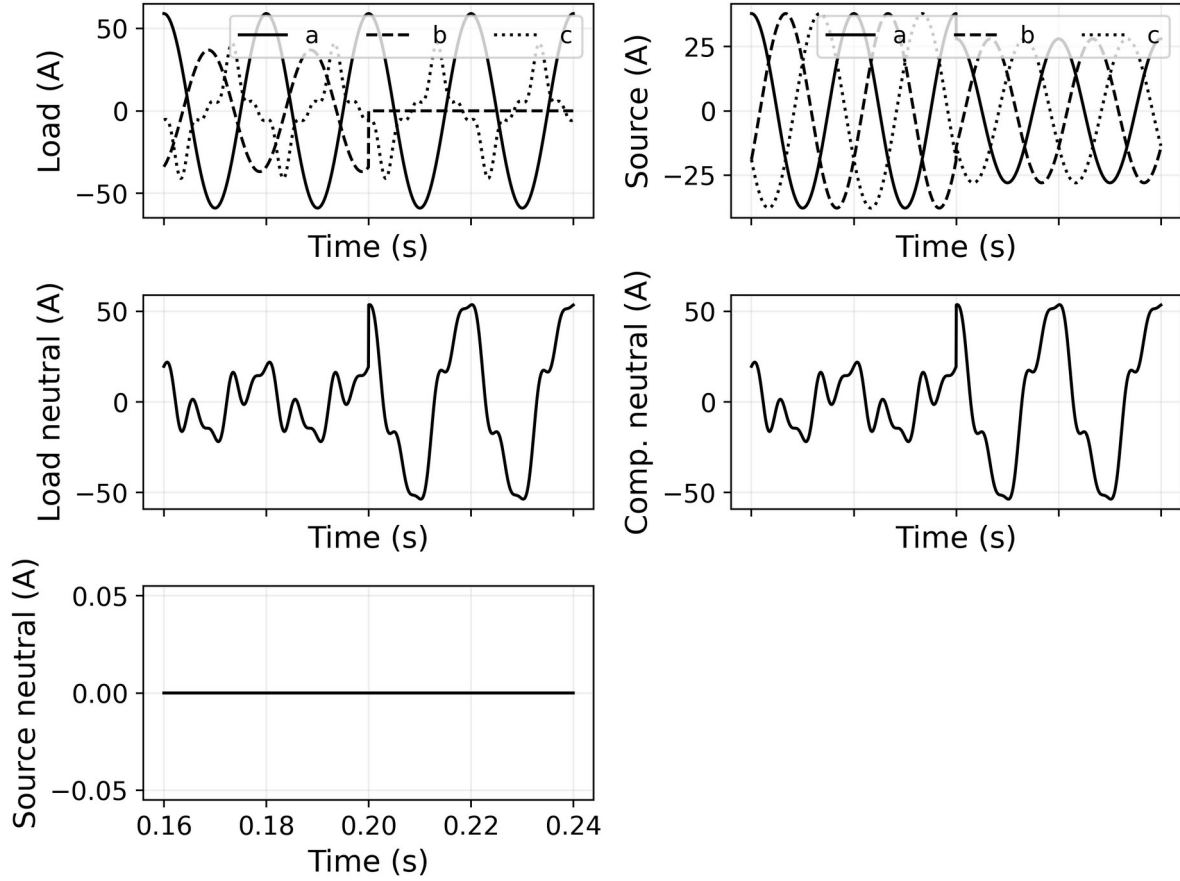


Figure 9.19. Ideal four-wire balancing references and neutral-current paths, using the explicitly prescribed phase-c harmonic load. Perfect tracking is assumed; the source neutral is zero by construction.

9.3.6 Worked examples: DSTATCOM

Example 9.1. UPF compensation and preliminary hardware sizing for a motor load

A 415 V, 50 Hz three-wire DSTATCOM corrects a 25 kW, 0.8 lagging-pf delta motor to unity source power factor. Find currents, star-equivalent resistance, PCC reactive duty, inductance and capacitance. Use $V_{dc,max} = 900$ V, 8% droop, $f_s = 10$ kHz, $m_{max} = 1$, $a = 1.2$, peak-to-peak ripple 15% of compensating RMS current, and a **declared 225 J transient real-energy budget**. Check fundamental modulation headroom and preliminary switch ratings.

Solution. $V_{ph} = 239.600$ V. Before compensation $I_s = 25\,000 / (\sqrt{3} 415 0.8) = 43.475$ A; afterward $I_s = 34.780$ A. The star-equivalent resistance is $V_{ph} / I_s = 6.889 \Omega$. The compensating current is $43.475(0.6) = 26.085$ A, giving PCC duty $3 V_{ph} I_c = 18.750$ kVA. Ripple is 3.913 A peak-to-peak; Equation (9.24) gives $L_r = 2.767$ mH. With $V_{dc,min} = 828$ V, $C = 450 / (900^2 - 828^2) = 3.617$ mF for the stated energy budget. For capacitive current the required converter fundamental is $V_c = V_{ph} + \omega L_r I_c = 262.27$ V RMS, below $828 / (2\sqrt{2}) = 292.74$ V available. The converter-terminal

fundamental duty is about 20.52 kVA, greater than the PCC reactive duty. Switch screens are $I_{sw}=1.25(3.913+\sqrt{2}26.085)=51.0$ A and $V_{sw}=990$ V; a 1200 V, 75 A class is a preliminary choice subject to switching, thermal and overload checks. $R_f=5\ \Omega$ and $C_f=6.366\ \mu\text{F}$ give a 5 kHz corner. For comparison, a 700 V bus with the same 8% droop falls to 644 V, below even the 677.7 V zero-drop SPWM bound, and a larger 8.61 mH inductor would need about 310 V converter RMS. A correct capacitor-energy calculation alone does not make a design feasible.

Example 9.2. Balancing a single-phase load connected across two lines

A three-phase three-wire DSTATCOM at 415 V, 50 Hz is used for load balancing and power-factor correction of a single-phase 50 kW unity-power-factor load connected between lines a and b. Calculate (a) the load line currents, (b) the supply line currents after compensation, (c) the DSTATCOM currents and (d) its kVA rating.

Solution. (a) $V_{ab}=415\angle 30^\circ$ V and $|Z_L|=415^2/50\,000=3.44\ \Omega$, so $I_{La}=120.48\angle 30^\circ$ A, $I_{Lb}=-I_{La}=120.48\angle -150^\circ$ A and $I_{Lc}=0$. (b) The balanced UPF source currents carry 50 kW: $I_s=50\,000/(3\times 239.6)=69.56$ A, so $I_{sa}=69.56\angle 0^\circ$, $I_{sb}=69.56\angle -120^\circ$ and $I_{sc}=69.56\angle 120^\circ$ A. (c) From $I_c=I_L-I_s$: $I_{ca}=69.56\angle 60^\circ$ A, $I_{cb}=69.56\angle 180^\circ$ A and $I_{cc}=69.56\angle -60^\circ$ A. (d) $S_{DST}=239.6\times 3\times 69.56=50.0$ kVA. The DSTATCOM rating equals the load rating. This is a general result for a single-phase resistive load balanced by an ideal compensator: the negative-sequence power to be circulated equals the load power, the same conclusion reached for the passive Steinmetz circuit, now realized with a converter.

Example 9.3. Four-wire unbalanced star load realized as a balanced UPF load

A three-phase four-wire unbalanced star-connected load with $Z_a=9.0+j3.0$ pu, $Z_b=3.0+j1.5$ pu and $Z_c=7.5+j1.5$ pu (base impedance $9.15\ \Omega$) is supplied at 415 V, 50 Hz. A four-leg DSTATCOM realizes it as a balanced UPF load. Find (a) the source currents, (b) the DSTATCOM phase currents, (c) its neutral current, (d) its kVA rating and (e) the equivalent per-phase resistance.

Solution. The load currents are $2.760\angle -18.43^\circ$, $7.807\angle -146.57^\circ$ and $3.424\angle 108.69^\circ$ A. Total real power is 3104.89 W. (a) The balanced UPF source current is 4.319 A per phase. (b) Subtraction gives compensator currents $1.912\angle -152.84^\circ$, $4.391\angle -172.66^\circ$ and $1.173\angle -25.10^\circ$ A. (c) The neutral return is $5.354\angle -158.85^\circ$ A and is supplied by the compensator. (d) The sum of phase AC duties is $239.600(1.912+4.391+1.173)=1.791$ kVA. (e) The equivalent resistance is $239.600/4.319=55.47\ \Omega$. The compensator circulates reactive and unbalanced current without average ideal real-power export.

The 1.791 kVA value is the **sum of phase AC duties**, not the installed nameplate of a conventional equal-leg bridge. Rating all three phase legs for the largest 4.391 A gives $3V_{ph}(4.391)=3.156$ kVA. The phase-current maximum is 6.210 A peak before ripple, and the neutral is 5.354 A RMS or 7.572 A peak. Neutral terminal potential does not remove neutral-switch blocking voltage, conduction loss or switching stress; each leg requires its own hardware check.

Example 9.4. Load balancing and zero voltage regulation on a weak feeder

A 415 V, 50 Hz supply has a feeder impedance of $1+j3\ \Omega$ per phase. A single-phase load $Z_a=10+j5\ \Omega$ is connected between phase a and neutral; phases b and c are unloaded. A four-leg DSTATCOM at the load bus balances the load and regulates the load-bus voltage to 239.6 V per phase. Find (a) the load-bus voltage without compensation, (b) the source currents, (c) the DSTATCOM currents and (d) its rating.

Solution. (a) Without compensation, phase a sees $V_{La}=239.6Z_a/(Z_a+Z_s)=196.9\angle -9.46^\circ$ V, an 18 % sag in that phase. (b) With $V_L=239.6$ V, $I_{La}=239.6/(10+j5)=21.43\angle -26.57^\circ$ A and

$P_L = 21.43^2 \times 10 = 4592.7$ W, or 1530.9 W per phase, so $I_p = 6.389$ A. In Equation (9.7), $A = 245.99$ and $B = 19.17$, giving $10 I_q^2 - 1437.6 I_q + 3470.0 = 0$ and $I_q = 2.456$ A. Thus $I_{sa} = 6.389 + j2.456 = 6.845 \angle 21.02^\circ$ A, and I_{sb}, I_{sc} have the same magnitude displaced by $\mp 120^\circ$. The source EMF then leads the load bus by 5.18° . Because the source current leads the bus voltage, 588.4 var per phase flows from the load bus back toward the source, and this reactive flow through X_s is what lifts the bus voltage. (c) $I_{ca} = I_{La} - I_{sa} = 17.56 \angle -43.29^\circ$ A, $I_{cb} = -I_{sb} = 6.845 \angle 81.02^\circ$ A and $I_{cc} = -I_{sc} = 6.845 \angle -38.98^\circ$ A. The neutral leg carries $21.43 \angle -26.57^\circ$ A, the whole load current returning through the compensator. (d) $S_{DST} = 239.6 \times (17.56 + 6.845 + 6.845) = 7.49$ kVA, plus a 21.4 A neutral leg. The feeder drop is $|Z_s I_s| = 21.6$ V, but because the source current leads, the drop is almost perpendicular to the load voltage and $|V_s| = |V_L|$: the ZVR principle of Figure 9.6(c).

The 7.487 kVA sum is an operating phase-duty measure. An equal-leg bridge rated at the largest phase current, 17.557 A RMS, requires $3 V_{ph}(17.557) = 12.620$ kVA before margins. The neutral carries 21.431 A RMS or 30.307 A peak, greater than the largest phase-leg peak of 24.830 A. Four-leg thermal and switch selection cannot be inferred from the summed VA alone.

Example 9.5. Power-factor correction rating

A 415 V, 50 Hz system feeds a 45 kW star-connected load at 0.72 lagging power factor. Find the source current before and after UPF compensation, the load reactive power, the DSTATCOM current and kVA, and the equivalent resistance after compensation.

Solution. Before: $I_s = 45\,000 / (\sqrt{3} \times 415 \times 0.72) = 86.95$ A. After: $I_s = 45\,000 / (\sqrt{3} \times 415) = 62.60$ A, a 28 % reduction. $Q_L = 45 \tan(\cos^{-1} 0.72) = 43.37$ kvar. $I_{DST} = 43\,373 / (\sqrt{3} \times 415) = 60.34$ A and $S_{DST} = 43.37$ kVA. $R_{eq} = 239.6 / 62.60 = 3.83 \Omega$ per phase.

Example 9.6. Interfacing inductor

A DSTATCOM operates with $V_{dc} = 900$ V, $f_s = 4$ kHz, $m = 1$, $a = 1.2$ and a permitted ripple of 10 % of its 50 A rated current. Find L_r .

Solution. $\Delta i_{cr} = 5$ A, and Equation (9.24) gives $L_r = \sqrt{3} \times 900 / (12 \times 1.2 \times 4000 \times 5) = 5.41$ mH.

Example 9.7. Harmonic compensation current and rating

A 415 V nonlinear load draws 40 A fundamental (in phase with the voltage) with 5th and 7th harmonics of 20 % and 15 % of the fundamental. Find the RMS load current, the harmonic current, the DSTATCOM current for sinusoidal source current and its rating.

Solution. $I_L = 40 \sqrt{1 + 0.2^2 + 0.15^2} = 41.23$ A and $I_H = 40 \sqrt{0.2^2 + 0.15^2} = 10.0$ A, giving 25% current THD. With unity fundamental displacement, ideal compensation needs $I_{DST} = 10.0$ A and $S_{DST} = 3(239.6)(10.0) = 7.19$ kVA. True load power factor is $1 / \sqrt{1 + 0.25^2} = 0.970$. Adding a plain capacitor cannot cancel the specified harmonic current; a passive harmonic filter would require a separate tuning and resonance study.

Example 9.8. Rating for several loads

A 415 V system supplies 30 kW at 0.8 lagging and 20 kW at 0.9 lagging. Find the DSTATCOM rating for UPF and the source current after compensation.

Solution. $P = 50$ kW and $Q = 30 \times 0.75 + 20 \times 0.4843 = 32.19$ kvar, so $S_{DST} = 32.19$ kVA. After compensation $I_s = 50\,000 / (\sqrt{3} \times 415) = 69.56$ A.

Example 9.9. Clarke transform verification

With balanced voltages $v_a = V_m \cos \omega t$, $v_b = V_m \cos(\omega t - 120^\circ)$, $v_c = V_m \cos(\omega t + 120^\circ)$, a balanced linear load draws $i_a = 10 \cos \omega t$, $i_b = 10 \cos(\omega t - 120^\circ)$, $i_c = 10 \cos(\omega t + 120^\circ)$ A. Using Equation (9.13), find i_α and i_β at $\omega t = 0$ and the instantaneous powers.

Solution. At $\omega t = 0$: $i_a = 10$ and $i_b = i_c = -5$ A, so $i_\alpha = (2/3)(10 + 2.5 + 2.5) = 10$ A and $i_\beta = 0$. Throughout the cycle $i_\alpha = 10 \cos \omega t$ and $i_\beta = 10 \sin \omega t$. With the stated in-phase voltages, $p = 10 V_m$, $q = 0$ and $P_{3\phi} = 15 V_m$. Current data alone, without a voltage reference, could not establish zero reactive power: a circular current locus shows balanced sinusoidal current, not unity power factor.

Example 9.10. Complete three-phase pq compensation calculation

An ideal balanced three-wire 50 Hz supply has $v_k = 100 \cos \theta_k$ V with $\theta_a = \theta$, $\theta_b = \theta - 2\pi/3$, $\theta_c = \theta + 2\pi/3$. The load currents are $i_{L,k} = 10 \cos(\theta_k - \pi/6) + 2 \cos 5\theta_k + \cos 7\theta_k$ A (peak values). Derive the shunt current reference that cancels harmonics and fundamental reactive current, evaluate it at $\theta = \pi/6$ and check the source current and powers.

Solution. The fifth harmonic is negative sequence and the seventh positive sequence for these phase shifts. The Clarke transform gives $v_\alpha = 100 \cos \theta$, $v_\beta = 100 \sin \theta$, $i_{L,\alpha} = 10 \cos(\theta - \pi/6) + 2 \cos 5\theta + \cos 7\theta$ and $i_{L,\beta} = 10 \sin(\theta - \pi/6) - 2 \sin 5\theta + \sin 7\theta$. From Equation (9.14), $p = 866.025 + 300 \cos 6\theta$ and $q = -500 - 100 \sin 6\theta$, so $\tilde{p} = 866.025$, $\tilde{q} = -500$, $\tilde{p} = 300 \cos 6\theta$ and $\tilde{q} = -100 \sin 6\theta$.

Set $p_C = \tilde{p}$ and $q_C = q$. At $\theta = \pi/6$: $(v_\alpha, v_\beta) = (86.603, 50)$ V, $(p_C, q_C) = (-300, -500)$ and $v_\alpha^2 + v_\beta^2 = 10000$. Equation (9.15) gives $i_{C,\alpha} = [86.603(-300) - 50(-500)]/10000 = -0.098$ A and $i_{C,\beta} = [50(-300) + 86.603(-500)]/10000 = -5.830$ A. The inverse Clarke transform gives $(i_{C,a}, i_{C,b}, i_{C,c}) = (-0.098, -5.000, 5.098)$ A. The load currents at the same instant are $(7.402, -5.000, -2.402)$ A, so $i_s = i_L - i_C = (7.5, 0, -7.5)$ A. Over the whole cycle $i_{s,k} = 8.660 \cos \theta_k$ A: balanced sinusoids in phase with their voltages, 6.124 A RMS.

At this instant the physical powers are $P_L = 1.5 \times 566.025 = 849.04$ W, $P_s = 1.5 \times 866.025 = 1299.04$ W and $P_C = 1.5 \times (-300) = -450$ W, so $P_s + P_C = P_L$. The filter absorbs 450 W at this instant and returns it at others, with zero average exchange. The load's fundamental reactive power is $Q_L = -1.5 \tilde{q} = 750$ var. The RMS compensating phase current is $\sqrt{(5^2 + 2^2 + 1^2)}/2 = 3.873$ A, including the reactive fundamental and both harmonics; harmonic-only compensation would use $q_C = \tilde{q}$ and leave the reactive current in the source.

9.4 Dynamic Voltage Restorer (DVR)

A DVR is a series-connected VSC that inserts a controlled voltage between the supply and a sensitive load, so that the load voltage stays at its reference while the supply voltage sags, swells, becomes unbalanced or distorted. The first utility DVR entered service in 1996 on the Duke Power system at Anderson, South Carolina (12.47 kV), and the device has since become a common way to protect a large sensitive load, such as a semiconductor fab or a paper machine, against sags caused by faults elsewhere on the network [13]. The series connection is what makes it economical: a DVR supplies only the missing voltage, so its rating is a fraction of the load rating for sags of practical depth.

9.4.1 Components and operating principle

Figure 9.20 shows a complete DVR. Its main parts are:

- a voltage-source converter, usually a three-leg or three H-bridge IGBT inverter switched at several kHz;

- a DC link and energy source, which supplies the real power the injection requires (Section 9.4.2);
- an injection (series) transformer in each phase, which isolates the converter from the line, matches its voltage and current to the line, and blocks zero sequence where required;
- a passive filter, normally an L_f - C_f low-pass filter on the converter side of the transformer (line-side filters avoid transformer harmonic heating but expose the filter to fault current);
- a bypass switch (antiparallel thyristors or a mechanical switch) that short-circuits the transformer's line winding during downstream faults, overloads or maintenance; and
- a controller with voltage and current sensors, a PLL and disturbance detection.

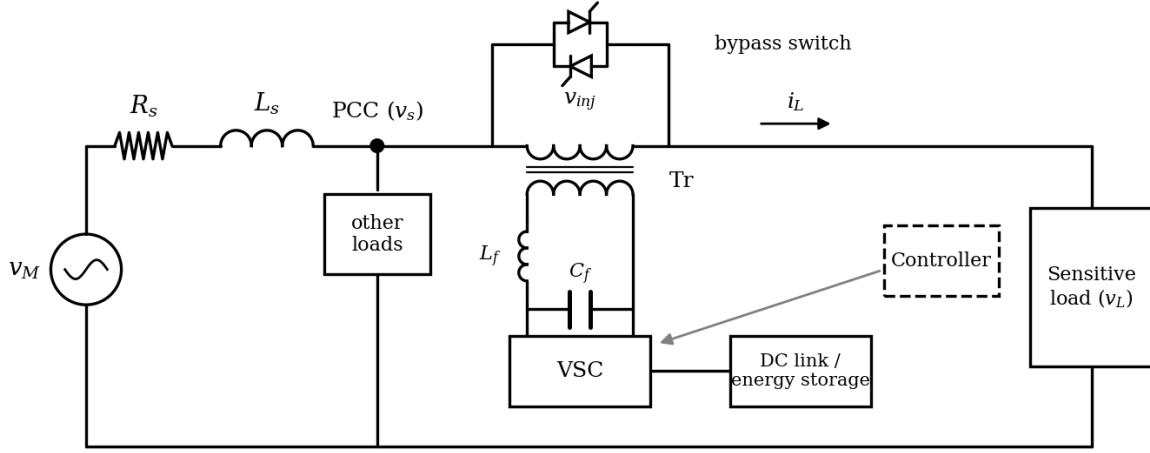


Figure 9.20. Complete DVR arrangement: injection transformer, LC filter, VSC, DC link or energy storage, bypass switch and controller.

With transformer and filter quantities referred to the line side, the governing ideal phasor relationship is

$$V_L = V_s + V_{inj} \quad (9.27)$$

where V_L , V_s and V_{inj} are RMS voltage phasors at the same terminals and positive injection is a rise from source to load, the voltage-rise convention of Chapter 8. Sag depth is $X = 1 - |V_s|/V_{Lc}$, where V_{Lc} is the rated (target) load voltage. The real and reactive power the DVR exchanges with the line are, per phase,

$$P_{DVR} + jQ_{DVR} = V_{inj} I_{DVR}^* \quad (9.28)$$

and for a standalone DVR the series current equals the load current, $I_{DVR} = I_L$.

A DVR has standby, injection and protection modes. In standby a rated bypass or an appropriate converter zero state carries current with minimal injection. After detecting a disturbance, injection builds up within the measured response time of the chosen controller and filter. For a downstream fault beyond the permitted duty, coordinated protection bypasses the series winding and blocks the converter. The switching sequence must be specified; a blocked converter alone does not provide a fault-current path.

9.4.2 Classification and energy-source topologies

DVRs are classified, like the DSTATCOM, by converter type (CSC or VSC), by supply system (single-phase, three-phase three-wire and three-phase four-wire) and by bridge topology. A single-phase DVR may use a half bridge or a full bridge (Figure 9.21). For the same DC voltage the full bridge produces twice the AC output voltage; the half bridge needs two series capacitors and a doubled DC voltage, but only two switches. A three-phase DVR uses either a three-leg VSC feeding three injection transformers with isolated windings (Figure 9.22) or three H-bridges. Three H-bridges, or a four-leg converter, allow independent phase voltages including zero sequence, which a three-leg bridge with delta-connected windings cannot produce. Multilevel and

cascaded structures reduce device voltage stress and dv/dt at higher distribution voltages but need more devices, isolated supplies or capacitor balancing. Transformerless DVRs remove the injection transformer but transfer isolation, common-mode voltage and fault-withstand requirements to the converter and its insulation.

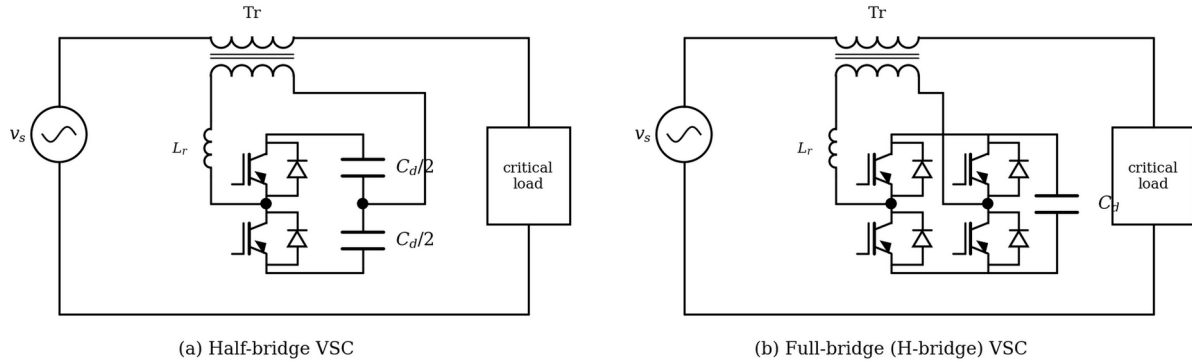


Figure 9.21. Single-phase DVR: (a) half-bridge VSC, (b) full-bridge VSC.

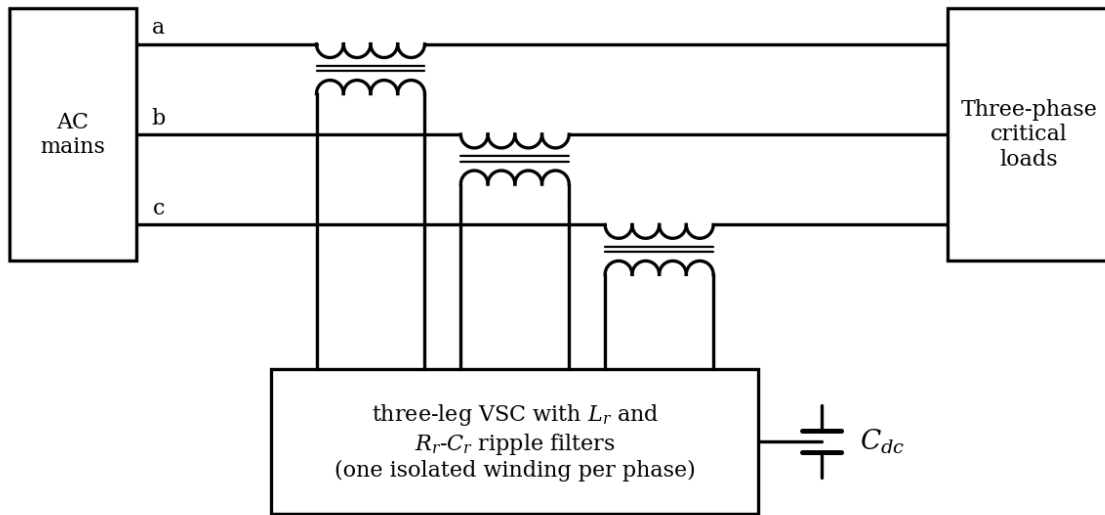


Figure 9.22. Three-phase three-wire DVR: three injection transformers driven by a three-leg VSC with ripple filters.

The most important classification for design is by the source of the injected energy (Figure 9.23) [14]:

- **Supply-side rectifier support.** A rectifier can replenish the DC link while the remaining supply provides sufficient input voltage, current and power. Sustained sag support is limited by rectifier/DC-converter capability, source capacity, thermal duty and modulation headroom. Increasing capacitance alone only extends finite ride-through when input power is inadequate; it cannot correct an indefinite power deficit. Deep-sag designs may need a boost stage or independent storage.
- **No storage, load-side (right-shunt) rectifier.** The rectifier is fed from the restored load voltage, so its input stays at rated value. The extra rectifier current flows through the injection transformer, and the scheme forms a positive-feedback loop that must be controlled carefully.
- **Stored energy (BESS, supercapacitor, flywheel).** A battery or other store, usually through a DC/DC converter, supplies real power. The DVR can use in-phase injection (minimum voltage rating) and can also preserve phase, but the store must be sized for the worst sag depth and duration (Section 9.4.5).
- **Capacitor-supported (self-supported).** A DC capacitor can buffer finite transient real energy. For sustained ideal operation without replenishment, average series real-power export must be zero, so the fundamental injection is perpendicular to the actual series current. This geometry restores only

the sag range derived in Section 9.4.3. Practical operation absorbs enough real power to cover losses, and must also satisfy phase-transition, voltage and current limits [15].

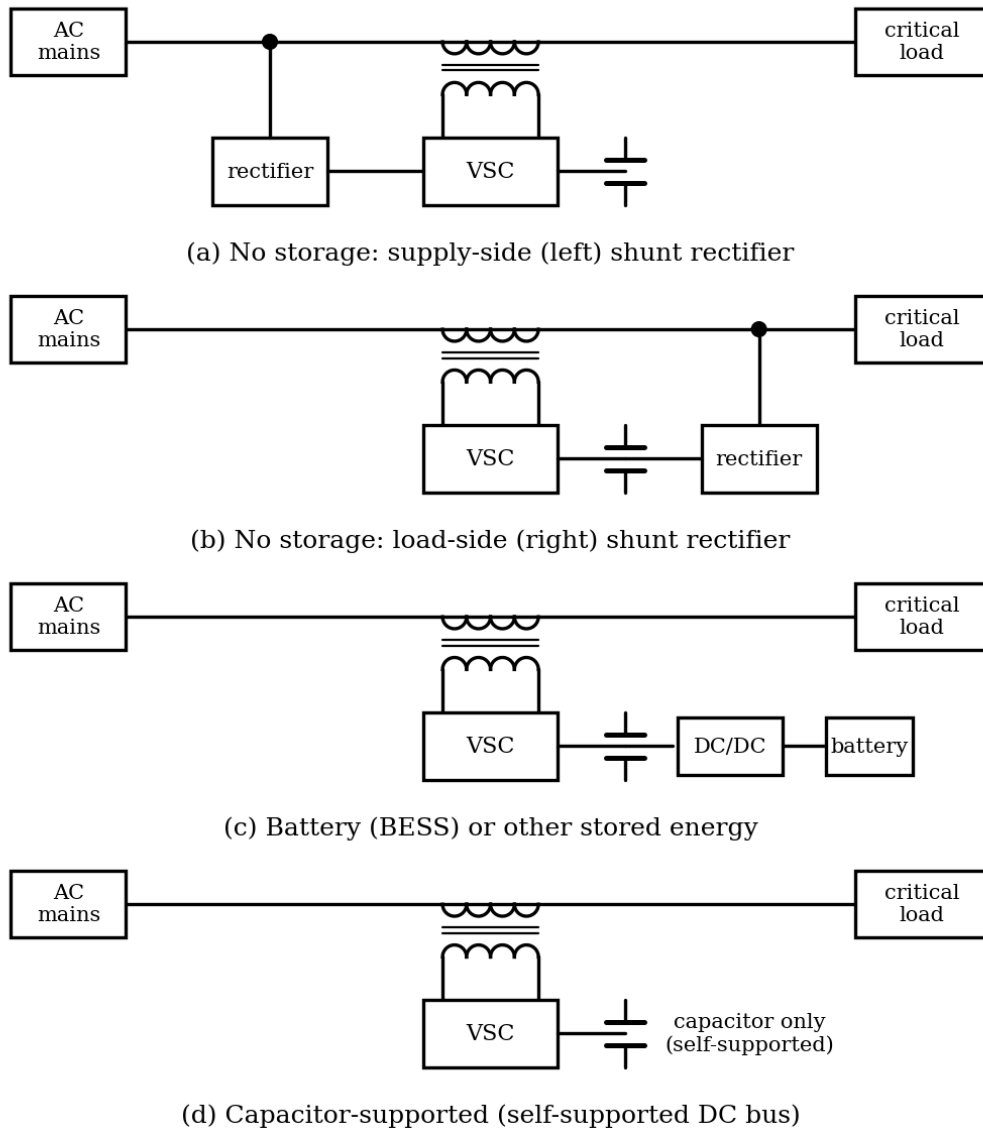


Figure 9.23. DVR energy-source options: (a) supply-side rectifier, (b) load-side rectifier, (c) battery (BESS) with DC/DC converter, (d) capacitor-supported.

Nielsen and Blaabjerg's comparison [14] concluded that, for the same power rating, a supply-side rectifier scheme without storage is simple and cheap and handles long shallow sags well, while a stored-energy DVR is the better choice when deep sags must be corrected and the stored energy needed is modest. Two DVRs with the same series-converter kVA can therefore have very different ride-through durations (Table 9.8).

Table 9.8. DVR energy-source and topology choices.

Choice	Advantage	Limitation that must be checked
Capacitor-supported	Simple DC side	Short real-power duration and DC-voltage droop
BESS-supported	Longer real-power support	Battery power capability, state of charge and converter losses

Choice	Advantage	Limitation that must be checked
Supercapacitor-supported	High pulse power and cycle life	Lower energy density and wide voltage variation
Supply-side rectifier	No dedicated storage for moderate sags	Source must retain enough voltage and power
Multilevel VSC	Lower device stress and better waveform	Balancing, device count and protection complexity
Transformerless DVR	Reduced magnetic hardware	Isolation, common-mode and fault-duty challenges

9.4.3 Compensation strategies

The DVR can reach the target load-voltage magnitude with many different injections, because only $|V_L|$ is specified. Figure 9.24 shows the strategies in common use for a sag that also carries a phase jump.

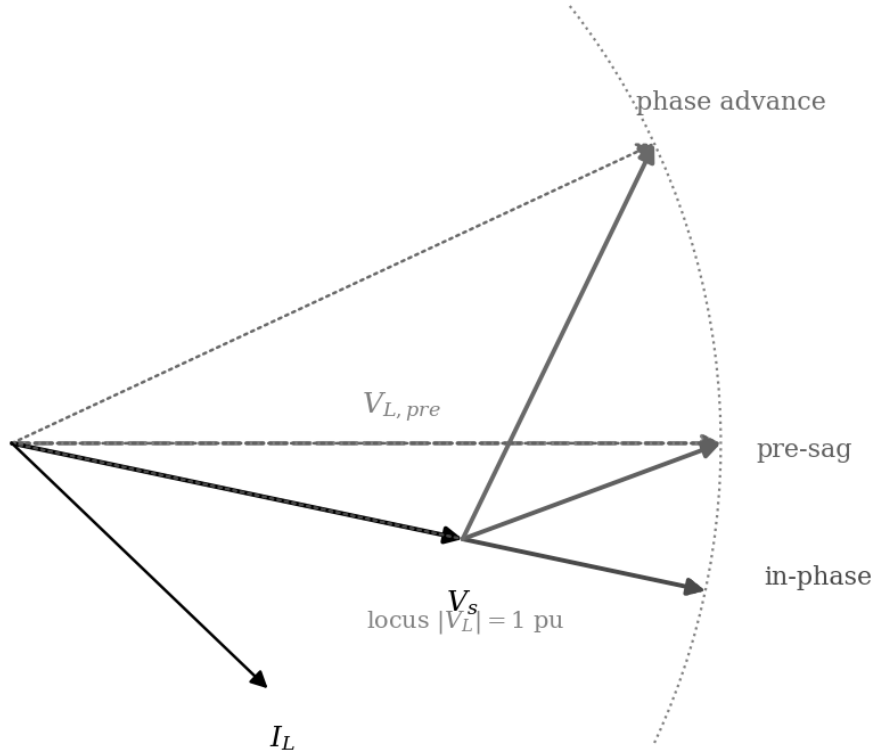


Figure 9.24. DVR compensation strategies for a sag with a phase jump: in-phase, pre-sag and phase-advance injection (the quadrature strategy is treated separately below).

In-phase injection. The injected voltage is in phase with the disturbed supply voltage. With V_s as reference, the injection is a signed scalar,

$$V_{inj} = V_{Lc} - |V_s| = V_{Lc} X \quad (9.29)$$

which is the minimum injection magnitude of any strategy, so it gives the smallest voltage rating and the smallest transformer. For a standalone DVR carrying I_L at load power-factor angle ϕ ,

$$P_{DVR} = (V_{Lc} - |V_s|) I_L \cos \phi, Q_{DVR} = (V_{Lc} - |V_s|) I_L \sin \phi \quad (9.30)$$

For a balanced three-phase sag this means $P_{DVR} = X P_L$: the DVR supplies exactly the fraction of load power that the source can no longer deliver. During a swell ($X < 0$) the signs reverse and the DVR absorbs real power, which needs a regenerative path, a store that can accept charge or a dissipative chopper. In-phase injection does not preserve the pre-sag phase.

Pre-sag compensation. Phase-sensitive loads, such as line-commutated converters and some drives, are disturbed by a phase jump as well as by a magnitude drop. The DVR then restores the pre-sag phasor itself,

$$V_{inj} = V_{L,pre} - V_s \quad (9.31)$$

Pre-sag compensation restores the pre-disturbance **rotating waveform trajectory**, not a stationary angle. At event detection t_0 , store θ_0 and pre-event frequency ω_{pre} ; use $\theta_{ref}(t) = \theta_0 + \omega_{pre}(t - t_0)$ throughout the event. This generally needs greater injection and energy than in-phase restoration [16]. On recovery, ramp the phase and magnitude and relock the PLL without a discontinuity.

Phase-advance (energy-optimized) compensation. If the load tolerates a phase shift, the load voltage can be rotated ahead of the supply so that the injection contains a larger reactive part and less real power, extending ride-through for a given store [17]. The price is a larger injection voltage and a phase transition at the start and end of the event.

Quadrature (zero-real-power) compensation. With fundamental injection perpendicular to series current, $P_{DVR} = \Re\{\underline{V}_{inj} \dot{\underline{I}}_{DVR}\} = 0$ ideally. A self-supported converter still needs energy for losses and transient or oscillating power; quadrature is its sustained lossless operating condition, rather than a statement that the capacitor is unnecessary. A standalone DVR carries load current, so injection is perpendicular to $\underline{I}_L$, not automatically to $\underline{V}_s$. In the right-shunt UPQC the series current is the source current, giving a different geometry.

Exact feasibility of zero-real-power restoration

Choose V_s real and let $V_L = V_{Lc} e^{j\theta_L}$ and $I_L = |I_L| e^{j(\theta_L - \phi)}$. Expanding $\Re\{(V_L - V_s) \dot{\underline{I}}_L\} = 0$ gives

$$\cos(\theta_L - \phi) = \frac{V_{Lc} \cos \phi}{|V_s|} \quad (9.32)$$

A real solution requires the right-hand side not to exceed one, so a standalone DVR can restore a sag without real power only if

$$|V_s| \geq V_{Lc} \cos \phi \Leftrightarrow X \leq X_{max} = 1 - \cos \phi \quad (9.33)$$

Physically, the source must still deliver the full load real power $V_{Lc} I_L \cos \phi$ through the same current magnitude I_L , and it can deliver at most $|V_s| |I_L|$. Figure 9.25 plots this limit. A load at 0.85 power factor can have a 15 % sag corrected by a capacitor-supported DVR, and a unity-power-factor load cannot have any sag corrected without real power. At the limit, $\theta_L = \phi$, the load current comes into phase with V_s , the injection is perpendicular to V_s and its magnitude is $V_{Lc} \sin \phi$. A formula giving the limit as $1 - \sin \phi$ appears in some teaching material; it confuses the limiting injection $V_{Lc} \sin \phi$ with the limiting supply voltage and must not be used.

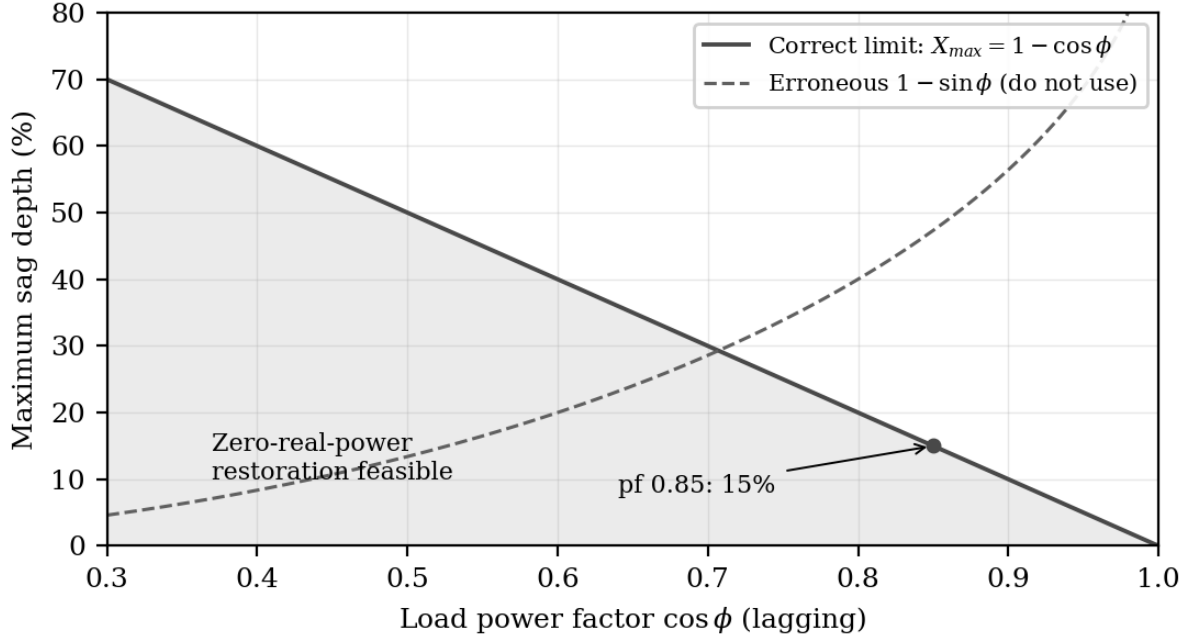


Figure 9.25. Exact zero-real-power feasibility limit of a standalone DVR versus load power factor.

Where the sag is feasible, both roots must be checked:

$$\theta_L = \phi \pm \cos^{-1} \left(\frac{V_{Lc} \cos \phi}{|V_s|} \right), V_{inj} = V_{Lc} e^{j\theta_L} - |V_s| \quad (9.34)$$

For a 230 V, 0.8 lagging-pf load with $|V_s| = 207$ V the roots are 9.604° and 64.136° , with injections of $43.17 \angle 62.73^\circ$ V and $232.83 \angle 117.27^\circ$ V (Figure 9.26). Both give zero real power; the smaller one is preferred if its load-phase shift is acceptable.

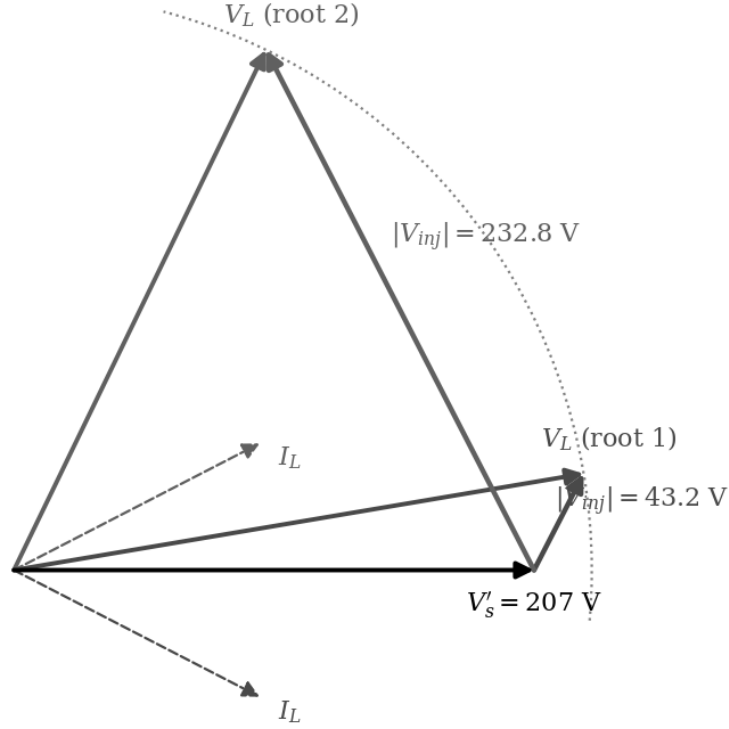


Figure 9.26. The two zero-real-power roots for a 230 V, 0.8 pf load and a 207 V supply.

General-angle injection. For a specified injection magnitude $U = |V_{inj}|$, the law of cosines on the triangle of Equation (9.27) gives the injection angle measured from V_s :

$$\cos \psi = \frac{V_{Lc}^2 - |V_s|^2 - U^2}{2|V_s|U} \quad (9.35)$$

For $U > 0$, a real geometric solution requires $|V_{Lc} - V_s| \leq U \leq V_{Lc} + V_s$, equivalently the right-hand side of Equation (9.35) must lie between -1 and $+1$. Both signs of ψ then satisfy the magnitude triangle, subject to allowable load phase shift and converter limits. The absolute value matters for a swell as well as a sag. At $U = 0$, restoration is possible only when $V_s = V_{Lc}$ with the required phase already present; no injection angle is defined. Selecting an angle is a feasibility calculation, not optimization without an explicit objective such as minimum energy, VA or phase shift.

Unbalanced sags

A single-line-to-ground or line-to-line fault elsewhere sags the three phases unequally, introduces negative-sequence voltage and can add a phase jump. Restoring each phase magnitude independently, with the fault's phase relationships still in place, would not reconstruct a balanced set. Using the symmetrical components of Chapter 2 with $V_0 = 0$ (an assumption that must be checked, because a grounded-wye winding can support zero sequence),

$$V_a = V_p + V_n, V_b = a^2 V_p + a V_n, V_c = a V_p + a^2 V_n \quad (9.36)$$

and the required injections for the balanced target $V_{Lc} < 0^\circ$, $V_{Lc} < -120^\circ$, $V_{Lc} < 120^\circ$ are

$$V_{inj,k} = V_{k,target} - V_{s,k}, k = a, b, c \quad (9.37)$$

If $V_0 \neq 0$ it must also be subtracted, and the DVR's bridge, transformer connection and neutral path must be able to synthesize zero sequence. Figure 9.27 shows the case $V_p = 0.85$ pu, $V_n = 0.15$ pu, worked in Example 9.38. Unbalanced injection carries a second-harmonic real-power pulsation (visible in Figure 9.32), which adds DC-link ripple even when the average real power is modest.

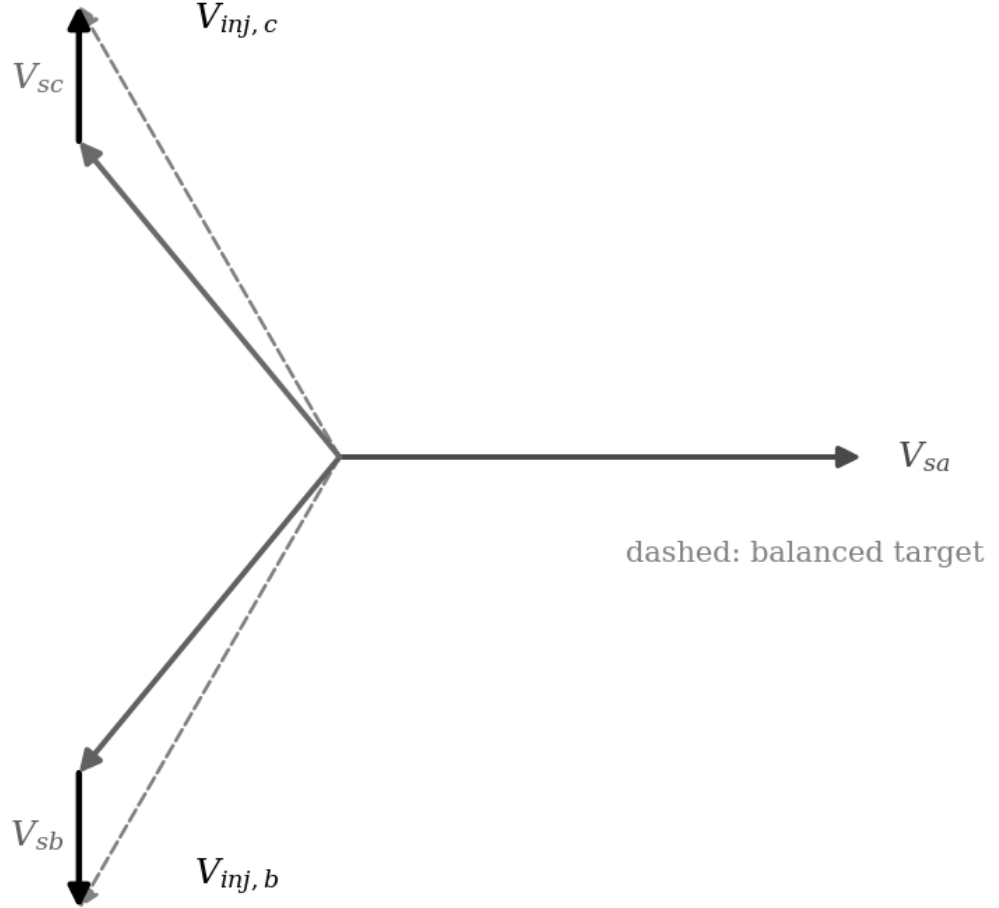


Figure 9.27. Sequence-defined unbalanced sag ($V_p = 0.85$ pu, $V_n = 0.15$ pu) and the per-phase injections that restore the balanced target.

Voltage harmonics and notches

A DVR with enough bandwidth can also inject the negative of supply-voltage harmonics, $v_{inj,h} = -v_{s,h}$, and fill in commutation notches, so the load sees a clean sinusoid. This is series active filtering (Section 9.6). The injection-transformer leakage inductance and the L_f - C_f filter limit the achievable bandwidth; selective harmonic controllers are used at medium voltage to keep the switching frequency low.

9.4.4 Control of the DVR

Figure 9.28 separates synchronization from voltage regulation. In-phase compensation tracks the positive-sequence supply angle. Pre-sag compensation holds the pre-event frequency estimate and advances its angle with $\theta_{ref}(t) = \theta_0 + \omega_{pre}(t - t_0)$. In a dq frame rotating with this chosen reference, $v_{Ld}^* = V_m$ and $v_{Lq}^* = 0$, giving Equation (9.38). Freezing the numerical angle would command DC-like stationary phase references rather than a 50 Hz voltage.

$$\mathbf{v}_{inj,d}^i = V_m - v_{sd}, \mathbf{v}_{inj,q}^i = -v_{sq} \quad (9.38)$$

Transforming back gives $\mathbf{v}_{inj,abc}^i = \mathbf{v}_L^i - \mathbf{v}_s$, which corrects magnitude, phase, unbalance (if negative sequence is handled, for example by a second frame rotating at $-\omega$) and harmonics in one step. A feedback loop on the measured load voltage removes the drops across the filter and transformer leakage. The injected voltage is produced by a PWM voltage controller, often with an inner loop on filter-capacitor current for active damping of the L_f - C_f resonance.

For a capacitor-supported DVR, a DC-voltage PI adds a small real-power component so that the converter absorbs its own losses:

$$v_{loss}(n) = v_{loss}(n-1) + K_{p1}[v_{de}(n) - v_{de}(n-1)] + K_{i1}v_{de}(n), v_{de} = v_{dc}^i - v_{dc} \quad (9.39)$$

The loss PI output in Equation (9.39) is an absorbing voltage amplitude. When the DC voltage is too low, its voltage vector must oppose the load-current vector so that $\Re\{\mathbf{V}_{loss}\mathbf{I}_L^i\} < 0$. A separate quadrature component regulates load-voltage magnitude. The complete command must still satisfy phasor feasibility, voltage limits and controlled phase transitions; a positive d-axis voltage absorbs real power only when it opposes the load current.

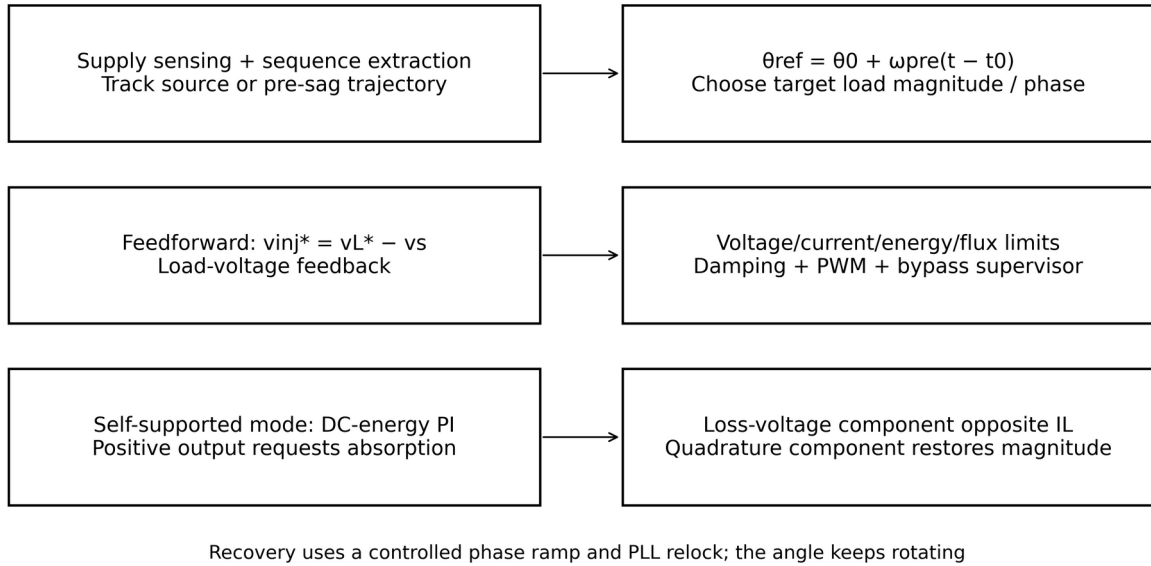


Figure 9.28. SRF-based DVR control with feedforward, load-voltage feedback and (for capacitor-supported DVRs) a DC-voltage loop.

Detection and restoration must be timed separately. Total response includes sensing, detection, reference calculation, computation/PWM delay and output-filter build-up. A sliding half-cycle or full-cycle RMS detector smooths noise, but threshold crossing depends on event depth, inception angle and window length; it need not wait for a completely replaced window. Instantaneous waveform-error and sequence/dq methods can be faster, but their delays depend on filtering, synchronization and thresholds [34]. An algorithm name alone does not imply a 1 to 2 ms response. Report worst-case detection latency and transient load-voltage deviation for the specified event set. Include voltage/current/DC-energy limits, anti-windup, recovery ramps and bypass logic. Because transformer flux is the voltage integral, insertion angle and residual flux can cause saturation. Flux estimation and controlled first-cycle volt-seconds reduce this risk within a verified design envelope.

9.4.5 Energy storage and DC-link requirements

For a storage-supported standalone DVR, injected positive real power and converter losses deplete the store. For arbitrary phase injections the average converter AC real power is $P_{DVR} = \sum_k \text{Re}\{V_{inj,k} \dot{I}_k\}$; for balanced operation it is $3|V_{inj}||I|\cos\gamma$, where γ is the angle between injection and current, and in an in-phase standalone DVR $\gamma = \phi$. For positive, approximately constant delivered power and a stated conversion efficiency η ,

$$E = \frac{P_{DVR}}{\eta} t_{sag} \quad (9.40)$$

The general balance, which also covers zero and negative real power, is

$$E_{drawn} = \int_0^t [P_{DVR}(\tau) + P_{loss}(\tau)] d\tau \quad (9.41)$$

Pure quadrature operation has $P_{DVR} = 0$ but still needs $\int P_{loss} dt$. During absorbing operation the signed power charges the store, and charging limits must be checked. For a capacitor allowed to fall from $V_{dc,max}$ to $V_{dc,min}$, the usable energy is $\frac{1}{2}C(V_{dc,max}^2 - V_{dc,min}^2)$, so the minimum capacitance is

$$C_{dc} = \frac{2E}{V_{dc,max}^2 - V_{dc,min}^2} \quad (9.42)$$

For an in-phase standalone DVR at fixed load power, $P_{DVR} = X P_L$, so $C_{dc} = 2X P_L t / [\eta(V_{dc,max}^2 - V_{dc,min}^2)]$ is linear in both depth and duration (Figure 9.29). Long events quickly make a pure capacitor impractical, which is why batteries, supercapacitors, flywheels or rectifier-supported links are used for longer ride-through.

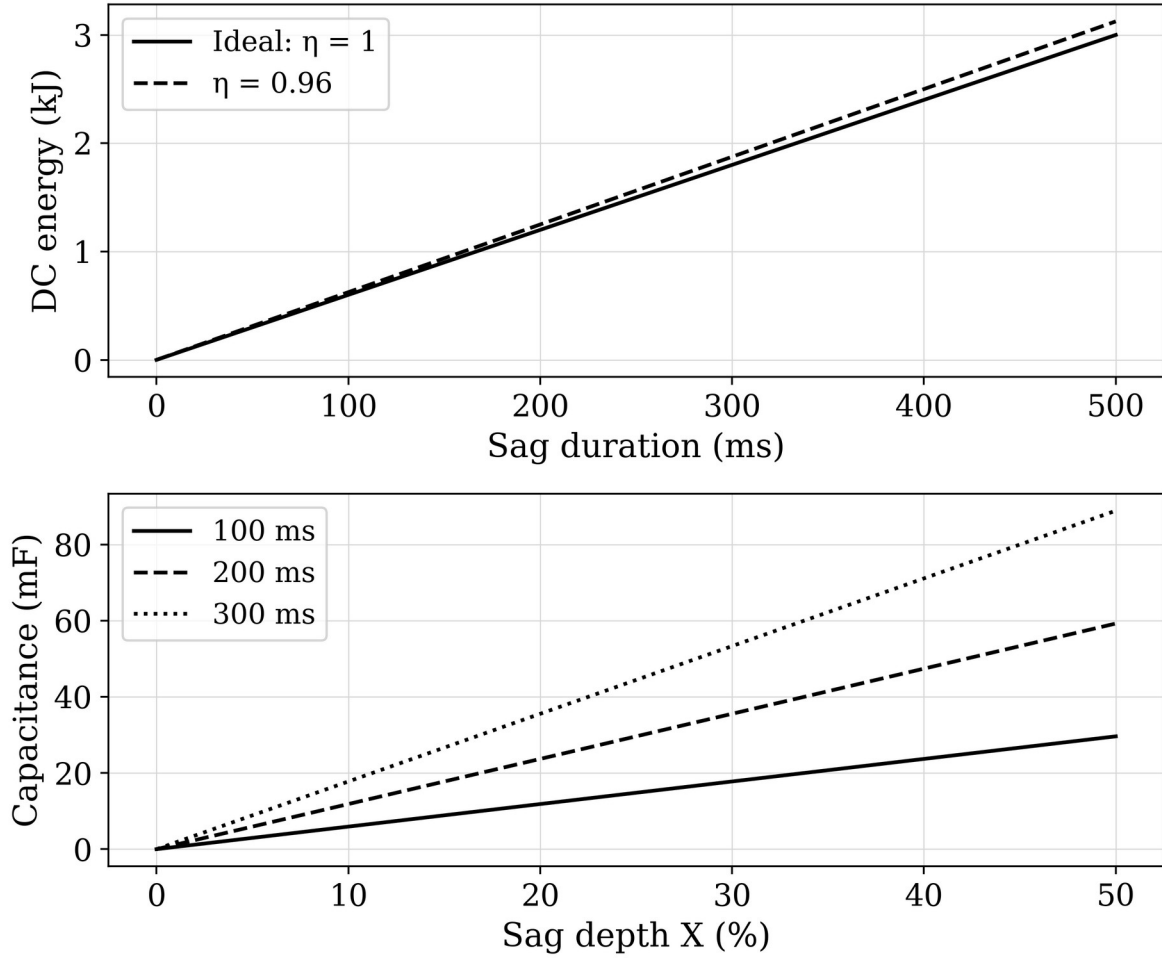


Figure 9.29. Upper: DC energy versus sag duration at 6 kW AC real power. Lower: energy-based capacitance versus sag depth for a 20 kW load and a 700-to-650 V DC window; the depth-duration curves assume ideal conversion.

Stored energy alone does not size a DVR. The store must provide the event energy, its interface must provide the event power, and the DC-link voltage must stay high enough for the converter to synthesize the commanded AC voltage. A battery with ample watt-hours may still be unsuitable if its allowable discharge current is too small, while a supercapacitor may supply the pulse power but need a wide DC/DC conversion range as its voltage falls. For repeated sags, recharge time and thermal accumulation must also be checked.

9.4.6 Design of a DVR

The design covers the voltage, current and VA ratings of the VSC, the injection transformer, the DC-bus voltage and capacitance, the interfacing inductance and the ripple filter [2].

The injection voltage is set by the worst disturbance and strategy. In-phase restoration needs $|X|V_{Lc}$ for a balanced event; sequence voltage and phase jumps add vector requirements. The right-shunt UPQC-Q needs $\sqrt{V_{Lc}^2 - V_s^2}$, whereas a standalone zero-power DVR uses **both** roots of Equation (9.34) and normally selects the smaller feasible injection if its phase shift is acceptable. The larger root is not an automatic rating requirement.

Current rating. A standalone DVR carries the full load current, $I_{DVR} = I_L$, and its kVA rating is $S_{DVR} = 3V_{inj}I_L$ for a three-phase unit.

The injection transformer has continuous line-current, compensation-voltage, insulation and short-duration fault duties. For an ideal transformer the two winding VA products are equal, with $n = V_{\text{converterwinding}}/V_{\text{linewinding}}$ and $I_{\text{converter}} = I_{\text{line}}/n$. The converter terminal voltage additionally includes its filter-inductor drop, so converter VA need not equal transformer injection VA. Flux is the time integral of winding voltage: worst insertion angle, residual flux, harmonics and unbalance must be checked, or the controller must limit first-cycle volt-seconds.

DC-bus voltage. For a three-leg VSC with sinusoidal PWM, $V_{dc} \geq 2\sqrt{2}V_{VSC}/m$, where V_{VSC} is the converter-side phase RMS voltage. For a full bridge, $V_{dc} \geq \sqrt{2}V_{VSC}/m$.

The DC capacitor buffers the actual real-energy imbalance and switching/unbalanced-power ripple. Use Equation (9.42) with a declared ΔE_{real} over the support delay. The product $3V_{inj}I_L\Delta t$ is an apparent-power-based budget, not an identity for DC energy. Verify the minimum bus voltage against the converter **terminal** voltage, including filter drops; a BESS or rectifier does not remove that requirement.

Interfacing inductance. Equation (9.43) below gives the converter-side inductance using line-side peak-to-peak ripple.

$$L_{r,\text{converter}} = n\sqrt{3}mV_{dc}/(12af_s\Delta i_{\text{line}}), L_{r,\text{line}} = L_{r,\text{converter}}/n^2 \quad (9.43)$$

This follows from $\Delta i_{\text{converter}} = \Delta i_{\text{line}}/n$ and Equation (9.24). Every inductance, voltage and ripple specification must identify its winding side; including n does not make the result a line-referred inductance.

A DVR commonly uses an output LC filter, with transformer leakage and active or passive damping included in its resonance study. An optional series-RC shunt branch suppresses switching ripple and has the corner-frequency relationship of Equation (9.25). These filters have different functions and must not be interchanged without checking attenuation, reactive loading and damping.

9.4.7 Protection

The series injection transformer carries downstream fault current. A rated bypass and converter blocking sequence must operate within the device and transformer withstand envelope; a crowbar may provide backup. Distinguish remote supply disturbances that the DVR can compensate from local faults, loss of source and events outside its energy/current limits. Voltage and current magnitudes alone may not establish fault direction, so use the required directional, differential or coordinated relay logic. Check interactions with upstream protection, transformer inrush, bypass thermal duty and safe maintenance isolation.

9.4.8 Defined linear tracking illustrations

Figures 9.30 to 9.32 use a reproducible linear model at 415 V, 50 Hz. Target load voltage is $V_m \cos \theta_k$ and injection command is $v_{inj,k}^i = v_{\text{target},k} - v_{s,k}$. Injection tracking is $H(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$ with $\omega_n = 2\pi(800)$ rad/s and $\zeta = 0.7$, initially at rest. A star RL load has $Z = V_{ph}/[20000/(3V_{ph})]$, $R = 0.8 Z$ and $L = 0.6 Z/\omega$; each phase obeys $L\dot{i} + Ri = v_L$ with nominal steady-state initial current. State-space integration uses exact zero-order-hold transitions at 50 μ s intervals. There is no detection delay, PWM, saturation, storage model or inverse-frequency-response compensation.

The source has a balanced 0.7 pu sag from 0.10 to 0.20 s and a 1.2 pu swell from 0.30 to 0.40 s. The finite tracking response gives a visible insertion transient and a small steady amplitude/phase error. Consequently the calculated mean AC export is about 5.0 kW during the sag and -3.5 kW during the swell, rather than the ideal 4.8 kW and -3.2 kW: the 5° phase lag of H at 50 Hz turns part of the injection

toward the lagging load current. Figure 9.31 shows insertion detail. These linear waveforms demonstrate bandwidth effects, not guaranteed load ride-through.

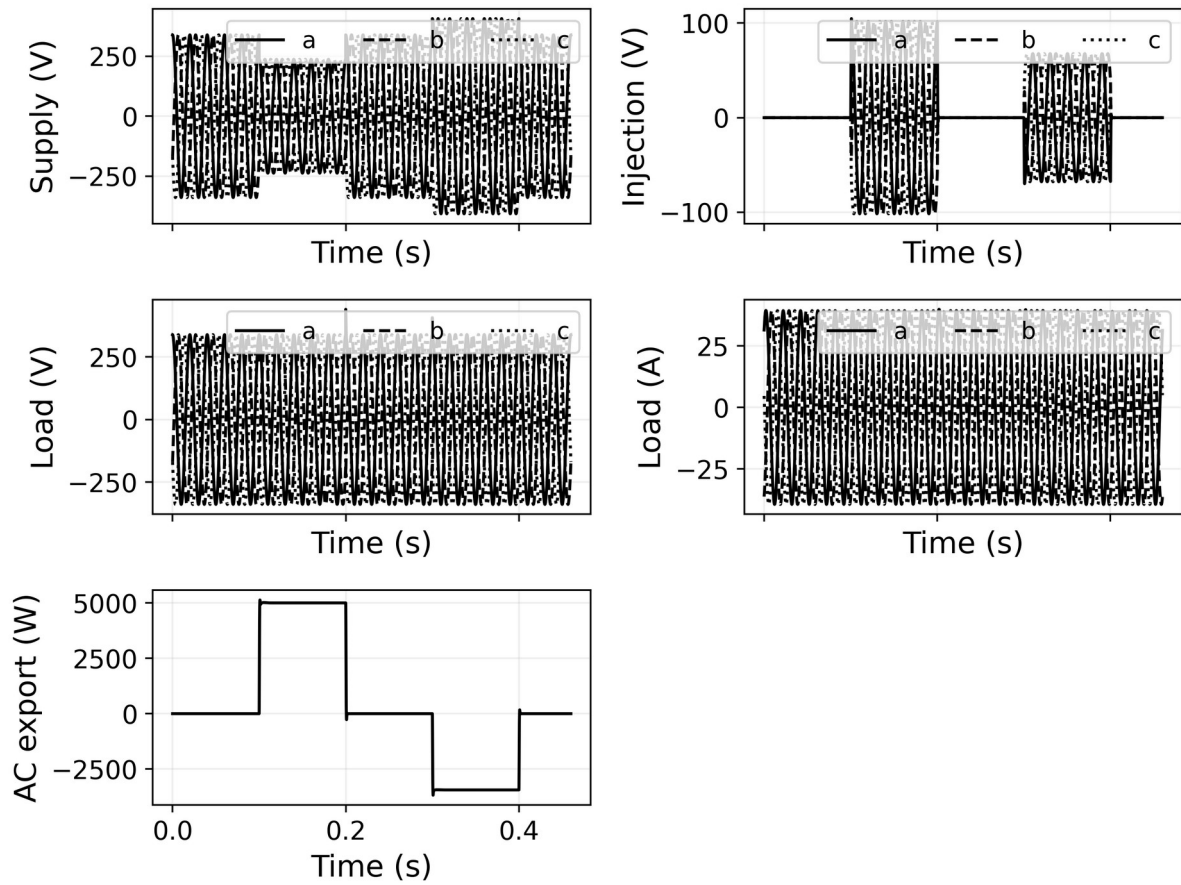


Figure 9.30. Defined second-order DVR injection-tracking model during a 30% sag and a 20% swell. Load current is computed from the RL circuit; instantaneous AC export is positive. Model parameters are given in Section 9.4.8.

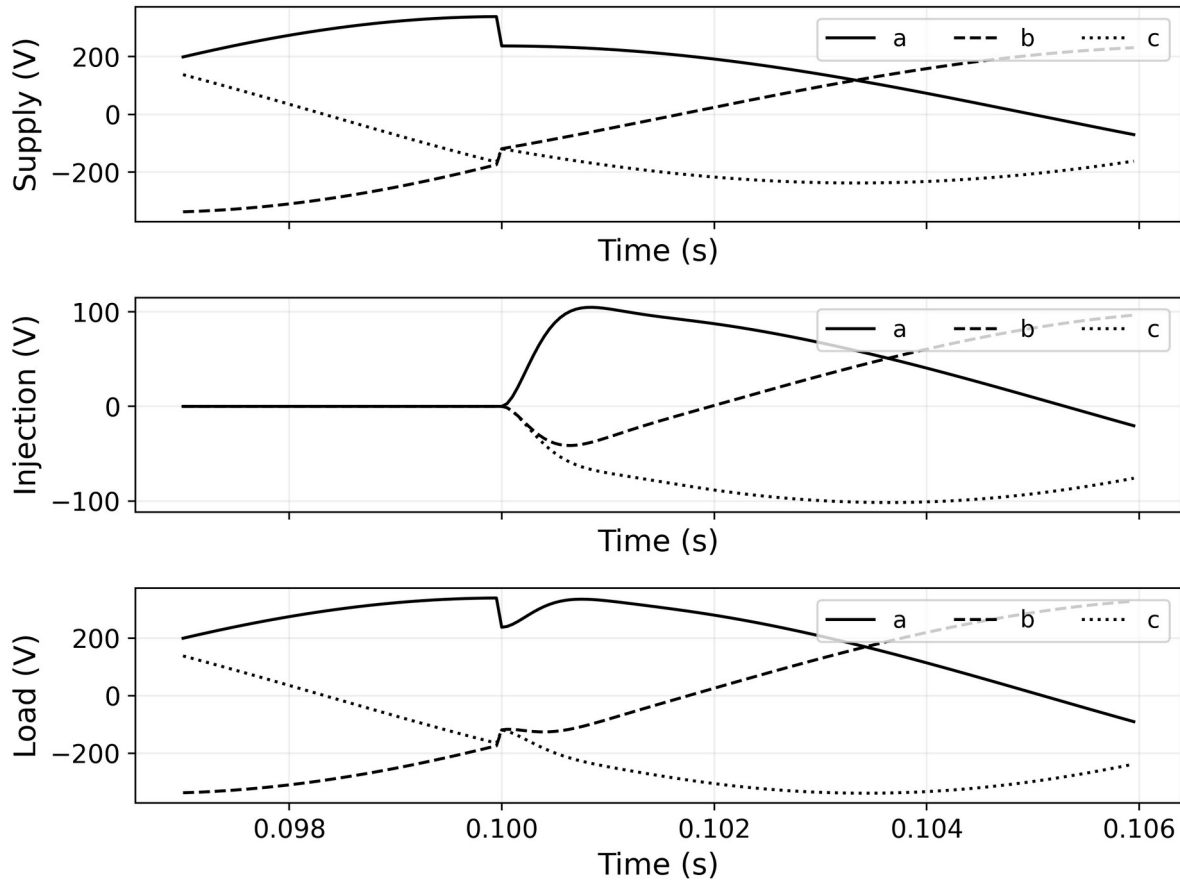


Figure 9.31. Sag-onset detail for the specified 800 Hz, damping-0.7 tracking model. The finite insertion transient and residual tracking error are retained.

For Figure 9.32 the event from 0.10 to 0.20 s contains 0.8 pu positive sequence and $0.15 \angle 30^\circ$ pu negative sequence. From 0.30 to 0.40 s add $0.10 V_m \cos 5\theta_k + 0.07 V_m \cos 7\theta_k$. The same second-order injection tracker and RL load are used. The negative sequence produces a twice-frequency power pulsation. Harmonic cancellation is incomplete because H has gain and phase error at 250 and 350 Hz; the mean injection power need not be exactly zero with finite tracking and a frequency-dependent load. Ideal negative-harmonic injection is the limiting reference, not the plotted realized response.

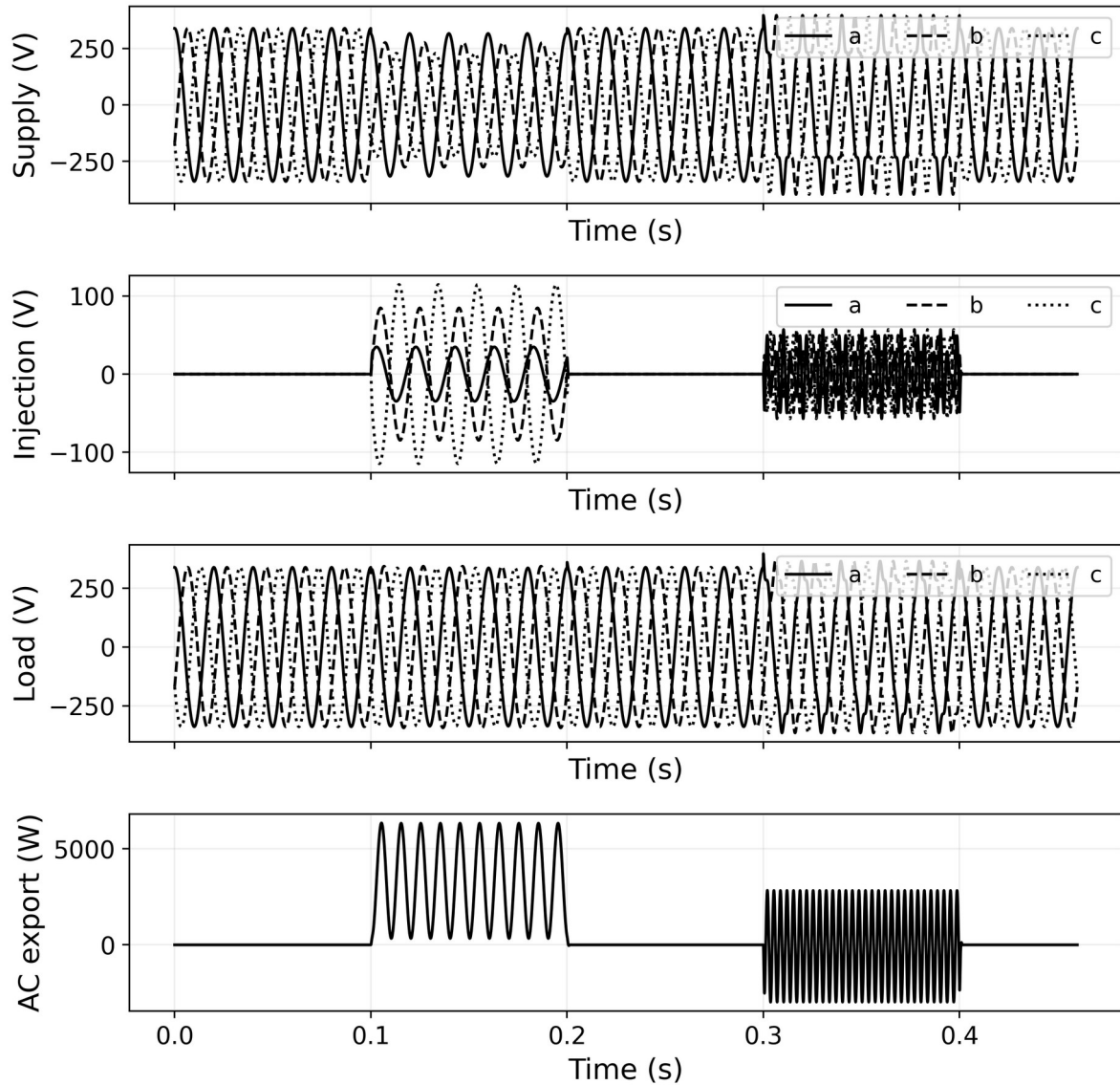


Figure 9.32. Defined linear DVR model under sequence-unbalanced supply and fifth/seventh voltage harmonics. Residual load-voltage distortion and power ripple reflect the finite tracking transfer function.

9.4.9 The interline DVR (IDVR)

An IDVR shares a DC link between series compensators on different feeders (Figure 9.33), following the interline power-balance principle of Chapter 8. When one feeder sags, the DVR on a healthy feeder absorbs real power from its own line (by injecting a small voltage component in phase opposition to that feeder's current), and the sagged feeder's DVR exports it. The healthy feeder's load voltage must remain acceptable while this happens, so the healthy feeder is not an unlimited source. The common link still needs capacitance for switching ripple, unbalanced power pulsation and transient mismatch.

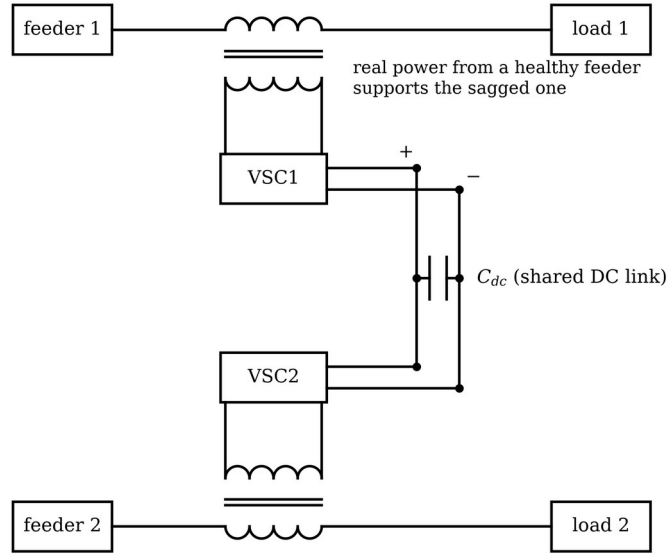


Figure 9.33. Interline DVR: series converters on separate feeders sharing a DC link.

9.4.10 Worked examples: DVR

Example 9.11. Single-phase BESS-supported DVR with in-phase injection

A single-phase 5 kVA, 230 V, 50 Hz, 0.85 lagging pf critical load is protected by a BESS-supported DVR using in-phase injection. The supply sags by 25 %. Find the injected voltage, the DVR current, its kVA rating and the real power drawn from the battery.

Solution. $V_s = 230 \times 0.75 = 172.5$ V and $V_{inj} = 57.5$ V. $I_{DVR} = I_L = 5000/230 = 21.74$ A, so $S_{DVR} = 57.5 \times 21.74 = 1.25$ kVA. From Equation (9.30), $P_{DVR} = 57.5 \times 21.74 \times 0.85 = 1063$ W and $Q_{DVR} = 658$ var.

These are ideal AC and battery powers with unity conversion efficiency. At efficiency η , battery discharge power is $1062.5/\eta$ W; ripple, reserve and DC/DC current limits require separate checks.

Example 9.12. Capacitor-supported DVR with a unity-power-factor load

A single-phase capacitor-supported DVR is proposed for a 10 kVA, 230 V, unity-pf load facing a 30 % sag, using an injection of $\sqrt{230^2 - 161^2} = 164.25$ V at right angles to the supply voltage. Check whether this injection really needs no real power.

Solution. With $V_s = 161$ V and $V_{inj} = j 164.25$ V, $V_L = 230 \angle 45.57^\circ$ V. The unity-pf load draws $43.48 \angle 45.57^\circ$ A, giving $P_{DVR} = \Re \{ V_{inj} I_L^* \} = 5.10$ kW. The zero-power condition is perpendicularity to **load current**, not supply voltage. Equation (9.33) gives no feasible sag at unity load pf. A capacitor with usable voltage window supports this power for $t = C(V_{max}^2 - V_{min}^2)/(2 \times 5100)$ seconds, before losses. Without C and its voltage limits, no fraction-of-cycle duration can be stated.

Example 9.13. Maximum sag for a capacitor-supported DVR

A capacitor-supported DVR protects a 415 V (239.6 V phase) load at 0.85 lagging pf. Find the maximum sag it can correct without real power, the injection at that limit and the phase shift of the load voltage.

Solution. From Equation (9.33), $X_{max} = 1 - 0.85 = 15$ %, so $|V_{s|min}| = 203.7$ V. At the limit $\theta_L = \phi = 31.79^\circ$, the injection is perpendicular to V_s and $|V_{inj}| = 239.6 \sin 31.79^\circ = 126.2$ V. The load voltage leads the

supply by 31.79° . The value 0.5268 pu is $\sin \phi$, the injection magnitude in per unit; mistaking it for the minimum supply voltage gives an incorrect “47 % maximum sag”.

Example 9.14. DVR rating for different injection schemes

A three-phase 415 V, 30 kVA, 0.8 pf critical load experiences a balanced 35 % sag. The phase current is 41.74 A. Compare in-phase injection with zero-real-power injection.

Solution. $|V_s| = 0.65 \times 239.6 = 155.74$ V. In-phase: $V_{inj} = 83.86$ V and $S_{DVR} = 3 \times 83.86 \times 41.74 = 10.50$ kVA (35 % of the load rating), with $P_{DVR} = 0.35 \times 24 = 8.4$ kW. Zero real power: by Equation (9.33) a 0.8 pf load allows at most a 20 % sag, so a 35 % sag is infeasible for a standalone capacitor-supported DVR. The figure of 182.1 V (22.8 kVA) obtained from $\sqrt{V_{Lc}^2 - |V_s|^2}$ is the injection perpendicular to the supply voltage. It is the correct UPQC-Q series voltage (Section 9.5), where the series current is in phase with V_s , but it is not a zero-power solution for a standalone DVR carrying a lagging load current.

Example 9.15. Swell compensation and power direction

A single-phase DVR protects a 2.5 kVA, 230 V, unity-pf load against a 20 % swell. Find the injection, the kVA rating and the power direction.

Solution. $V_s = 276$ V, so $V_{inj} = 230 - 276 = -46$ V (in phase opposition). $I_L = 10.87$ A and $S_{DVR} = 500$ VA. $P_{DVR} = -46 \times 10.87 = -500$ W; the DVR absorbs 500 W, which must charge the store or be dissipated in a chopper.

Example 9.16. BESS energy for an extended sag

A three-phase 415 V, 15 kVA, 0.8 pf critical load is protected by a BESS-supported in-phase DVR against a 30 % sag lasting 500 ms. Find the load power, the battery power and the energy per event.

Solution. $P_L = 12$ kW. $V_{inj} = 71.88$ V and $I = 20.86$ A, so $P_{BESS} = 3 \times 71.88 \times 20.86 \times 0.8 = 3.60$ kW, which is $0.3 P_L$. $E = 3.60 \times 0.5 = 1.80$ kJ, or 0.50 Wh. The energy is tiny, but the battery must deliver 3.6 kW for half a second: DVR storage is sized by power and discharge rate at least as much as by energy, which is why supercapacitors are attractive.

At 95% efficiency the required battery power is $3.60/0.95 = 3.789$ kW and event energy is 1.895 kJ. The 3.60 kW and 1.80 kJ values describe ideal AC injection and lossless storage, respectively.

Example 9.17. Standalone DVR energy storage

A balanced three-phase storage-supported DVR in in-phase mode protects a 415 V, 20 kW, 0.9 lagging pf load against a 30 % sag for 200 ms. Find the phase voltage, series current, injected and sagged phase voltages, delivered real power and ideal energy.

Solution. $V_{ph} = 239.60$ V and $I_L = 20000/(\sqrt{3} \times 415 \times 0.9) = 30.92$ A. $U = 0.30 V_{ph} = 71.88$ V and $|V_s| = 167.72$ V. $P_{DVR} = 3 U I_L \cos \phi = 0.30 P_L = 6000$ W and the VA duty is 6667 VA. Ideal energy: $E = 6000 \times 0.2 = 1200$ J. Source real power is 14 kW and DVR export 6 kW, totalling the 20 kW load demand.

Example 9.18. Loss-inclusive capacitance

For Example 9.17, with the DC bus permitted to droop from 700 V to 650 V and $\eta = 0.96$, find the DC-link capacitance and compare with the ideal value.

Solution. $E = (6000/0.96) \times 0.2 = 1250$ J, 4.2 % above 1200 J. From Equation (9.42), $C_{dc} = 2 \times 1250/(700^2 - 650^2) = 37.04$ mF (ideal 35.56 mF). These are energy-based preliminary values;

ESR, ripple current, voltage derating, tolerance and lifetime must also be checked. For a 240 ms event the ideal values become 1440 J and 42.67 mF, and 44.44 mF at 96 % efficiency.

Example 9.19. Transformer, filter and DC-window sizing for a BESS-supported DVR

Design a three-phase DVR for a 415 V, 20 kVA, 50 Hz critical load. The supply varies from -30% to $+20\%$ with up to 6 % unbalance; the switching frequency is 10 kHz. Use in-phase injection, a converter-side voltage of 50 V, $m=1$, $a=1.2$ and a 2 % line-side ripple.

Solution. Balanced sag injection is 71.88 V per phase; adding a 6% negative-sequence allowance conservatively gives 86.26 V, so select 90 V line winding. Line current is 27.824 A and transformer injection duty 7.513 kVA. A 50 V converter winding gives $n=50/90=0.55556$ and converter current 50.083 A. Select a **600-to-570 V DC window**. With $\Delta i_{line}=0.02(27.824)=0.55648$ A, Equation (9.43) gives $L_{converter}=7.205$ mH and $L_{line}=23.344$ mH. The conservative normal-current terminal-voltage bound is $50+\omega L_{converter}(50.083)=163.36$ V RMS; at 1.2 times current it is 186.03 V. Both are below $570/(2\sqrt{2})=201.53$ V available, with further harmonic and transient margins still to be checked. The normal-current terminal VA upper bound is 24.55 kVA, distinct from transformer injection duty. A declared 1.503 J delay-energy budget gives $C=2(1.503)/(600^2-570^2)=0.0856$ mF; a 100 μ F, 800 V capacitor is preliminary before ripple, tolerance and lifetime checks. For a 5 Ω RC branch, 6.366 μ F gives a 5 kHz corner; an LC output filter needs separate resonance/damping design. The 171.1 V injection perpendicular to the sagged supply gives standalone zero real power **only at pf=0.7**. For lower pf the feasibility condition may hold, but both roots of Equation (9.34) must be recalculated.

9.5 Unified Power Quality Conditioner (UPQC)

A UPQC combines series and shunt converters on a common DC link [4], [18]. The series converter regulates protected-load voltage and the shunt converter shapes source current while regulating DC energy. These objectives are achieved only within their joint voltage, current, bandwidth and energy envelope. Their coupled real-power balance determines ratings, control priorities and sag capability. The back-to-back arrangement resembles the UPFC of Chapter 8, with distribution power-quality objectives.

9.5.1 Structure and power flow

The series VSC is voltage-controlled through injection transformers; the shunt VSC is current-controlled through interfacing inductors. Both need ripple filters, and both share the DC capacitor. With $P_{into,k}$ defined as the real power absorbed from the AC system by converter k (positive into the converter), the DC-link energy obeys

$$\frac{dW_{dc}}{dt} = \frac{d}{dt} \left(\frac{1}{2} C_{dc} V_{dc}^2 \right) = \sum_k P_{into,k} - P_{loss} \quad (9.44)$$

At constant average DC energy, $\sum P_{into,k} = P_{loss}$. Equivalently, with AC exports positive, $P_{se} + P_{sh} + P_{loss} = 0$. When the series converter exports real power to boost a sagged supply, the shunt converter must draw the same power from the AC system, so in a feasible grid-fed steady state the real power circulates: source to shunt converter, through the DC link, to series converter and back into the line. This is why a UPQC can correct long sags without a large energy store, provided the sagged source can still supply the load power at a higher current. The capacitor only buffers transient mismatch and ripple; it is not a source of sustained real power.

9.5.2 Classification

UPQCs are classified by converter type (CSC or VSC), by supply system (single-phase two-wire, three-phase three-wire and three-phase four-wire), by the location of the shunt converter and by how the series converter

shares real and reactive power (Figure 9.34) [4]. Extensions include the interline UPQC (series and shunt converters on different feeders), the multi-converter UPQC and UPQCs with distributed generation or storage on the DC link.

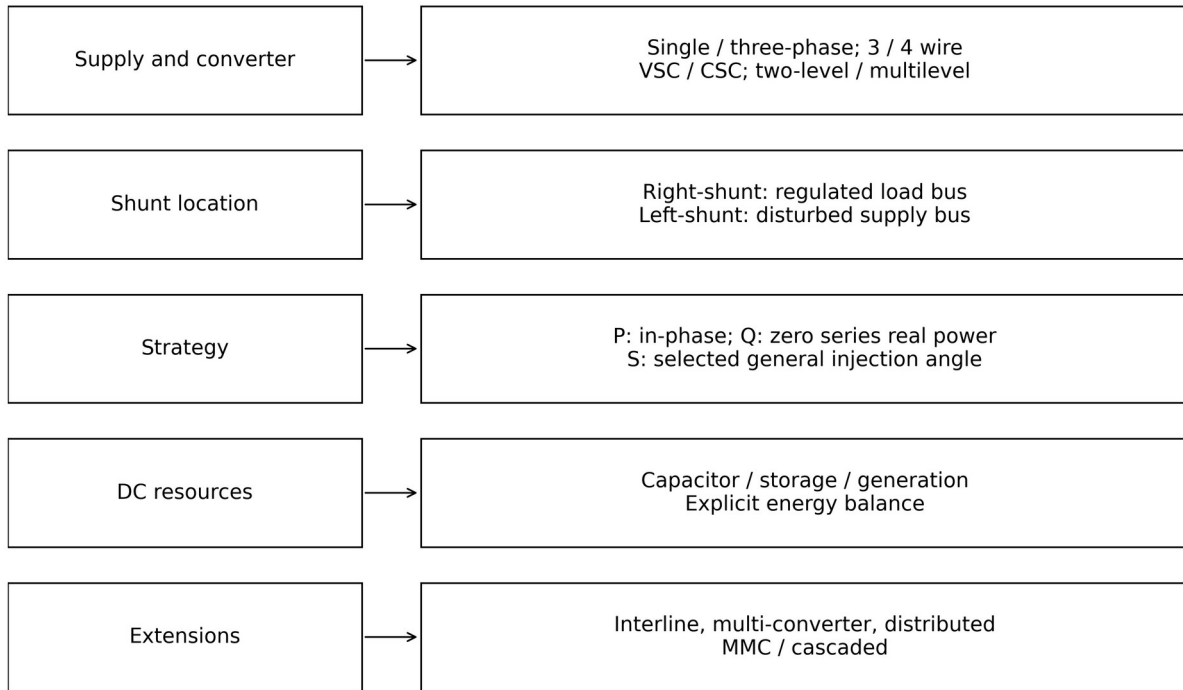


Figure 9.34. Classification of UPQC by converter structure, connection topology, supply system and compensation objective.

Right-shunt and left-shunt UPQC. In the right-shunt UPQC (Figure 9.35(a)) the shunt converter is on the load side of the series converter. The series converter then carries the source current, which the shunt converter has made sinusoidal and in phase with the supply, and the shunt converter sees the regulated load voltage; during a sag it draws the series converter's real power from the restored bus, so the source current rises as the source voltage falls. In the left-shunt UPQC (Figure 9.35(b)) the shunt converter is on the supply side: the series converter carries the load current, with its harmonics and reactive component, and the shunt converter sees the disturbed supply voltage. The right-shunt arrangement is usually preferred because the shunt converter works at a clean voltage, its current harmonics do not flow through the series transformer and the series converter carries a well-defined sinusoidal current. The left-shunt arrangement is used when the shunt converter must also compensate other loads at the PCC. The two arrangements can deliver similar external objectives but do not have identical converter ratings, so comparisons must be made at the same load, sag, source power factor and control objective.

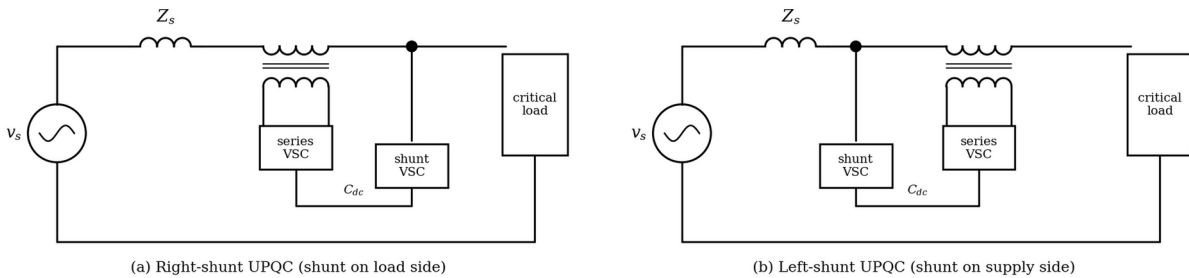


Figure 9.35. (a) Right-shunt UPQC, (b) left-shunt UPQC.

Single-phase and three-phase UPQCs. Balanced three-phase phasor sizing can use a per-phase equivalent, but switching and control implementations are not obtained simply by repeating a single leg. A single-phase full bridge needs a complete return path; a three-wire three-phase UPQC uses three-leg converters; a four-wire

UPQC needs a rated neutral path for the shunt converter, such as a fourth leg or a split DC link (Figure 9.36). The neutral leg must be rated from its current, since its terminal voltage can be small while its conduction loss is large. Single-phase units have twice-line-frequency DC-energy ripple, and unbalanced three-phase operation also produces ripple; both enter the capacitor sizing.

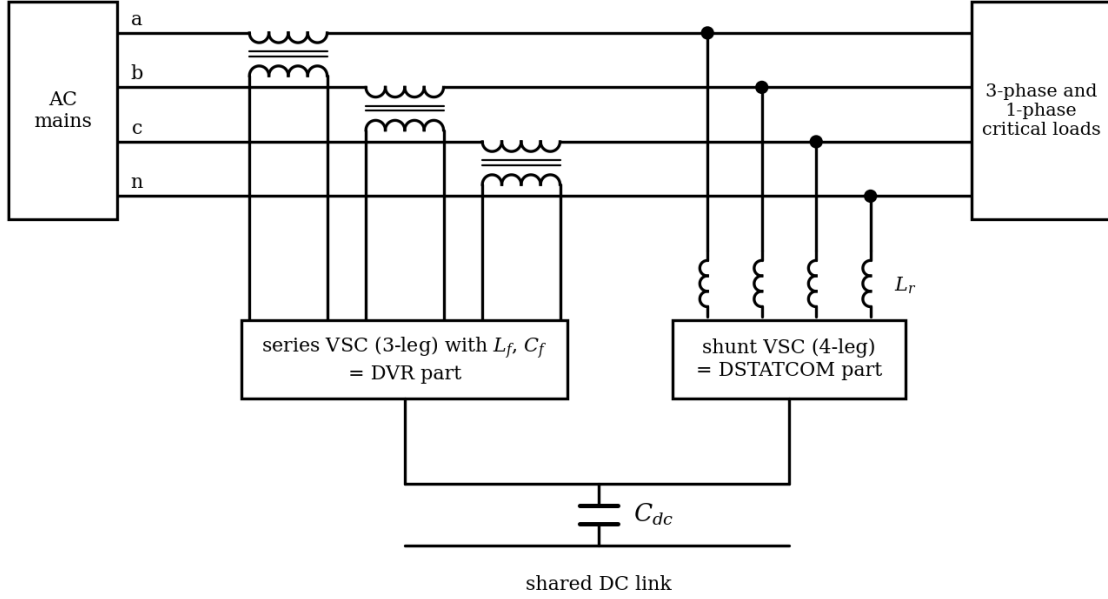


Figure 9.36. Three-phase four-wire right-shunt UPQC: three-leg series VSC (DVR part) and four-leg shunt VSC (DSTATCOM part) on a shared DC link.

UPQC-P, *UPQC-Q* and *UPQC-S*. These names describe operating strategies, not different hardware. They state how the series converter injects its voltage relative to the series current:

- *UPQC-P*: injection in phase with the supply voltage and the (unity-pf) source current, so the series converter handles only real power. Its injection voltage is the smallest possible, but the shunt converter must circulate that real power.
- *UPQC-Q* in the lossless right-shunt arrangement: injection is perpendicular to unity-pf source current, so the series duty is reactive. This geometry corrects sags but cannot reduce a swell. A left-shunt zero-power arrangement has a different series-current reference and can correct suitable swells (Section 9.5.4).
- *UPQC-S*: injection at a chosen angle, so the series converter handles both real and reactive power. Khadkikar and Chandra [19] used this to let the series converter share the load reactive power during normal operation as well as during sags and swells; a minimum-VA angle can also be chosen.

The derivations below use the right-shunt topology, so the series current equals the source current, unless stated otherwise.

9.5.3 Phasor analysis and converter ratings

Take the sagged supply voltage $V'_s = V_{Lc}(1 - X)$ as real reference. With unity source power factor and a lossless UPQC, the source supplies exactly the load real power $P_L = V_{Lc} I_L \cos \phi$:

$$I_s = \frac{P_L}{V'_s} = \frac{I_L \cos \phi}{1 - X} \quad (9.45)$$

The source current rises as the supply sags. For a balanced three-phase system use per-phase quantities, or $I_s = P_{L,3\phi} / (\sqrt{3} V'_{s,LL})$. Figure 9.37 shows the right-shunt operating modes at a 30% sag.

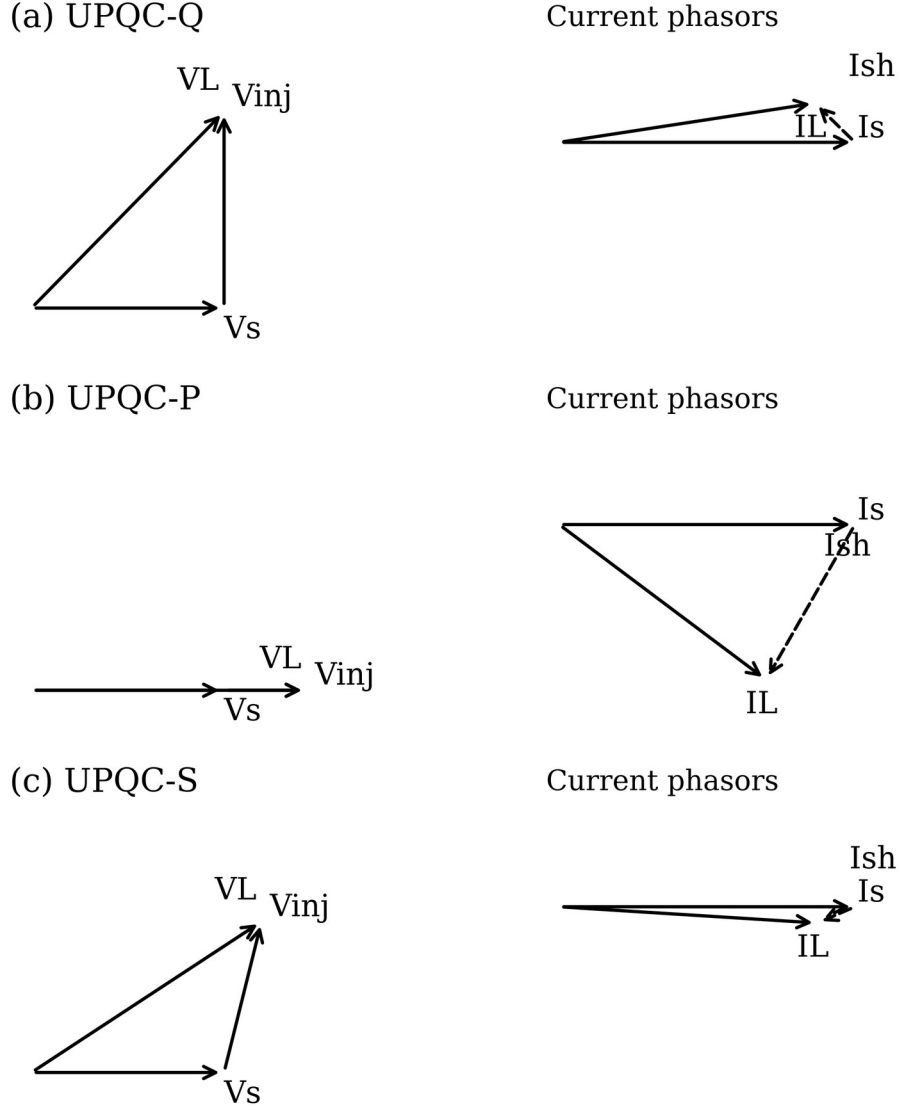


Figure 9.37. Right-shunt UPQC phasors at a 30% sag and 0.8 load power factor: (a) Q, (b) P and (c) S with 130/230 pu injection. Left column: voltages on target load-voltage base. Right: currents on load-current base. Voltage and current scales are separate; dashed I_{sh} runs from the source-current tip to the load-current tip.

UPQC-Q. The injection is perpendicular to I_s , hence to V'_s , so $V_{Lc}^2 = |V'_s|^2 + U^2$:

$$U = |V_{inj}| = \sqrt{V_{Lc}^2 - |V'_s|^2}, \delta = \cos^{-1}(1 - X) \quad (9.46)$$

where δ is the angle by which the load voltage leads the supply. The series converter's duty is entirely reactive:

$$S_{se} = U I_s = V_{Lc} I_L \cos \phi \tan \delta, P_{se} = 0 \quad (9.47)$$

Writing V'_s in place of V_{Lc} in the last form is a common slip that underestimates the rating by the factor $1 - X$. The load current lags the rotated load voltage by ϕ , so it lies at $\delta - \phi$ from V'_s , and the shunt converter supplies the difference $I_{sh} = I_L - I_s$:

$$I_{sh} = I_L \frac{\sqrt{(1 - X)^2 + \cos^2 \phi - 2 \cos \phi \cos(\phi - \delta)(1 - X)}}{1 - X} \quad (9.48)$$

with $S_{sh} = V_{Lc} I_{sh}$. When $X = 1 - \cos \phi$, $\delta = \phi$ and $I_{sh} = 0$: the rotation of the load voltage has brought the load current into phase with the supply, so the series converter alone supplies all the load's reactive power. At other sag depths the shunt current is nonzero, and its reactive direction reverses across that point.

UPQC-P. The injection is in phase with V'_s , $V_{inj} = V_{Lc} X$ (signed, negative for a swell). The series duty is entirely real:

$$P_{se} = V_{Lc} X I_s = V_{Lc} X I_L \cos \phi / (1 - X), S_{se} = |P_{se}| \quad (9.49)$$

The factor $1/(1 - X)$ is easily forgotten but essential: the series converter carries the increased source current, not the load's active current. The load voltage stays in phase with the supply, so the shunt current has a reactive part (the load's own) and an active part that returns the series converter's real power:

$$|I_{sh}| = I_L \sqrt{\sin^2 \phi + \cos^2 \phi \left(\frac{X}{1 - X} \right)^2} \quad (9.50)$$

UPQC-S. For a specified injection magnitude U , the angle measured from V'_s follows from Equation (9.35), and

$$P_{se} = U I_s \cos \psi, Q_{se} = U I_s \sin \psi, \delta = \text{atan2}(U \sin \psi, V'_s + U \cos \psi) \quad (9.51)$$

The shunt current is again $I_{sh} = I_L \angle (\delta - \phi) - I_s$. A minimum-combined-VA design varies the restored load angle δ and minimizes

$$J(\delta) = I_s |V_{Lc} e^{j\delta} - V'_s| + V_{Lc} |I_L e^{j(\delta - \phi)} - I_s| \quad (9.52)$$

subject to each converter's voltage, current and VA limits, DC-energy balance and the permissible phase shift. Figure 9.38 plots J for a 20 % sag on a 230 V, 25 A load: at 0.8 pf the minimum is the quadrature point ($\delta = 36.87^\circ$, 3450 VA), while at 0.95 pf it lies at $\delta = 10.5^\circ$ with 3132 VA.

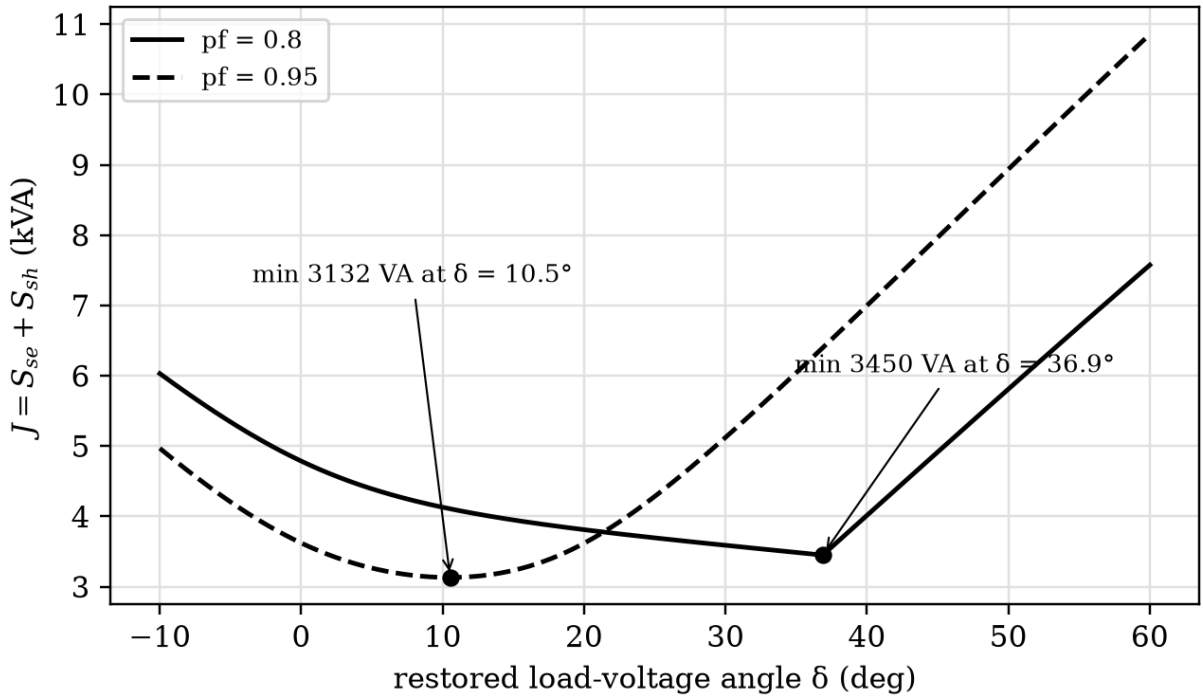


Figure 9.38. Combined operating VA of a UPQC versus restored load-voltage angle for a 20 % sag (230 V, 25 A).

Operating duty and installed rating. The numbers above are operating duties at one sag depth. The installed rating of each converter must cover its own maximum voltage and current over every specified condition, and

those maxima do not occur at the same point. In particular, at nominal supply voltage the series converter is idle and the shunt converter supplies the whole load reactive current $I_L \sin \phi$. Figure 9.39 shows the series, shunt and total duties versus sag depth for a 230 V, 25 A, 0.8 pf load: in UPQC-Q the shunt duty falls from 3450 VA at nominal voltage to zero at a 20 % sag and rises again, while the series duty grows steadily; in UPQC-P the series duty is small but the shunt duty never falls below the nominal 3450 VA. A UPQC-Q designed for a 20 % sag therefore needs about 3450 VA in each converter, 6900 VA in total, although the operating total during the sag is only 3450 VA (Example 9.20). Figure 9.40 gives the series-converter VA of both strategies over a wider range of sags. Harmonic and unbalance compensation add RMS current and voltage that must also enter the thermal ratings, so fundamental kVA alone understates the installed requirement.

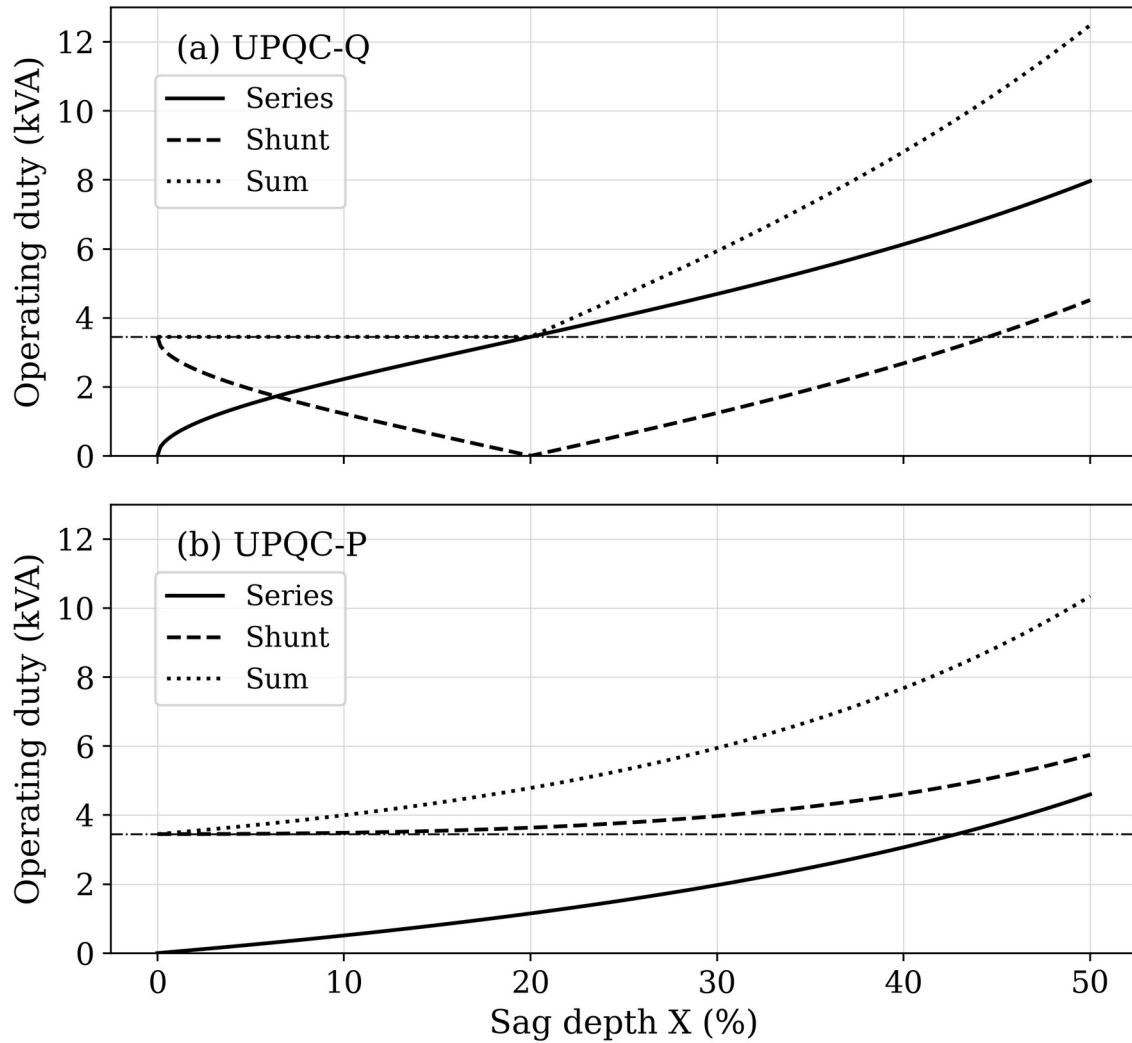


Figure 9.39. Series, shunt and total operating duties of a 230 V, 25 A, 0.8 pf right-shunt UPQC: (a) Q, (b) P. The dash-dot horizontal line marks the nominal-voltage shunt duty of 3.45 kVA. Converter nameplates must cover each duty envelope separately.

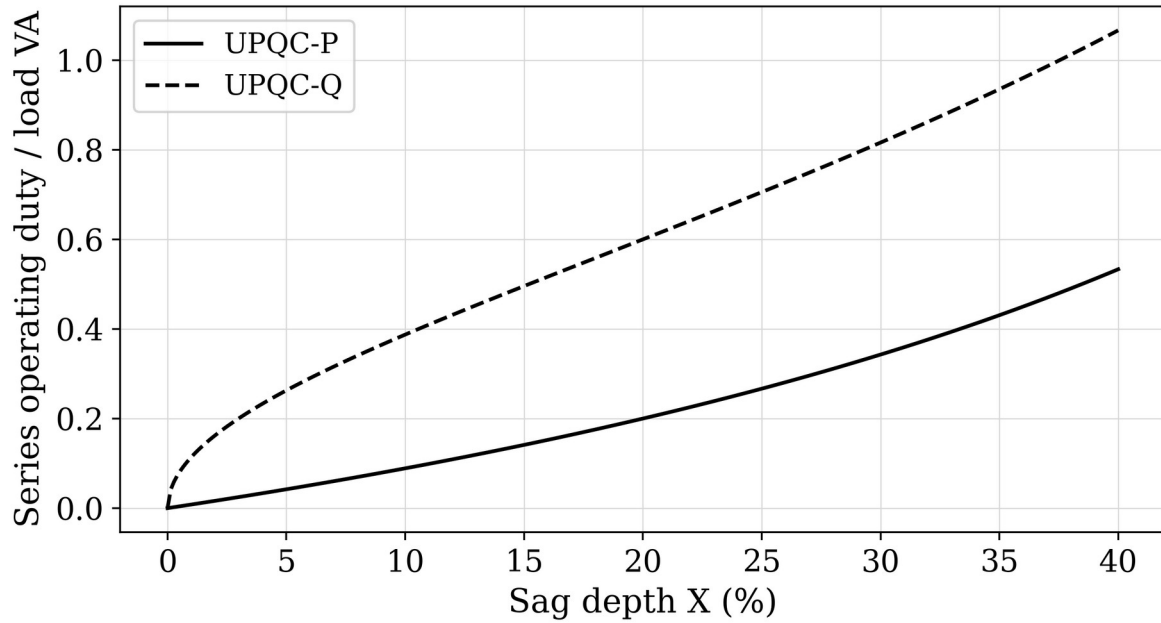


Figure 9.40. Normalized series-converter VA versus balanced sag depth for UPQC-P and UPQC-Q at 0.8 load power factor. These are screening results, not complete converter ratings.

Table 9.9. Injection strategies: current reference and design limits.

Strategy	Right-shunt UPQC ($I_{se} = I_s$)	Standalone DVR ($I_{se} = I_L$)
Quadrature	Sag only; zero series real power; $U = \sqrt{V_{Lc}^2 - V_s'^2}$	Must be perpendicular to I_L ; feasible only if $X \leq 1 - \cos \phi$; check both roots
In-phase	Minimum injection; shunt circulates the series real power	Real power $X P_L$ from storage; swell needs an absorption path
General angle (UPQC-S)	Shares P and Q; optimize a stated objective	Choose phase shift, voltage and energy together
Pre-sag	Restore pre-sag phasor; check both converters	Larger voltage and energy than in-phase

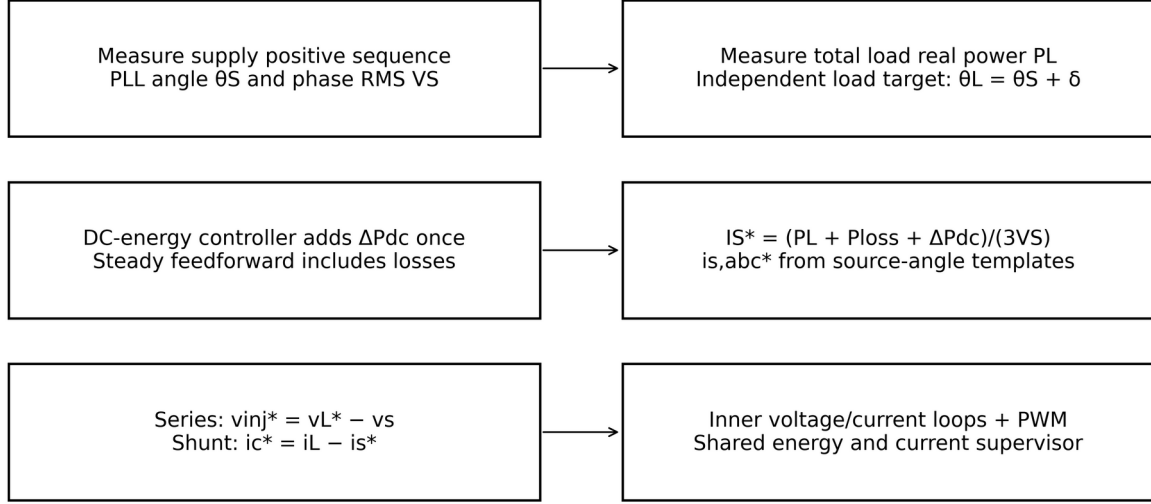
9.5.4 Left-shunt UPQC analysis

In a left-shunt UPQC the series converter carries load current. Zero series real power therefore requires injection perpendicular to I_L , governed by Equations (9.32) and (9.34). Let $\beta = \phi - \theta_L$; the shunt current is the signed reactive component, with magnitude $I_L |\sin \beta|$ and nonnegative apparent duty $S_{sh} = V_s' I_L |\sin \beta|$. Unlike the right-shunt geometry, suitable swells can be corrected without series real power by rotating load voltage behind supply voltage (Example 9.26).

9.5.5 Control of the UPQC

Figure 9.41 shows the usual coordinated structure. The series converter uses the DVR controller of Section 9.4.4: an SRF voltage reference built from the PLL angle, $v_{inj}^i = v_L^i - v_s$, with load-voltage feedback and PWM voltage control. The target may preserve the pre-sag phase or deliberately rotate the load voltage (UPQC-Q, UPQC-S), and the synchronization and reference extraction must reflect that choice. Unbalanced regulation needs phase or sequence controllers rather than one positive-sequence dq magnitude.

Generate the source reference from **total measured load real power in the supply-voltage frame**, rather than blindly scaling the load d-axis current. For balanced positive-sequence supply phase RMS V_s , $I_s^i = (P_L + P_{loss} + \Delta P_{dc}) / (3V_s)$. For amplitude-invariant peak dq variables aligned to the source, $i_{sd}^i = 2(P_L + P_{loss} + \Delta P_{dc}) / (3v_{sd})$ and $i_{sq}^i = 0$. Here ΔP_{dc} is the transient energy-controller correction, included once; feedforward supplies steady load power and estimated losses. Use the source PLL angle to synthesize source currents and $i_c^i = i_L - i_s^i$. The load target may have $\theta_L = \theta_s + \delta$, so i_{Ld} in that rotated frame is not interchangeable with source-frame active current. Coordinate saturation and DC feedback to avoid counting series power twice.



$$\text{For peak dq variables: } i_{sd}^* = 2(P_L + P_{loss} + \Delta P_{dc}) / (3v_{sd})$$

Figure 9.41. Coordinated UPQC control: series voltage loop, shunt current loop and common DC-link regulation.

The two controllers act on different variables but share the DC link, and poorly separated bandwidths make them fight: a fast shunt DC-voltage loop may react to the very transient power that the series loop is deliberately producing. A robust hierarchy (Figure 9.42) uses fast series-voltage and shunt-current loops, slower DC-link energy control, and a supervisor that selects UPQC-P, UPQC-Q or an optimized operating point. Current and voltage saturation must be coordinated as well. When the source voltage becomes very low, holding rated load voltage can demand both excessive series voltage and excessive source current, so the supervisor should reduce the restoration target or shed noncritical load before semiconductor limits are reached. This matters most in microgrids, where source strength can be low and PLL dynamics significant.

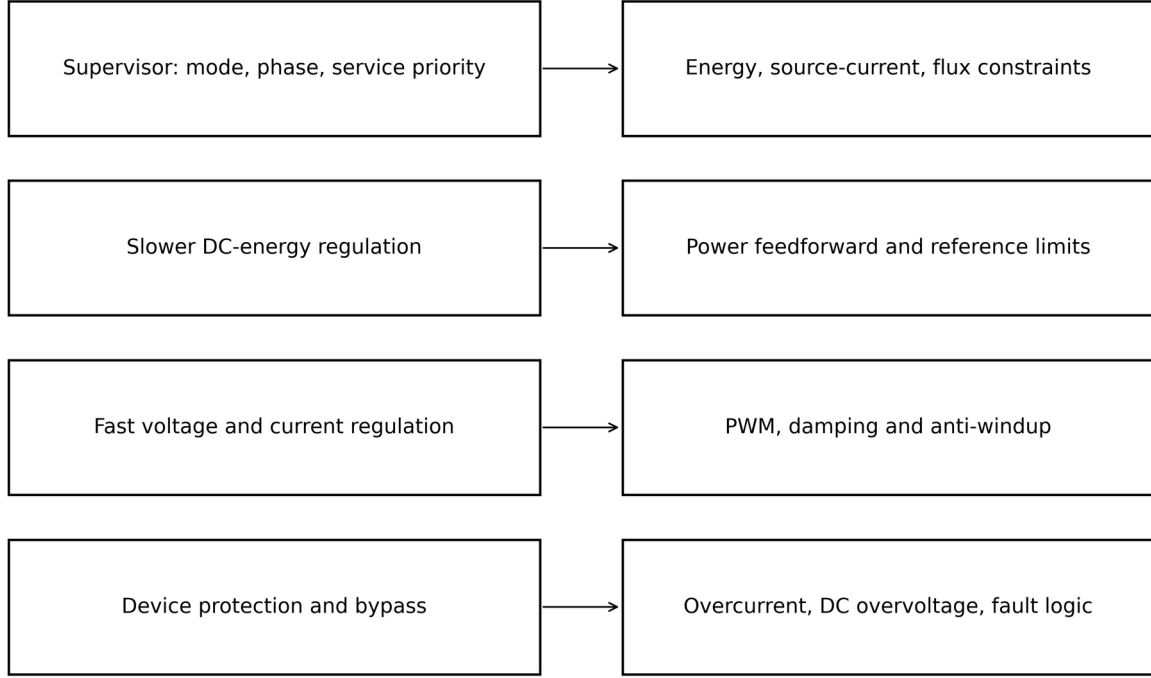


Figure 9.42. Layered control structure of a practical UPQC, from switching-level limits to the supervisory mode selection.

9.5.6 Design of a three-phase four-wire UPQC

The design combines the DSTATCOM design of Section 9.3.4 with the DVR design of Section 9.4.6, using one DC bus for both [2].

The common DC window must satisfy both converters. The series transformer ratio is $n = V_{\text{converterwinding}}/V_{\text{linewinding}}$; for $n > 1$ it steps **down from converter to line**. For the right-shunt case, maximum series line current occurs at the deepest sag, $I_s = P_L/[3(1-X)V_{ph}]$. If ripple is a fixed percentage of line current, its smallest permitted absolute value occurs at the largest swell. Equation (9.53) gives converter-side inductance; $L_{\text{line}} = L_{\text{converter}}/n^2$ and $\Delta i_{\text{converter}} = \Delta i_{\text{line}}/n$. Converter-terminal voltage includes filter drop and must be checked separately from winding injection voltage.

$$L_{DVR, \text{converter}} = n\sqrt{3}mV_{dc}/(12af_s\Delta i_{\text{line}}), L_{DVR, \text{line}} = L_{DVR, \text{converter}}/n^2 \quad (9.53)$$

9.5.7 Defined energy and power-circulation illustration

Figure 9.43 uses an ideal series-voltage reference with finite source-amplitude response. Supply is 415 V, 50 Hz with factor $s = 0.75$ from 0.10 to 0.20 s and 1.2 from 0.28 to 0.36 s, otherwise 1. Load voltage is the ideal target $v_{Lk} = V_m \cos \theta_k$; injection is $v_{Lk} - v_{sk}$. Load real power is 27.5 kW, $\text{pf} = 0.8$, with the same prescribed 13.1% harmonic spectrum as Section 9.3.5 and load peak $A = P_L/[1.5V_m 0.8]$. Source current is $B \cos \theta_k$. With $C = 0.02$ F, $V_{dc,0} = 800$ V and $W_0 = CV_{dc,0}^2/2$, set $B^{\dot{}} = [P_L + 50(W_0 - W)]/(1.5V_ms)$, $\dot{B} = (B^{\dot{}} - B)/0.001$ and $\dot{W} = 1.5V_msB - P_L$. Initialize at nominal steady state and integrate at 50 μs by forward Euler. Reconstruct plotted DC voltage from the full instantaneous sum of converter AC exports. This assumes exact series tracking and ideal shunt reference tracking, apart from the declared 1 ms amplitude response. PWM, PLL dynamics, losses, limits and hardware protection are omitted.

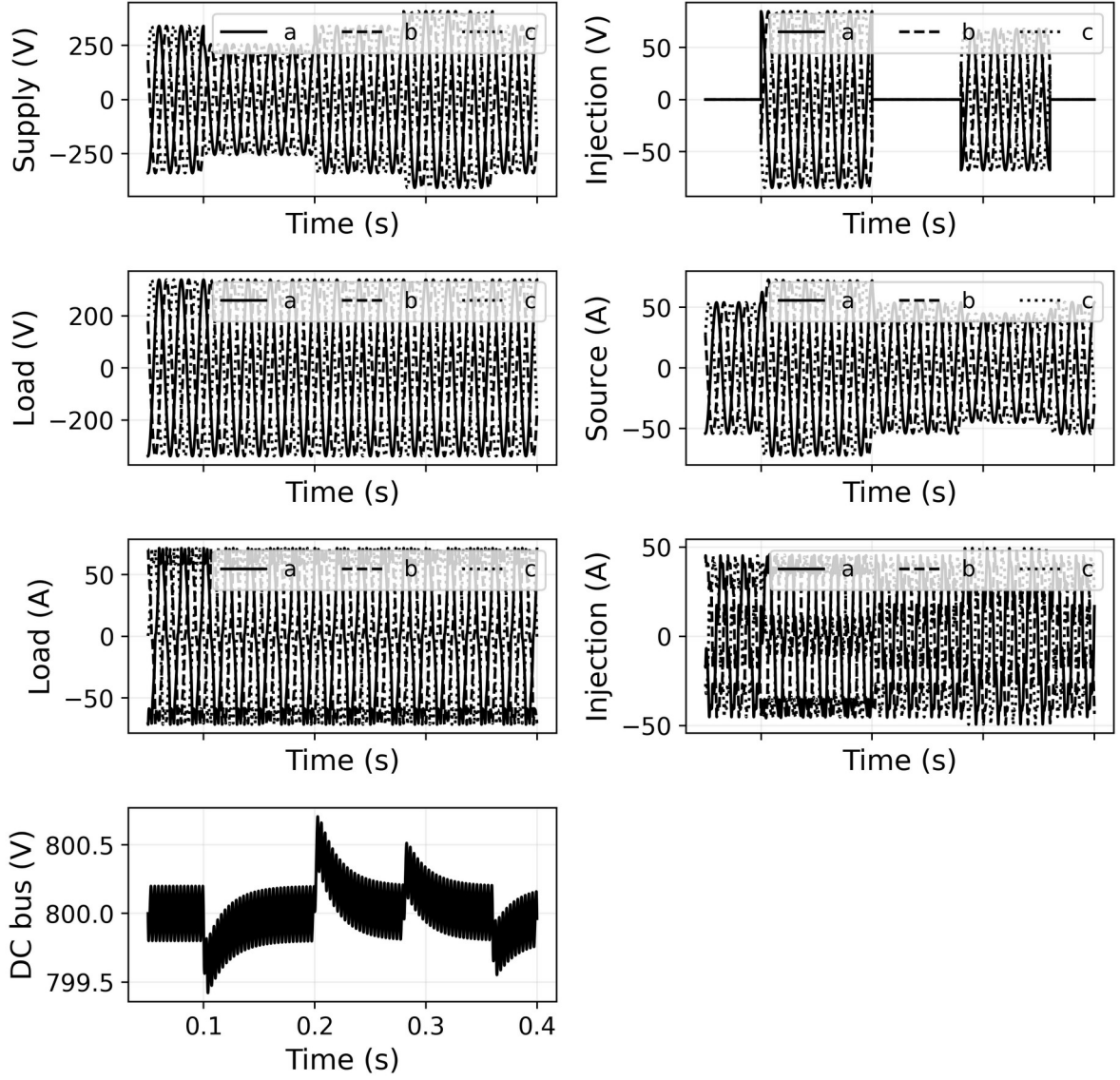


Figure 9.43. Defined reduced right-shunt UPQC-P model: supply, injection, ideal load voltage, source current, nonlinear load current, shunt injection and reconstructed DC voltage. This is an energy illustration with prescribed tracking, not switched-converter validation.

Compute $p_{se} = \sum v_{inj,k} i_{s,k}$ and $p_{sh} = \sum v_{Lk} (i_{Lk} - i_{s,k})$. Their exact sum is $p_{load} - p_{source}$, so $\dot{W}_{dc} = -(p_{se} + p_{sh})$ in this lossless model. At settled sag operation the series exports $27.5(0.25/0.75) = 9.167$ kW and the shunt absorbs it; during the swell series export is $27.5(-0.2/1.2) = -4.583$ kW. Figure 9.44 uses a centered one-cycle moving average for display, not a causal controller filter. Transition imbalance is buffered by the DC capacitor.

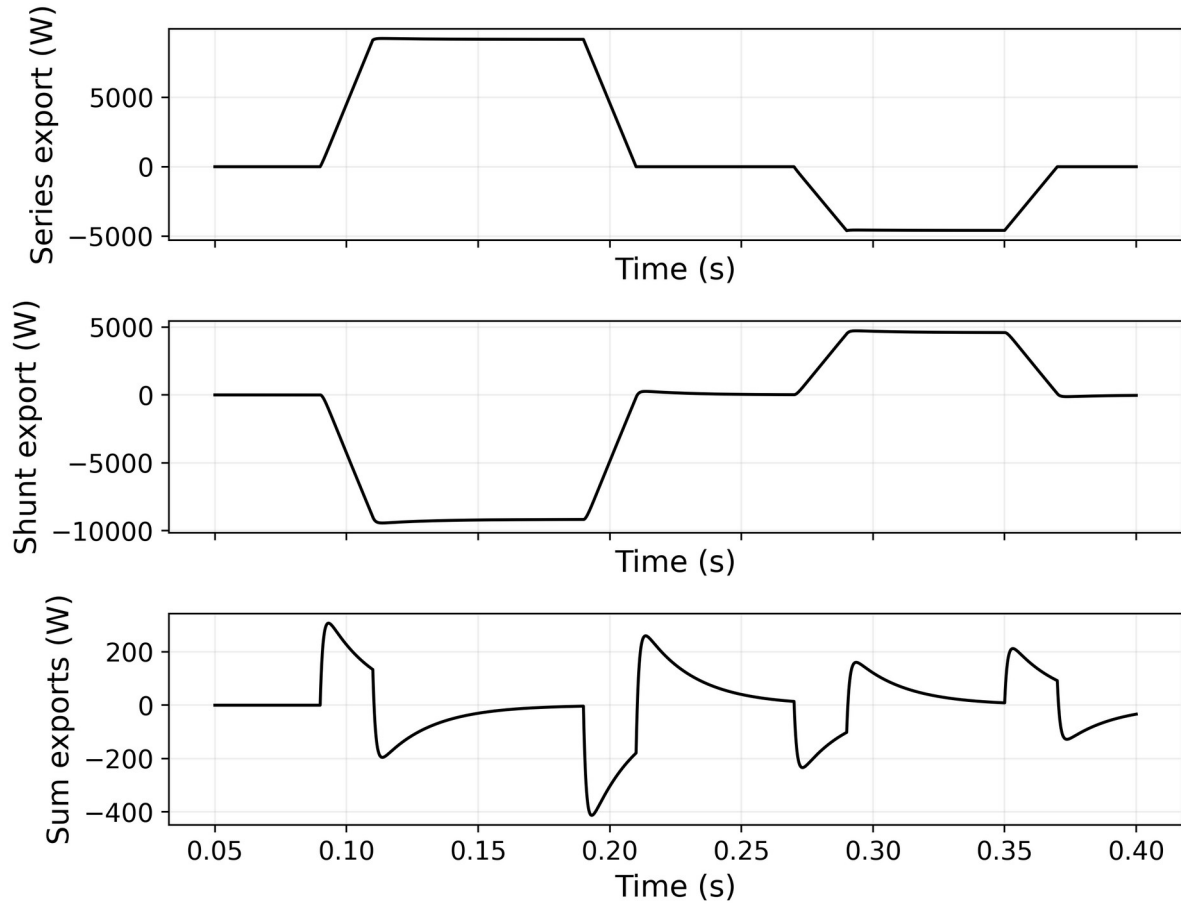


Figure 9.44. Centered one-cycle-average series and shunt AC exports and their sum for the defined UPQC model. Positive sum depletes DC energy; negative sum charges it. Edge samples of the averaging window are omitted.

9.5.8 Worked examples: UPQC

Example 9.20. Right-shunt UPQC-Q: ratings during sag and at nominal voltage

A single-phase right-shunt UPQC-Q supplies a 230 V, 50 Hz, 25 A, 0.8 lagging pf load and keeps the source at unity power factor. The supply sags by 20 %. Find the voltage, current and VA of each converter during the sag, and the installed ratings when nominal operation is included.

Solution. $P_L = 230 \times 25 \times 0.8 = 4600$ W. Nominal: $V_{inj} = 0$; the shunt converter supplies $I_L \sin \phi = 15$ A at 230 V, 3450 VA. Sag: $V'_s = 184$ V and $I_s = 4600/184 = 25$ A. From Equation (9.46), $U = \sqrt{230^2 - 184^2} = 138$ V and $\delta = 36.87^\circ$; $S_{se} = 138 \times 25 = 3450$ VA, all reactive. Since $\delta = \phi$, Equation (9.48) gives $I_{sh} = 0$. Installed: series 138 V, 25 A, 3450 VA; shunt 230 V, 15 A, 3450 VA; total 6900 VA. The operating total during the sag is only 3450 VA, but each converter's rating is set by a different condition.

Example 9.21. UPQC-Q at a shallower sag

A lossless right-shunt UPQC-Q protects a 230 V, 20 A, 0.8 pf load during a 10 % sag. Find the series and shunt duties.

Solution. $V'_s = 207$ V, $P_L = 3680$ W and $I_s = 17.78$ A. $U = \sqrt{230^2 - 207^2} = 100.25$ V, so $S_{se} = 1782.3$ VA. $\delta = 25.84^\circ$, $I_L = 20 \angle -11.03^\circ$ A relative to V'_s and $|I_{sh}| = |I_L - I_s| = 4.251$ A, so $S_{sh} = 977.7$ VA. Total 2760.0 VA. For comparison, a standalone DVR protecting the same load at the same sag (Figure 9.26) needs only 863 VA at zero real power, because it carries a different current with a different geometry. Neither result supports a general claim that one arrangement always needs the larger rating.

Example 9.22. Right-shunt UPQC-P

For the load and sag of Example 9.20, find the converter duties in UPQC-P mode.

Solution. $V_{inj}=230 \times 0.2=46$ V and $I_s=25$ A, so $S_{se}=P_{se}=1150$ VA. With $I_L=20-j15$ A, $I_{sh}=I_L-I_s=-5-j15$ A and $|I_{sh}|=15.81$ A, matching Equation (9.50): $25\sqrt{0.36+0.64 \times 0.0625}$. $S_{sh}=230 \times 15.81=3636.6$ VA and $P_{sh}=-1150$ W, returning the series converter's export. Total 4786.6 VA. The series duty is a third of the UPQC-Q value, but the total is larger, because the shunt converter carries both the load's reactive current and the circulated real power.

Example 9.23. UPQC-Q at a 25 % sag

A right-shunt UPQC-Q supplies 230 V, 20 A, 0.8 pf. The supply sags by 25 %. Find all ratings.

Solution. $P_L=3680$ W, $V_s'=172.5$ V and $I_s=21.33$ A. $U=152.13$ V and $S_{se}=3245.5$ VA. $\delta=41.41^\circ$, $\cos(\phi-\delta)=0.99686$, and Equation (9.48) gives $I_{sh}=2.111$ A and $S_{sh}=485.5$ VA during the sag. At nominal voltage the shunt converter needs $20 \times 0.6=12$ A, 2760 VA, which governs. Installed total: $3245.5+2760=6005.5$ VA.

Example 9.24. UPQC-S with a limited series voltage

A single-phase right-shunt UPQC protects a 230 V, 25 A, 0.8 pf load against a 30 % sag. The series converter is rated for 130 V. Find the injection angle, its real and reactive powers, the load-voltage rotation and the shunt duty, and compare with UPQC-Q and UPQC-P.

Solution. $V_s'=161$ V and $I_s=4600/161=28.57$ A. From Equation (9.35), $\cos \psi = (230^2 - 161^2 - 130^2) / (2 \times 161 \times 130) = 0.2408$ and $\psi = 76.07^\circ$. $P_{se}=130 \times 28.57 \times 0.2408=894.3$ W, $Q_{se}=3605.0$ var and $S_{se}=3714.3$ VA. $\delta = \sin^{-1}(130 \sin 76.07^\circ / 230) = 33.27^\circ$. $I_{sh}=|25 \angle -3.60^\circ - 28.57|=3.95$ A, so $S_{sh}=907.7$ VA and the total is 4621.9 VA. UPQC-Q at the same sag: $U=164.25$ V, $S_{se}=4692.9$ VA, $S_{sh}=1242.9$ VA, total 5935.9 VA. UPQC-P: $U=69$ V, $S_{se}=1971.4$ VA, $S_{sh}=3973.5$ VA, total 5945.0 VA. The UPQC-S angle reduces both the series voltage (130 V instead of 164 V) and the total operating VA, by about 22 %.

Example 9.25. Left-shunt UPQC-Q during a sag

A left-shunt UPQC-Q supplies 230 V, 25 A, 0.8 pf with a 10 % sag. The series converter injects perpendicular to the load current, and the shunt converter keeps unity pf at the source. Find both duties during the sag and the installed ratings.

Solution. $V_s'=207$ V. From Equation (9.34), $\cos(\theta_L - \phi) = 184/207 = 0.8889$, so $\theta_L = 9.60^\circ$ (smaller root) and $V_{inj}=43.17 \angle 62.73^\circ$ V. The series converter carries $I_L=25$ A: $S_{se}=1079.2$ VA, purely reactive. $\beta = \phi - \theta_L = 27.27^\circ$, $I_s=4600/207=22.22$ A and $I_{sh}=25 \sin 27.27^\circ = 11.45$ A at 207 V, $S_{sh}=2370.8$ VA. At nominal voltage the shunt converter needs 15 A at 230 V, 3450 VA. Installed: $1079.2+3450=4529.2$ VA.

Example 9.26. Left-shunt UPQC-Q during a swell

For the left-shunt UPQC of Example 9.25, the supply swells by 10 %. Find the duties.

Solution. $V_s'=253$ V and $\cos(\theta_L - \phi) = 184/253 = 0.7273$, so $\theta_L - \phi = \pm 43.34^\circ$. The root $\theta_L = -6.47^\circ$ (load voltage lagging the supply) gives $V_{inj}=35.65 \angle -133.34^\circ$ V; the other root needs 311.6 V. $S_{se}=35.65 \times 25=891.2$ VA. $\beta = 43.34^\circ$, so $I_{sh}=25 \sin 43.34^\circ = 17.16$ A at 253 V and $S_{sh}=4341.2$ VA. Installed: shunt 253 V and 17.16 A (4341 VA), series 891 VA, total 5232 VA.

9.5.9 Advanced and multi-converter UPQC structures

The two-converter UPQC is the reference structure, but modern distribution systems motivate extensions (Table 9.10). UPQC-DG connects distributed generation or storage to the common DC link so that the converters perform power-quality compensation and controlled active-power exchange together. An interline or multi-converter UPQC coordinates more than one feeder through a common DC link. Open and distributed UPQCs separate the converter locations while keeping coordinated objectives. Modular multilevel and cascaded converters extend the voltage rating and improve waveform quality at medium voltage. None of these is automatically superior: adding a PV array, battery or extra feeder changes the DC-link power balance, protection, islanding behaviour and converter rating, so a design must state which services are simultaneous, which are mutually exclusive under current limit, and how energy management interacts with fast power-quality control [20].

Table 9.10. Extensions of the UPQC.

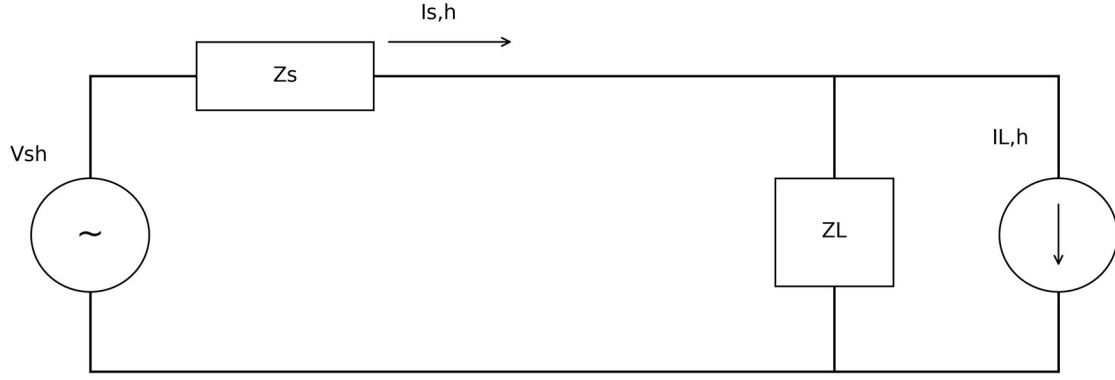
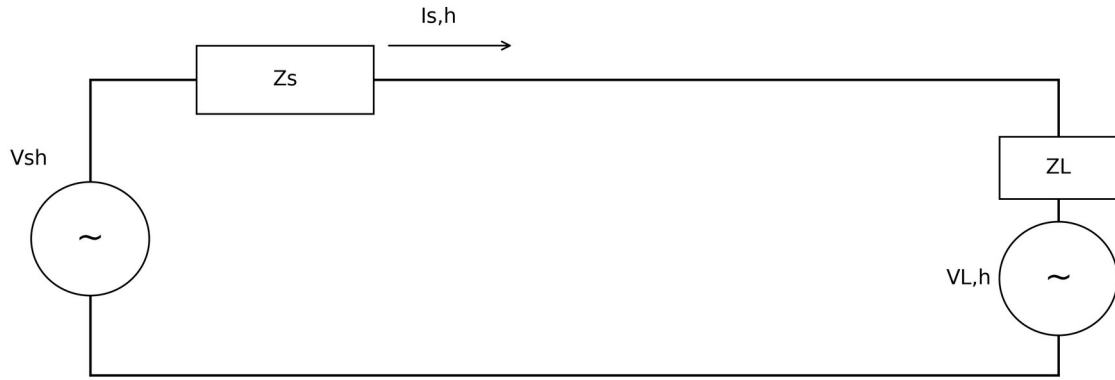
Extension	Added capability	New design issue
UPQC-DG / PV-UPQC	PQ compensation plus generation interface	DC-link energy management and curtailment
UPQC-BESS	Sag support, peak shaving, ride-through	State of charge, battery power limit and protection
Interline / MC-UPQC	Coordination of several feeders	Cross-feeder real-power balance and fault isolation
Open / distributed UPQC	Flexible placement, less common hardware	Communication, synchronization and protection
MMC / cascaded UPQC	Medium-voltage scalability and lower dv/dt	Cell balancing, stored energy and fault handling

9.6 Series and Hybrid Active Power Filters

Three ideas are often mixed together and should be kept apart. A shunt active filter creates a compensating current; a series active filter creates a compensating voltage; a hybrid filter deliberately assigns part of the harmonic spectrum to passive components so that the converter handles only a residual or only a small voltage. The engineering question is therefore not merely whether harmonics exist, but what kind of source produces them, where they originate and which impedance path the compensator must create. A DSTATCOM operated purely for harmonic cancellation is a shunt active power filter, and its pq and SRF control were developed in Section 9.3.3. This section builds the network-level theory that decides which filter arrangement suits a given load and supply [21], [22], [23], [24], [25].

9.6.1 Harmonic-source models

Peng's classification of harmonic sources is the starting point [23] (Figure 9.45). A thyristor rectifier with a large DC inductor draws a nearly rectangular current set by the DC-side inductance; seen from the AC side it is a harmonic *current source* I_{Lh} with a large parallel impedance Z_L (Norton form). A diode rectifier with a large DC capacitor, typical of drives, switch-mode supplies and EV chargers, draws a pulsed current whose shape is set by the AC impedance in front of it; seen from the AC side it is a harmonic *voltage source* V_{Lh} behind a small series impedance Z_L (Thevenin form). The supply is represented at each harmonic by a background voltage V_{sh} behind the source impedance $Z_s = R_s + jhX_s$.

(a) Norton load: $I_{L,h}$ positive from PCC toward load

(b) Thevenin load: positive node-voltage-rise convention

Figure 9.45. Harmonic-source models: (a) current-source type load (Norton), for which a shunt filter is natural; (b) voltage-source type load (Thevenin), for which a series filter is natural.

The distinction matters because the harmonic current that reaches the source depends on the ratio of the load's own impedance to the impedance the filter presents. A shunt path in parallel with a current-source load diverts the current; the same shunt path in parallel with a voltage-source load lowers the PCC impedance seen by the load and can actually increase the harmonic current it draws. A series impedance in front of a voltage-source load limits that current directly. The rest of the section quantifies these statements.

9.6.2 Passive and shunt active filtering of a current-source load

For the current-source limit ($|Z_L| \gg |Z_s|, |Z_F|$), a passive shunt branch of impedance Z_F gives the following harmonic source current by superposition. With finite parallel load impedance, replace Z_F by $Z_F \parallel Z_L$ in Equation (9.54).

$$I_{sh} = \frac{Z_F}{Z_s + Z_F} I_{Lh} + \frac{1}{Z_s + Z_F} V_{sh} \quad (9.54)$$

The filtering characteristic is set by the ratio $Z_F / (Z_s + Z_F)$, so it depends on the source impedance, which the designer does not control and which changes with network configuration. Two failure modes follow directly from Equation (9.54). Below its tuning frequency the filter is capacitive, and it forms a parallel resonance with the inductive source at which $|Z_s + Z_F|$ becomes small: both terms of Equation (9.54) are then amplified. And because Z_F is low at its tuning frequency, the filter also sinks background distortion V_{sh} from the rest of the network and can be overloaded by harmonics it was never intended to absorb. For a plain capacitor bank of reactive power Q_c on a bus of short-circuit power S_{sc} , the parallel resonance occurs near

$$h_r \approx \sqrt{\frac{S_{sc}}{Q_c}} \quad (9.55)$$

For a detuning reactor whose fundamental reactance ratio is $p = X_{L,1}/|X_{C,1}|$, the branch series resonance is $h_t = 1/\sqrt{p}$. Use p as a fraction: 7% means 0.07. The commonly selected tuning is below the lowest important characteristic harmonic; this choice does not replace a network frequency scan.

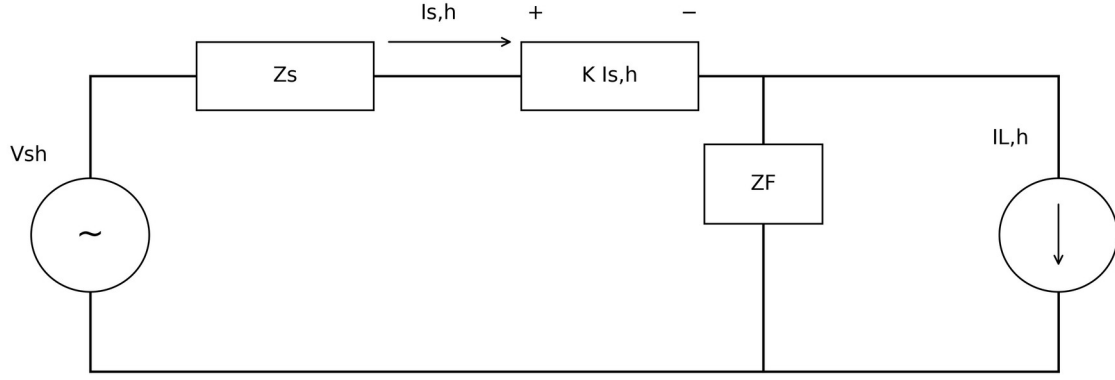
For a finite Norton load, let I_{Lh} denote the independent harmonic-demand source in parallel with Z_L . If the APF reference is specifically $I_{Ch} = G I_{Lh}$, circuit superposition gives

$$I_{sh} = (1 - G) \frac{Z_L}{Z_s + Z_L} I_{Lh} + \frac{1}{Z_s + Z_L} V_{sh} \quad (9.56)$$

Equation (9.56) cancels the independent Norton component as $G \rightarrow 1$, while leaving background-voltage current through finite Z_L . If feedback instead measures the **total** load harmonic current $I_{load,h} = I_{Lh} + V_{PCC,h}/Z_L$ and commands $I_{Ch} = G I_{load,h}$, KCL gives $I_{sh} = (1 - G)(Z_L I_{Lh} + V_{sh})/[Z_L + (1 - G)Z_s]$. In this ideal algebraic model both contributions vanish at $G=1$. These are different signal definitions; practical delay, finite gain and converter limits require a stability and rating check. The APF handles the actual compensating spectrum plus assigned reactive/unbalanced duty, with aggregate duty $S_{APF} = 3V_{ph} \sqrt{\sum_h I_{Ch}^2}$ for balanced harmonic compensation.

9.6.3 Series active filter: voltage cancellation and harmonic isolation

A series APF (Figure 9.46(a)) can serve two distinct objectives. For *voltage cancellation* it isolates the load from measured supply-voltage harmonics by commanding $v_{inj,h} = -v_{s,h}$ within its voltage, current and energy limits. The desired fundamental is estimated first: a PLL with positive-sequence extraction and filtering in a synchronous frame gives $v_{s,1}^{+i\hat{\omega}}$, and the reference becomes $v_{inj}^{\hat{i}} = v_{s,1}^{+i\hat{\omega} - v_s^{\hat{i}}}$. With a balanced fundamental supply the residue contains only harmonics; with an unbalanced supply it also contains unwanted sequence components and must not be called harmonics alone. More generally, any desired load waveform $v_L^{\hat{i}}$ gives $v_{inj}^{\hat{i}} = v_L^{\hat{i}} - v_s^{\hat{i}}$, consistent with Equation (9.27), which is why the DVR and the series APF are the same hardware with different references.



(a) Positive series harmonic drop; passive shunt return path

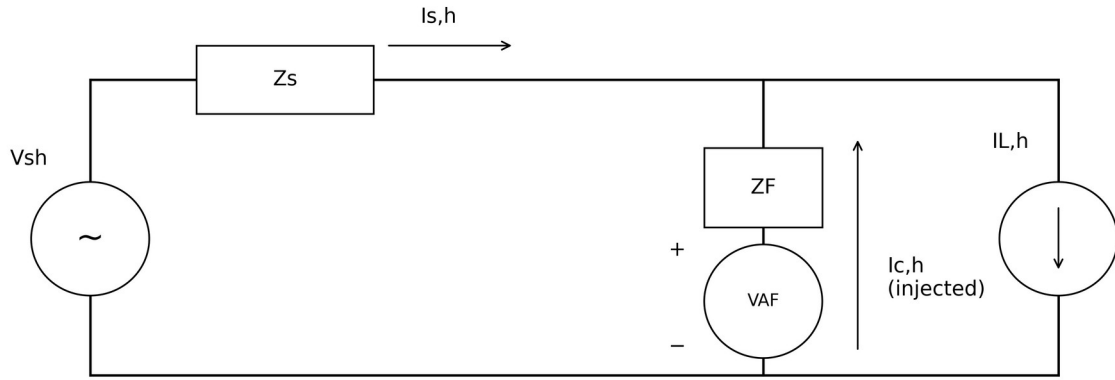
(b) $VAF = +K I_{s,h}$ is a rise from ground toward PCC

Figure 9.46. Harmonic equivalent circuits: (a) positive series-line voltage-drop law with a passive shunt return; (b) active voltage-rise stage in series with a passive shunt branch. Current arrows and active-stage polarity define the KCL/KVL derivation in Section 9.6.4; ZF represents the tuned LC branch impedance.

For *harmonic isolation* the series converter synthesizes a resistance at harmonic frequencies only. Its reference is derived from the detected harmonic source current, $v_c = K i_{sh}$ in the voltage-drop sense, with K in ohms (or per unit) and zero gain at the fundamental, so the converter presents a high series impedance to harmonics and none to the fundamental. Combined with a shunt passive filter, as proposed by Peng, Akagi and Nabae [22], the source harmonic current becomes

$$I_{sh} = \frac{Z_F}{Z_s + Z_F + K} I_{Lh} + \frac{1}{Z_s + Z_F + K} V_{sh} \quad (9.57)$$

When $K \gg |Z_s + Z_F|$, load-generated and background harmonic source currents are approximately suppressed by Equation (9.57), reducing sensitivity to source impedance. Positive real K damps the ideal harmonic loop; practical delay and frequency-dependent control can reduce or reverse this damping, so stability needs checking. Figure 9.47 is an explicitly stated ideal frequency scan. The series stage carries full line current, and $V_{ch} = K I_{sh}$ sets its harmonic voltage duty. A small rating follows only when the required harmonic voltage is small. A current-source harmonic load needs a local return path; the passive branch supplies it.

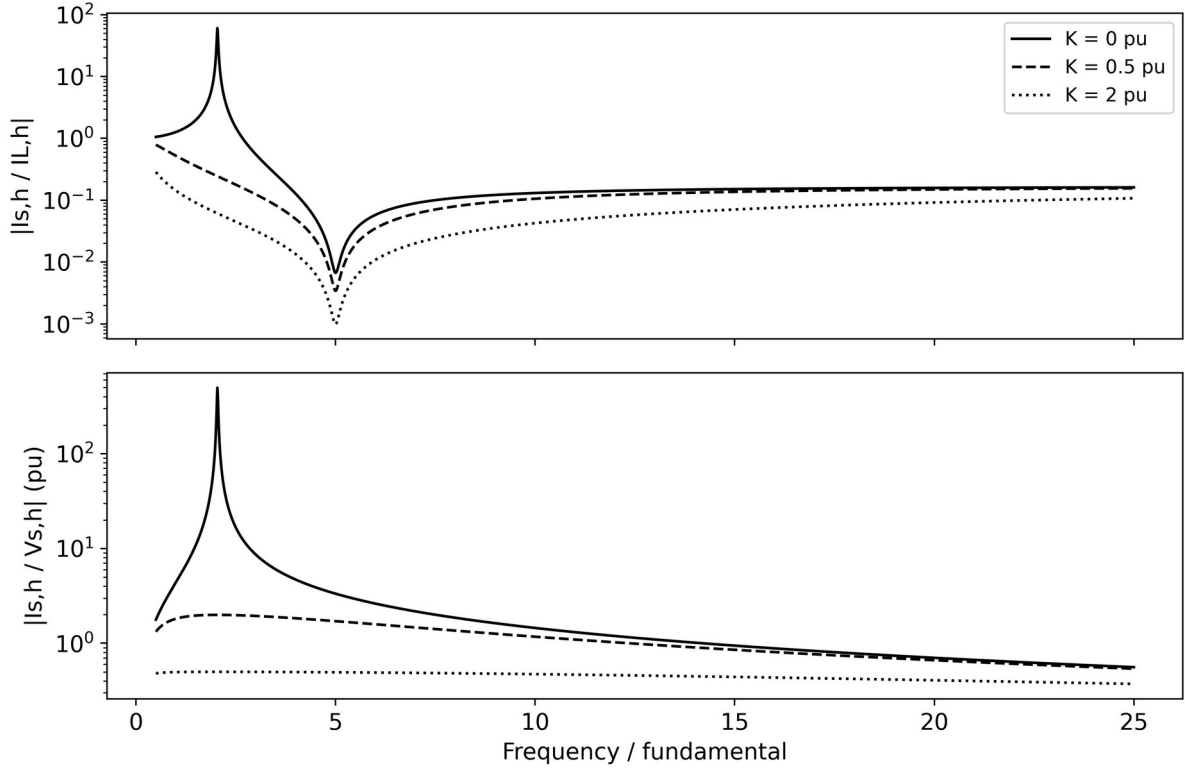


Figure 9.47. Analytical frequency scan using per-unit $Z_s = j0.06h$ and $Z_F = 0.002 + j[0.012h - 1/(3.33333h)]$, tuned at $h=5$ with branch $Q=30$. Source-current feedback gain is $K=0, 0.5$ or 2 pu. Upper panel: load-harmonic transmission; lower: background-voltage admittance. Constant harmonic K is an ideal scan assumption, not a realizable unlimited-bandwidth controller.

For a voltage-source load, signs depend on the chosen source polarity: if both V_{sh} and V_{Lh} are node-voltage rises and current is positive from supply toward load, $I_{sh} = (V_{sh} - V_{Lh}) / (Z_s + Z_L + K)$. A positive series harmonic voltage drop $K I_{sh}$ increases the denominator and suppresses current. The opposite current direction reverses the numerator but not the magnitude conclusion. A low load harmonic impedance can make simple shunt-current cancellation ineffective; topology must be selected with the harmonic equivalent circuit and its control loop, not rectifier labels alone.

9.6.4 Hybrid shunt filter with the active stage in series with an LC branch

Figure 9.46(b) places an active voltage stage in series with a tuned passive shunt branch [24]. Define I_{sh} toward the PCC, load harmonic demand I_{Lh} away from the PCC, and branch injection $I_{ch} = I_{Lh} - I_{sh}$ from ground toward the PCC. Let $V_{AF} = +K I_{sh}$ be a voltage rise from ground toward the PCC. Then $V_{PCC} = Z_F(I_{sh} - I_{Lh}) + K I_{sh}$ and $V_{sh} = Z_s I_{sh} + V_{PCC}$, hence $(Z_s + Z_F + K) I_{sh} = Z_F I_{Lh} + V_{sh}$, exactly Equation (9.57). Source-current feedback changes the **denominator while retaining Z_F in the numerator**; calling the branch simply $Z_F + K$ obscures this controlled-source coupling. The active stage carries the full branch current, including its fundamental capacitive component and harmonic current, giving Equation (9.58).

By contrast, passive-like branch-current feedback with $V_{AF} = -K I_{ch}$ replaces Z_F by $Z_F + K$ in **both numerator and denominator**. It is a different law and does not have the same isolation result. A series-line resistance law absorbs harmonic power $K |I_{sh}|^2$; the hybrid branch stage has AC export $\Re\{K I_{sh} I_{ch}^i\}$, which can have either sign. Its fundamental/DC regulator must manage that average power and losses. Virtual damping implies neither zero energy exchange nor stability under delay.

$$S_{AF} = V_{AF,rms} I_{F,rms}, I_{F,rms} = \sqrt{I_{F1}^2 + \sum_h I_{Fh}^2}, V_{AF,rms} \approx K \sqrt{\sum_h I_{sh}^2} \quad (9.58)$$

The series-branch active stage can have a small VA duty if the passive branch supports most fundamental voltage and the required controlled voltage remains small; its branch-current, insulation, loss and transient duties still need calculation. A separate small shunt APF in parallel with a passive filter carries only its assigned residual. These are different current paths, as Example 9.42 illustrates.

Example 9.27. Capacitor-bank resonance and detuning

An 11 kV bus has a short-circuit level of 100 MVA. A 3 Mvar power-factor-correction bank is to be added near a large six-pulse drive. Find the parallel resonant order and the order to which a 7 % detuning reactor tunes the branch.

Solution. $h_r = \sqrt{100/3} = 5.77$, near the fifth harmonic, so resonance risk needs checking; the exact fifth-harmonic amplification cannot be inferred from resonant order alone. A 7% reactor gives $h_t = 1/\sqrt{0.07} = 3.78$, below the fifth. The ideal branch is inductive above its tuning frequency and its parallel resonance with an inductive source lies below 3.78. Scan credible network configurations and check damping, harmonic loading and capacitor/reactor voltage duty.

Example 9.28. Series active filter with a shunt passive filter

At the fifth harmonic the supply has $Z_s = 0.02 + j0.30$ pu and the detuned passive branch has $Z_F = 0.02 + j0.05$ pu. The load harmonic-demand magnitude is $I_{L5} = 0.20$ pu and background supply harmonic voltage is $V_{s5} = 0.01$ pu; their relative phases are unspecified. Compare passive-only operation with a positive series harmonic voltage-drop law of $K=1$ pu, and estimate the worst-phase converter voltage and VA on the load base.

Solution. Passive only, from Equation (9.54): $|Z_F/(Z_s + Z_F)| = 0.0539/0.3523 = 0.153$, so the load contributes 0.031 pu, and the background contributes $0.01/0.3523 = 0.028$ pu. Depending on phase, the source 5th-harmonic current is up to 0.059 pu, and the filter is also absorbing distortion from the rest of the network. With $K=1$ pu, Equation (9.57) gives $0.0539/1.097 \times 0.20 = 0.0098$ pu and $0.01/1.097 = 0.0091$ pu, at most 0.019 pu in total, a reduction of about two thirds even with the filter detuned. The series-filter voltage is $V_{c5} = K I_{s5} \leq 0.019$ pu. Since it carries the full 1 pu line current, its rating is about $0.019 \times 1.0 = 1.9$ % of the load kVA, compared with roughly 20 % for a shunt APF that would have to carry the full 0.20 pu harmonic current.

9.6.5 Choosing among filter arrangements

A shunt APF and a DSTATCOM share the same hardware; the difference is the control objective. An APF emphasizes harmonic and unbalanced-current cancellation, while a DSTATCOM often includes fundamental reactive power and voltage regulation, and one converter can do both within its RMS current and bandwidth. Table 9.11 collects the trade-offs derived above.

Table 9.11. Filter arrangements compared.

Arrangement	Best-suited harmonic source	Dependence on Z_s	Resonance risk	Converter rating
Passive shunt filter	Current-source type, fixed spectrum	Strong (Equation (9.54))	High; sinks background distortion	None

Arrangement	Best-suited harmonic source	Dependence on Z_s	Resonance risk	Converter rating
Shunt APF	Current-source type, variable spectrum	Weak (Equation (9.56))	Filter/network/control impedance still needs checking	Full harmonic + reactive current
Series APF + passive shunt	Either type; weak or variable grids	Reduced in the stated high-gain harmonic model	Damped by K	Potentially small voltage duty; full line current, calculate actual envelope
Active stage in series with LC	Current-source type at medium voltage	Reduced for specified source-current feedback and sufficient stable gain	Damped by K	Potentially small voltage duty; full branch current, calculate actual envelope
Small shunt APF parallel to passive	Current-source type	Moderate	Must be checked	Residual harmonic current only

Solid-state transfer switches, static breakers and fault-current limiters also belong to the custom-power family, but their function is source transfer, interruption or current limitation rather than continuous waveform conditioning; they were treated in Chapter 8. A UPS differs from a DVR because it can supply the entire load during source loss, provided storage and isolation are adequate. A solid-state transformer can integrate voltage transformation, isolation, DC ports and power-quality control, but its architecture and protection go well beyond a conventional compensator (Table 9.12).

Table 9.12. Custom-power devices and what each does not replace.

Device	Primary action	Best-suited disturbance	What it does not replace
DSTATCOM / shunt APF	Inject current	Reactive, harmonic, unbalanced and neutral current	Series restoration of a missing source voltage
DVR / series APF	Inject voltage	Sag, swell, voltage unbalance and harmonics	Long interruption without an energy source
UPQC	Series voltage + shunt current	Combined voltage and current problems	UPS function unless storage and isolation are designed
SSTS	Transfer load between sources	Source disturbance with a healthy alternate feeder	Continuous harmonic compensation
FCL	Increase fault-path impedance	Excessive fault current	Voltage restoration
UPS	Supply load from stored or alternate energy	Interruption and ride-through	Low-cost feeder-wide PQ correction

9.7 IEEE Std 519-2022 Compliance at the Distribution Level

IEEE Std 519-2022 [26] is a system-level standard applied at the point of common coupling (PCC), not a statement that a particular converter has “good THD”. Its philosophy is a shared responsibility: the user limits the harmonic current injected at the PCC, and the system owner limits the voltage distortion delivered there. Current compliance is judged by total demand distortion and by individual harmonics referred to the maximum demand current, with limits that depend on the short-circuit ratio; voltage limits are checked separately. A device-level current THD measured at one operating point therefore cannot by itself establish compliance.

9.7.1 Definitions and the TDD-THD relation

The PCC is the point on the public system electrically nearest to the user’s installation where other customers are, or could be, connected. For harmonic currents I_h at the PCC,

$$TDD = \frac{\sqrt{\sum_{h=2}^{50} I_h^2}}{I_L} \times 100\% \quad (9.59)$$

Here I_L is the fixed RMS monthly-demand basis of IEEE Std 519-2022, not the instantaneous or fundamental current used elsewhere. Twelve monthly maximum 15- or 30-minute RMS demand currents are averaged. Equation (9.60) defines fundamental-normalized voltage THD.

For shorter operating history, use available monthly demand data; for a new installation use projected monthly apparent-power demand as specified in [26].

$$THD_v = \frac{\sqrt{\sum_{h=2}^{50} V_h^2}}{V_1} \times 100\% \quad (9.60)$$

THD uses the contemporaneous fundamental I_1 , whereas TDD uses the fixed I_L , so

The harmonic sums here extend through order 50 and exclude interharmonics; higher orders may also need assessment [26].

$$TDD = THD_i \frac{I_1}{I_L} \quad (9.61)$$

A lightly loaded drive can show a very high current THD while its TDD is small, because its harmonic currents are small in amperes (Figure 9.48). This is exactly why the standard uses TDD: it penalizes harmonic amperes, which create voltage distortion, rather than distortion percentages at light load.

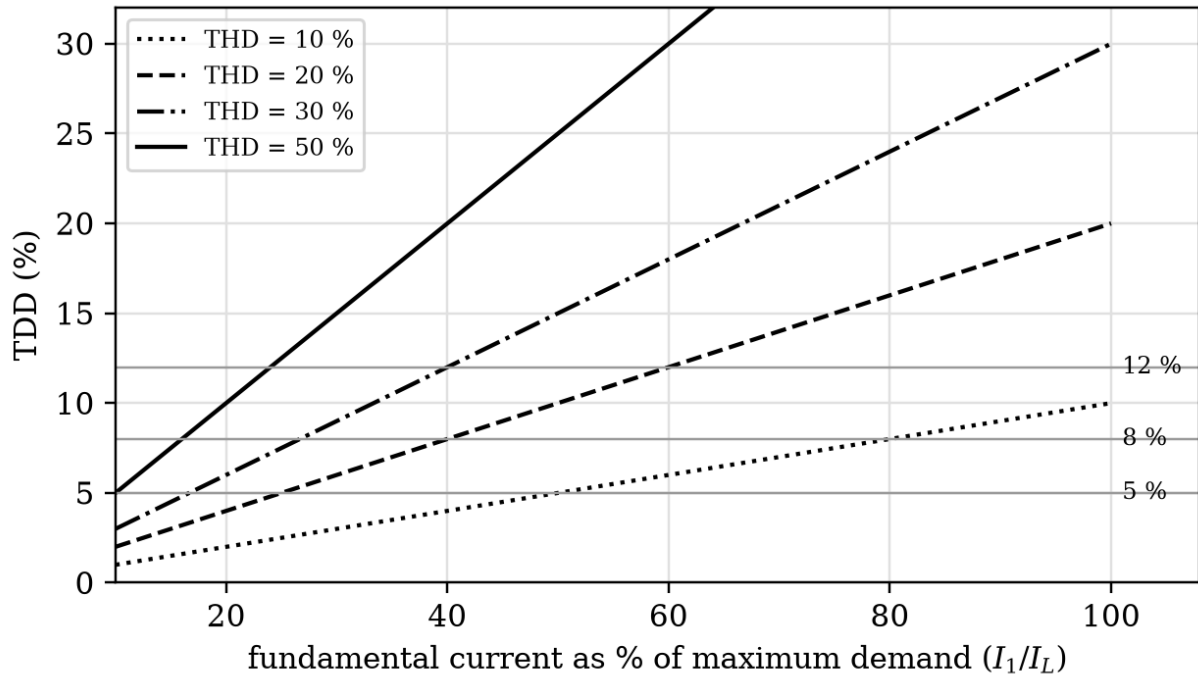


Figure 9.48. TDD versus loading for several values of current THD, with the 5 %, 8 % and 12 % TDD limits for $I_{sc}/I_L < 20$, 20 to 50 and 50 to 100.

9.7.2 Current and voltage limits

Table 9.13 gives the current limits for systems rated 120 V to 69 kV; the standard has separate tables for 69 to 161 kV and above 161 kV. The 2022 edition changed the treatment of even harmonics: in the 2014 edition all even harmonics were limited to 25 % of the odd-harmonic limits, whereas in 2022 only even harmonics of order 2, 4 and 6 are limited, to 50 % of the limit for their range, and even harmonics above the 6th share the limits of the odd harmonics in the same range.

Table 9.13. IEEE Std 519-2022 current distortion limits for systems from 120 V to 69 kV (percent of I_L ; even harmonics $h \leq 6$ are limited to 50 % of the values shown).

Table 9.13 summarizes load-installation limits. DC-offset-producing distortion is excluded; generation and mixed load/IBR installations require the applicability provisions of [26].

I_{sc}/I_L	$2 \leq h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h \leq 50$	TDD
below 20	4.0	2.0	1.5	0.6	0.3	5.0
20 to below 50	7.0	3.5	2.5	1.0	0.5	8.0
50 to below 100	10.0	4.5	4.0	1.5	0.7	12.0
100 to below 1000	12.0	5.5	5.0	2.0	1.0	15.0
1000 and above	15.0	7.0	6.0	2.5	1.4	20.0

Table 9.14. IEEE Std 519-2022 voltage distortion limits at the PCC.

Apply Table 9.14 to line-to-neutral harmonics on the rated PCC voltage basis. Equation (9.60) uses measured fundamental voltage, so record the denominator explicitly. The >161 kV row has a conditional HVDC exception in [26].

Bus voltage V	Individual harmonic (%)	THD (%)
$V \leq 1$ kV	5.0	8.0
$1 < V \leq 69$ kV	3.0	5.0
$69 < V \leq 161$ kV	1.5	2.5
$V > 161$ kV	1.0	1.5

9.7.3 Why current limits scale with the short-circuit ratio

The structure of Table 9.13 follows from the voltage drop that a harmonic current produces. With the source treated as inductive at the PCC, the fundamental short-circuit reactance is $X_1 = V_1 / I_{sc}$ per phase, so the harmonic reactance is $h X_1$ and

$$\frac{V_h}{V_1} = \frac{h X_1 I_h}{V_1} = h \frac{I_h}{I_{sc}} = \frac{h}{I_{sc}/I_L} \cdot \frac{I_h}{I_L} \quad (9.62)$$

For example, at $I_{sc}/I_L = 20$ this chapter uses the 20-to-below-50 row: the fifth-harmonic limit is 7%. Equation (9.62) predicts $V_5/V_1 = 5(0.07)/20 = 1.75\%$, below the 3% individual voltage limit at a medium-voltage PCC. This simplified estimate leaves headroom for other users but ignores resonance, background distortion and capacitor-bank interaction. Boundary classification and applicability must follow the invoked standard and utility requirements.

9.7.4 Statistical assessment

Formal assessment uses Class A measurement and harmonic grouping under IEC 61000-4-7 and IEC 61000-4-30. Aggregate 200 ms windows to 3 s (15 windows), then to 10 min (200 values); compare percentiles over the specified day/week [26].

IEC 61000-4-30:2025, edition 4, supersedes the 2015 edition and defines current power-quality measurement methods [27]. IEEE Std 519-2022 remains the explicitly invoked harmonic-limit edition here. A newer measurement-method reference does not silently change the harmonic limits or their assessment criteria.

$$X_{10min} = \sqrt{\frac{1}{N} \sum_{k=1}^N X_{3s,k}^2} \quad (9.63)$$

Current assessment uses daily 99th-percentile very-short-time values below twice the table limits, weekly 99th-percentile short-time values below 1.5 times the limits, and weekly 95th-percentile short-time values below the table limits. Voltage assessment uses daily 99th-percentile very-short-time values below 1.5 times its limits and weekly 95th-percentile short-time values below them. Figure 9.49 contains 1008 synthetic ten-minute samples generated as $\max(0.1, 3.4 + 0.75 z_k)$ percent, with independent standard-normal samples from NumPy default_rng(519); replace zero-based samples 150, 440 and 800 by 6.1, 6.6 and 6.2%. Linear-interpolated percentiles are 4.669% and 5.207%, so this record passes the 5% and 7.5% short-time TDD screens. It contains no three-second data, individual-harmonic or voltage checks, and therefore does not establish full compliance.

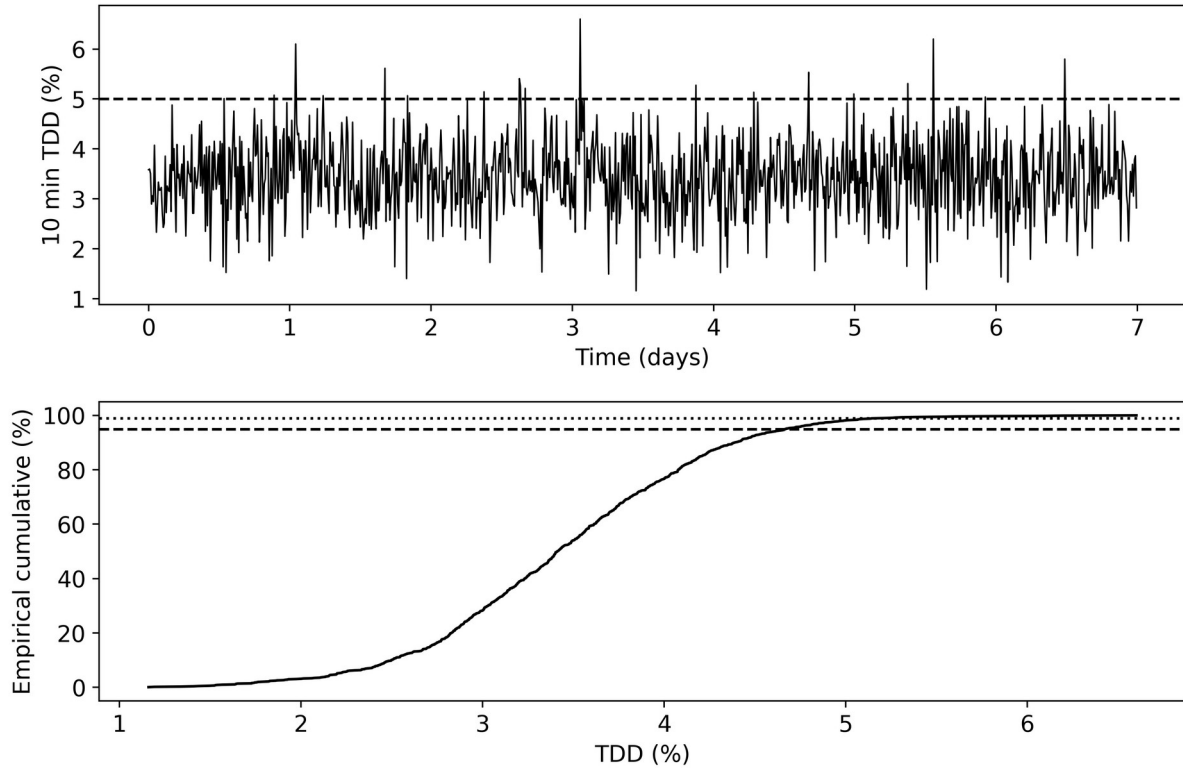


Figure 9.49. Reproducible synthetic week of 10-minute TDD and empirical cumulative distribution, with 95th/99th percentile guide lines. Computed percentiles are 4.669% and 5.207%; data-generation assumptions are stated in Section 9.7.4.

9.7.5 Several harmonic sources at one PCC

When several nonlinear loads share a PCC, their harmonic currents add as phasors, and their phase angles become increasingly random at higher orders. IEC TR 61000-3-6 [28] uses a general summation law,

$$I_h = \left(\sum_i I_{h,i}^\alpha \right)^{1/\alpha}, \alpha = \begin{cases} 1, & h < 5 \\ 1.4, & 5 \leq h \leq 10 \\ 2, & h > 10 \end{cases} \quad (9.64)$$

The exponent α represents a planning diversity assumption: 1 gives coherent arithmetic addition and 2 gives a root-sum-square estimate. Intermediate exponents estimate partial diversity. Actual harmonic currents add as complex phasors and can exceed a diversity estimate when operating phases align; arithmetic addition is a conservative upper bound on magnitudes, not automatically an error or oversizing. Measure or justify phase diversity and simultaneous loading before adopting the planning result as equipment duty.

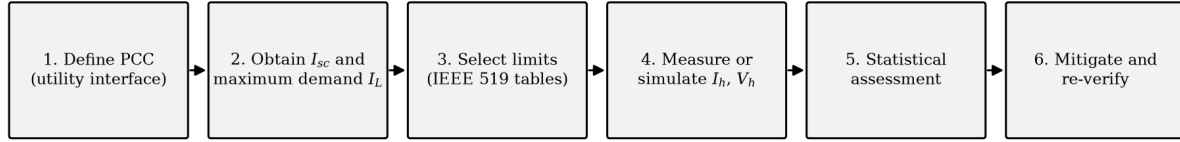
9.7.6 Sizing mitigation for compliance

A compensator is sized from the residual spectrum it must leave. If a shunt APF removes a fraction $(1 - k)$ of every harmonic uniformly, the residual is $k I_h$ and the required harmonic current rating is

Use Equation (9.65) only with nonzero harmonic RMS current and nonzero initial TDD. If the spectrum is zero or already satisfies all targets, choose $k=1$ and no harmonic-compensation duty. Every target and current limit must use the same demand-current base. Additional reactive, unbalance, neutral and transient duties are separate.

$$k = \min \{ 1, TDD_{target} / TDD, \min_{h: I_h > 0} (I_{h,lim} / I_h) \}, I_{APF} = (1 - k) I_H \quad (9.65)$$

A selective-harmonic design can do better by attenuating only the offending orders, at the price of a more complex controller. Figure 9.50 summarizes the compliance workflow. Interharmonics, rapidly varying distortion and transients need separate interpretation, because the standard is framed around steady-state limits.



Voltage limits (Table of voltage distortion) are checked separately from current limits at every step.

Figure 9.50. IEEE Std 519 compliance workflow at a distribution PCC.

Example 9.29. THD versus TDD at light load

A drive operating at 40 % of the plant's maximum demand current draws a current with 20 % THD. Find its TDD.

Solution. $I_1 = 0.40 I_L$, so $I_H = 0.20 I_1 = 0.08 I_L$ and TDD is 8%. If **these harmonic amperes remain unchanged** while the fundamental rises to I_L , both THD and TDD become 8%. If harmonics instead increase in proportion to the fundamental so that THD remains 20%, full-load TDD would become 20%. State which physical quantity is held constant when comparing load levels.

Example 9.30. Voltage distortion produced by a harmonic current spectrum

The installation of Example 9.43 ($I_{sc} = 14$ kA; $I_5 = 14$, $I_7 = 9$, $I_{11} = 5$, $I_{13} = 3.5$, $I_{17} = 2$ A) is connected at a medium-voltage PCC. Estimate the voltage distortion it causes, ignoring resonance and background distortion.

Solution. Assuming nominal fundamental PCC voltage and inductive source impedance, $V_5/V_1 = 5(14)/14000 = 0.50\%$, followed by 0.45%, 0.39%, 0.33% and 0.24%; $THD_V = 0.88\%$. These estimated contributions are below the medium-voltage screening limits. This does not exclude other harmonic orders, measurement uncertainty, background distortion or resonance, nor replace the complete statistical assessment.

Example 9.31. Sizing a shunt APF for compliance

A 415 V installation has $I_L = 500$ A and $I_{sc} = 8$ kA. Its spectrum is $I_5 = 40$, $I_7 = 25$, $I_{11} = 12$, $I_{13} = 8$, $I_{17} = 5$ and $I_{19} = 4$ A. Check compliance and size a shunt APF that brings the TDD to 4 % with uniform attenuation.

Solution. $I_{sc}/I_L = 16 < 20$, so the limits are 4 % (20 A) for $h < 11$, 2 % (10 A) for $11 \leq h < 17$, 1.5 % (7.5 A) for $17 \leq h < 23$, and 5 % TDD. The spectrum is 8.0, 5.0, 2.4, 1.6, 1.0 and 0.8 % of I_L ; the 5th, 7th and 11th exceed their limits, and $\sqrt{\sum I_h^2} = 49.74$ A gives TDD = 9.95 %. Cutting each harmonic only to its individual limit would leave $\sqrt{20^2 + 20^2 + 10^2 + 8^2 + 5^2 + 4^2} = 31.7$ A, or 6.34 % TDD, still failing, so the TDD governs. From Equation (9.65), $k = 4/9.95 = 0.402$; the residual spectrum (16.1, 10.1, 4.8, 3.2, 2.0 and 1.6 A) meets every limit (Figure 9.51). The APF must supply $I_{APF} = (1 - 0.402) \times 49.74 = 29.7$ A RMS of harmonic current, or $3 \times 239.6 \times 29.7 = 21.4$ kVA, before allowances for reactive power, ripple and overload.

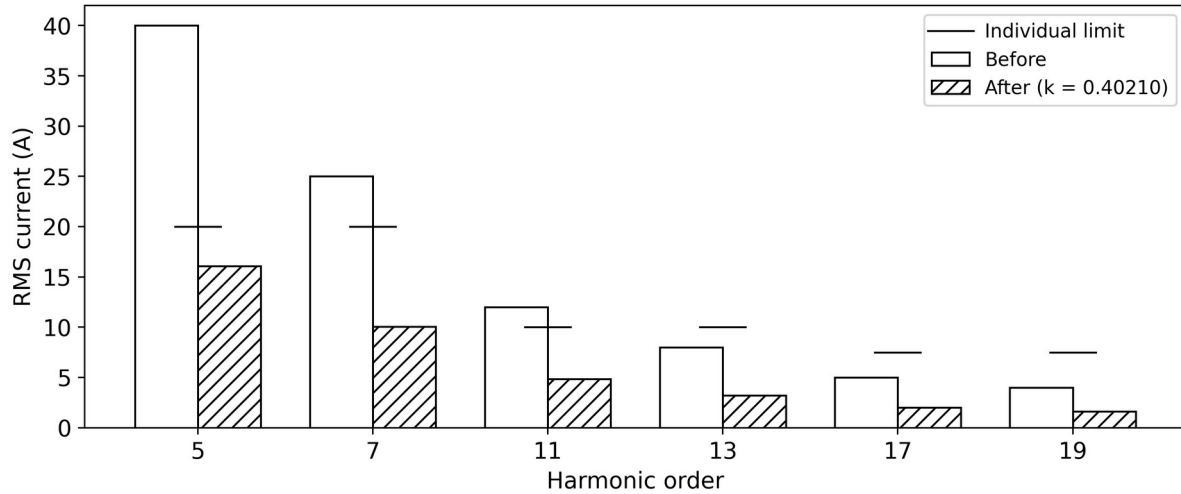


Figure 9.51. Harmonic spectrum of Example 9.31 before and after uniform APF attenuation, against the IEEE Std 519-2022 limits for $I_{sc}/I_L < 20$.

Example 9.32. Summation of two drives

Two drives at one PCC produce 5th-harmonic currents of 30 A and 20 A and 11th-harmonic currents of 8 A and 6 A. Estimate the combined values with Equation (9.64).

Solution. For the fifth harmonic the IEC planning estimate with $\alpha = 1.4$ is $(30^{1.4} + 20^{1.4})^{1/1.4} = 41.3$ A. For the eleventh with $\alpha = 2$, it is $\sqrt{8^2 + 6^2} = 10.0$ A. These are diversity estimates, not guaranteed upper bounds. Coherent currents could add to 50 A and 14 A, respectively. Arithmetic addition is 40% above the estimated eleventh-harmonic value, but is not unjustified oversizing if phase coherence or the design specification requires that case. Use measured phase angles, credible diversity and operating coincidences before choosing a filter rating.

9.8 Comparing and Selecting Custom Power Devices

Device selection should begin with the disturbed state variable and the required continuity of service. A voltage-quality problem upstream of a sensitive load points first toward a series compensator; a nonlinear or unbalanced load that pollutes the source points toward a shunt compensator; simultaneous voltage and current problems motivate a UPQC; a true interruption needs an energy-supported island or UPS function rather than a conventional conditioner. Converter rating, energy duration, fault duty and bypass philosophy are checked next, and only then is a control algorithm chosen. This section turns those statements into quantitative screening tools.

9.8.1 Functional comparison

Table 9.15 compares the three main devices by function, and Figure 9.52 arranges the candidate devices by the kind of disturbance and the side on which it originates. A DSTATCOM can move voltage through the feeder impedance but cannot guarantee restoration of a deep, externally imposed sag or an interruption. A DVR can compensate supply-voltage harmonics if designed for the bandwidth. A UPQC combines both services, subject to both converters' limits, the DC balance and the topology. The map is a first screen only: a DVR may suffice for a semiconductor process dominated by short sags, whereas the same plant may need a UPQC if its own drives also cause unacceptable source-current distortion; conversely, a UPQC where only displacement power factor is poor is unjustified if a capacitor bank or DSTATCOM meets the dynamic requirement.

Table 9.15. Comparison of DSTATCOM, DVR and UPQC.

Feature	DSTATCOM	DVR	UPQC
Connection	Shunt	Series (injection transformer)	Series and shunt, common DC link
Controlled quantity	Source current (and PCC voltage in ZVR)	Load voltage	Load voltage and source current
Main problems solved	Reactive power, unbalance, neutral current, current harmonics, flicker	Sag, swell, voltage unbalance, voltage harmonics, notches	All of these
Real-power source	None (losses only)	Storage, rectifier or none (quadrature, limited)	Grid via shunt converter
Typical rating	Load reactive and distortion power	Fraction of load ($X \times$ load for in-phase)	Series plus shunt; both converter maxima
Interruption ride-through	No	Only with storage and an island path	Only with storage, isolation and grid-forming control
Fault exposure	Shunt; current-limited	Carries downstream fault current; needs bypass	As DVR for the series part
Relative cost	Lowest per function	Moderate	Highest

interruption / severe event	UPS, island-capable storage (not a DVR)	SSCL / breaker (fault current)	UPS or storage-backed UPQC with isolation
harmonic / unbalance	DVR / series APF	shunt APF / DSTATCOM hybrid filter	UPQC or hybrid + DVR
fundamental (sag, swell, pf)	DVR (sag/swell) DSTATCOM (ZVR, weak feeder)	DSTATCOM (UPF, balancing)	UPQC
	supply-voltage disturbance	load-current disturbance	both

Figure 9.52. First-stage selection map for custom-power devices. Final selection still requires rating, protection, energy and economic studies.

9.8.2 Interruptions are an energy problem

A sag leaves some source voltage, so a grid-fed UPQC or a rectifier-supported DVR can draw the missing power from the remaining supply. An interruption leaves none. The protected load must then be fed entirely from storage, through a topology that isolates it from the dead network and forms its voltage and frequency. With usable stored energy E_{us} and load power P_L , the ideal ride-through time is

$$t_{rt} = \frac{\eta E_{us}}{P_L} \quad (9.66)$$

which for DC-link capacitors of practical size is a few milliseconds to a few hundred milliseconds. Problem 9.13 shows the scale. Longer interruptions require batteries, flywheels, a standby generator or a UPS, and once isolation and grid-forming control are added, the device is functionally a UPS rather than a power-quality conditioner.

9.8.3 Normalized converter ratings

Use a **positive-sequence fundamental** three-phase apparent-power base $S_1 = 3V_{ph}I_1$, where I_1 is positive-sequence fundamental RMS current. Use its displacement pf $\cos\phi$, aggregate harmonic fraction $h = I_H/I_1$ with $I_H = \sqrt{\sum_k I_{Hk}^2/3}$, and fundamental negative-sequence fraction $u = I_{neg}/I_1$. For orthogonal sequence/harmonic components the aggregate phase-current compensation duty is

$$S_{DST,aggregate}/S_1 = \sqrt{\sin^2\phi + u^2 + h^2} \quad (9.67)$$

Equation (9.67) is an aggregate equivalent duty, not the maximum unequal phase-leg rating, and excludes zero sequence and neutral hardware. With no zero sequence the total load RMS current gives $S_{rms} = S_1 \sqrt{1 + h^2 + u^2}$, so using measured total kVA as S_1 overstates the base. For a separate **DVR upstream of a load-side DSTATCOM**, source current is compensated to its active fundamental, so in-phase DVR duty is $S_{DVR}/S_1 = X \cos\phi$ and storage is $X P_L t / \eta$. A standalone DVR without downstream compensation instead carries full distorted load current, giving $S_{DVR}/S_1 = X \sqrt{1 + h^2 + u^2}$. These arrangements must not be conflated. A lossless right-shunt UPQC-P with balanced fundamental load and harmonic fraction h has

$$S_{UPQC-P}/S_1 = |X| \cos \phi / (1 - X) + \sqrt{\sin^2 \phi + h^2 + [X \cos \phi / (1 - X)]^2}, X < 1 \quad (9.68)$$

Figure 9.53 compares a specified separate arrangement, DVR upstream of a load-side DSTATCOM, with the right-shunt UPQC-P on the same fundamental base. At nominal voltage their shunt duties coincide. During a sag the UPQC source current increases to supply full load power from the depressed voltage; the separate storage-supported DVR supplies missing real power instead. The separate converter sum is smaller for positive sag depths in this ideal comparison, while it needs event storage. Installed voltage/current/thermal ratings, unequal phase currents, recharge, losses, fault duties and redundancy can change the engineering choice; no single sag depth marks a general crossover between the two arrangements.

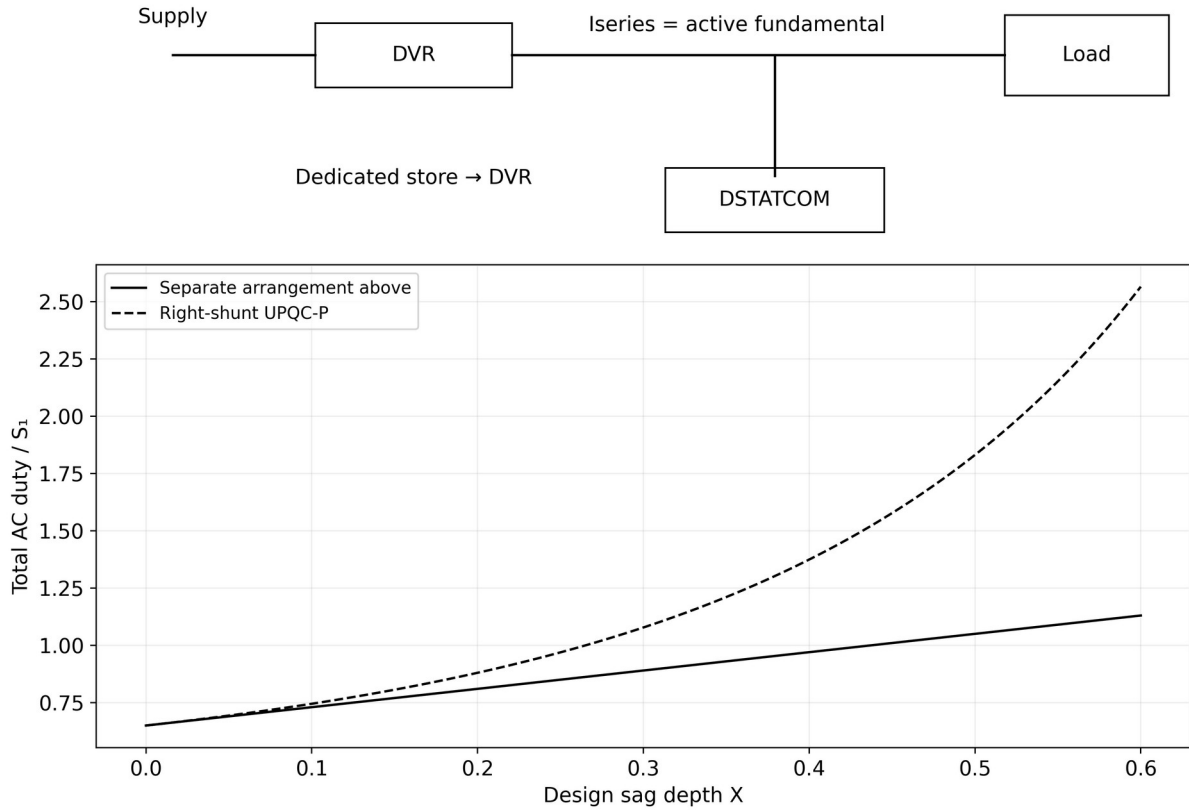


Figure 9.53. Aggregate AC operating duty on fundamental base S_1 versus sag depth, $\text{pf}=0.8$ and harmonic fraction $h=0.25$. Separate arrangement is DVR upstream of load-side DSTATCOM; ideal storage supplies missing power. Installed component ratings require additional checks.

9.8.4 Energy storage and ride-through

Figure 9.54 gives the exact ideal energy contours $E=XPLt$ for in-phase DVR injection, expressed per megawatt of real load power. Storage selection also needs pulse power, permitted voltage window, recharge, event repetition, lifetime, temperature and DC/DC conversion range. Duration alone does not decide between capacitor, supercapacitor, flywheel and battery storage. Overlay measured events only after checking converter voltage/current limits and the protected load's tolerance.

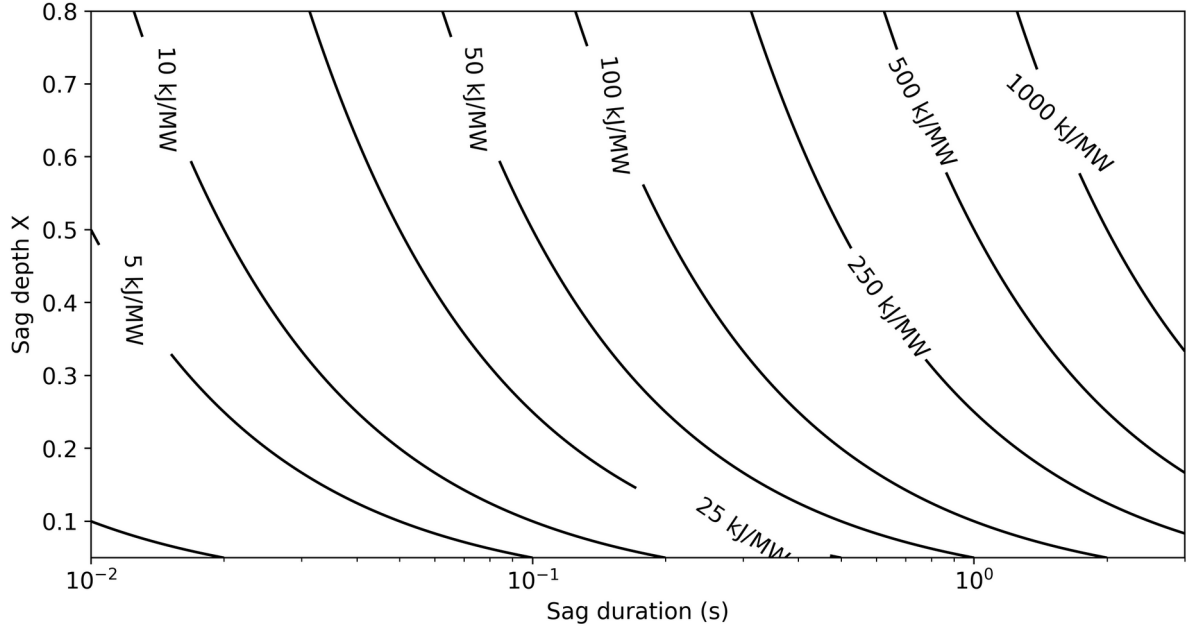


Figure 9.54. Ideal event-energy contours in kJ per MW of real load power, $E=XPLt$ with $\eta=1$. They are energy requirements, not storage-technology boundaries.

9.8.5 Economic evaluation

A custom-power device is justified when the losses it avoids exceed its life-cycle cost. The expected annual cost of power-quality events without mitigation is

$$C_{PQ} = \sum_i N_i p_{trip,i} C_{trip,i} \quad (9.69)$$

where N_i is the annual number of events of class i (from the audit), $p_{trip,i}$ the probability that such an event trips the process (from equipment tolerance, as in IEEE Std 1346 [29]) and $C_{trip,i}$ the cost of one trip, including scrap, restart and lost production. The device's annualized cost is

$$C_{ann} = CRF \cdot C_{cap} + C_{OM} + C_{loss}, \quad CRF = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (9.70)$$

Here C_{cap} is installed capital cost, r the annual discount rate, n the service life in years, C_{OM} annual maintenance and C_{loss} annual energy-loss cost. At $r=0$ the capital recovery factor has the limiting value $1/n$. Calculate C_{loss} from the converter loss map and operating hours rather than assuming a fixed percentage of kVA rating. Standby bypass reduces some DVR losses, but auxiliaries, storage maintenance and recharge losses may remain. Mitigation is economically favourable when $C_{PQ} - C_{PQ,res} > C_{ann}$ under the stated assumptions.

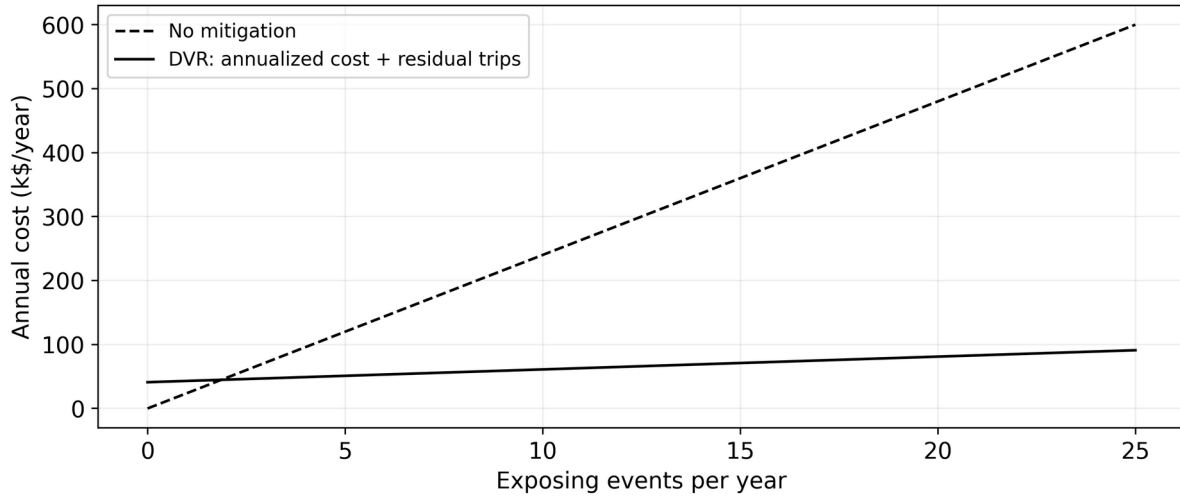


Figure 9.55. Annual unmitigated cost and annualized DVR plus residual-trip cost using the explicit assumptions of Example 9.34. No UPQC cost curve is inferred without a separate cost model.

9.8.6 Screening procedure and multi-criteria scoring

Figure 9.56 gives a first screening procedure; it is not a procurement rule. The essential first step is a power-quality audit that measures sag depth, duration and phase jumps, separates supply-voltage harmonics from load-current harmonics, checks TDD with the full method of Section 9.7 and identifies interruption exposure, which needs a separate energy path. When more than one device survives the screen, a weighted score $S_j = \sum_k w_k r_{jk}$, with weights w_k agreed in advance and ratings r_{jk} for each candidate j on criteria such as technical coverage, converter rating, energy, losses, footprint, redundancy and cost, makes the trade-off explicit and auditable (Example 9.35).

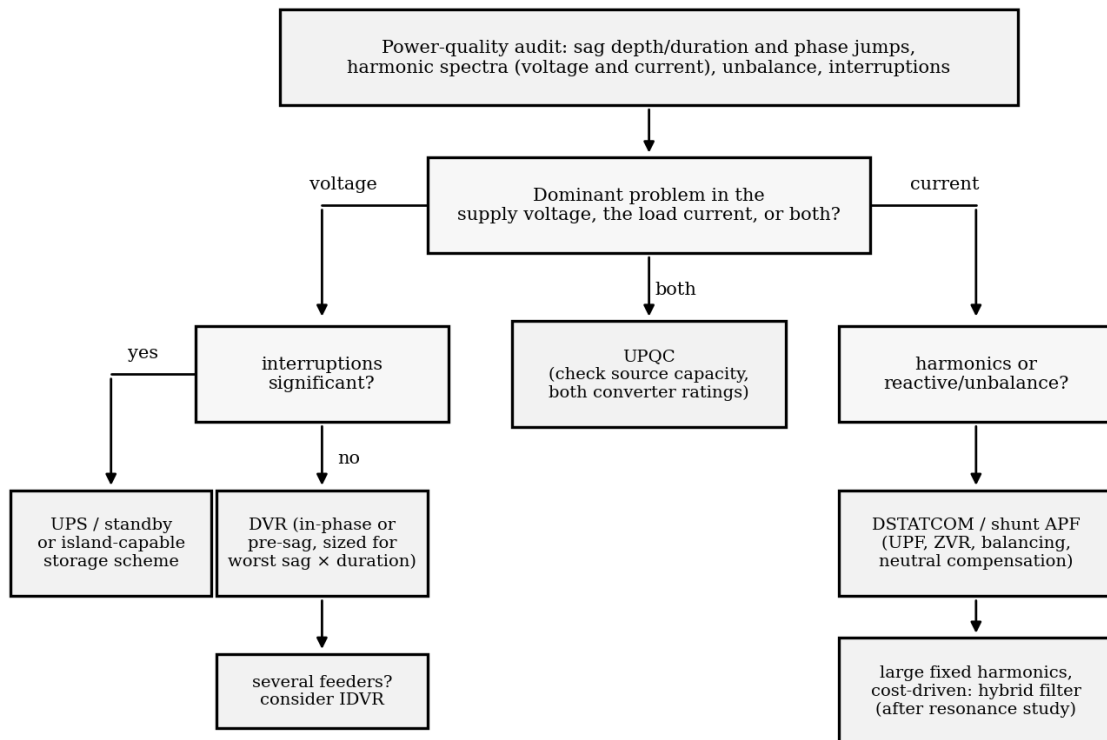


Figure 9.56. Screening procedure for selecting a custom power device.

Example 9.33. Normalized ratings of competing solutions

A plant has fundamental apparent power $S_1=500$ kVA, displacement $\text{pf}=0.8$ and $h=0.25$. Compare a DVR upstream of a load-side DSTATCOM with a right-shunt UPQC-P for 30% balanced sags lasting 300 ms. Use ideal lossless conversion and balanced fundamental current. State actual series-current paths.

Solution. $P_L=400$ kW. The load total RMS apparent power is $500\sqrt{1+0.25^2}=515.39$ kVA. (a) The DSTATCOM aggregate duty is $500\sqrt{0.6^2+0.25^2}=325$ kVA. Because it is downstream of the DVR, the DVR carries only compensated active fundamental current and needs $0.30(0.8)(500)=120$ kVA. Total operating duty is 445 kVA; ideal event storage is $0.30(400)(0.30)=36$ kJ. (b) Right-shunt UPQC-P: series duty $500(0.30)(0.8)/0.7=171.43$ kVA; shunt duty $500\sqrt{0.6^2+0.25^2+(0.30 \times 0.8/0.7)^2}=367.44$ kVA; total 538.87 kVA, about 21.1% above 445 kVA. Steady sag power comes from the grid, but transient capacitance is still needed. A standalone DVR carrying the uncompensated distorted load would need $0.30(515.39)=154.62$ kVA; adding an independently screened shunt duty is not the same coordinated current-path calculation. Final selection needs installed ratings, storage power, losses, protection and life-cycle cost.

Example 9.34. Economic justification of a DVR

A plant experiences 20 exposing sags per year. Without mitigation 60 % of them trip the process at 40 k\$ per trip. A DVR costing 300 k\$ (life 15 years, discount rate 8 %, O&M 6 k\$ per year, negligible standby losses) reduces the trip probability to 5 %. Is it justified?

Solution. $C_{PQ}=20 \times 0.6 \times 40=480$ k\$ per year; the residual is $20 \times 0.05 \times 40=40$ k\$. From Equation (9.70), $CRF=0.08 \times 1.08^{15}/(1.08^{15}-1)=0.1168$, so $C_{ann}=0.1168 \times 300+6=41.0$ k\$ per year. The net annual benefit is $480-40-41=399$ k\$, and the simple payback is $300/(480-40-6)=0.69$ year. The break-even exposure is $41.0/(0.55 \times 40)=1.9$ events per year (Figure 9.55). The result is driven by the trip cost and the audit's event count, which is why both must be measured rather than assumed.

Example 9.35. Multi-criteria screening

A semiconductor plant suffers frequent utility-side sags and significant harmonic distortion from its own drives, with comparable production losses from each. A second plant has rare, shallow sags but continuous harmonic distortion well above its TDD limit. Screen the technology for each, and set up a weighted score for the first plant.

Solution. For the first plant both problems are present and significant, so a UPQC deserves serious consideration: its shared DC link lets the series converter's sag energy come from the grid through the shunt converter rather than from a separately sized store. With weights of 0.35 (technical coverage), 0.20 (converter rating), 0.15 (storage), 0.15 (losses and footprint) and 0.15 (cost), and ratings from 1 to 5 of 5, 3, 5, 3, 3 for the UPQC and 5, 4, 2, 3, 3 for a DVR plus a shunt APF, the scores are 4.0 and 3.75. The UPQC leads narrowly, so the decision is sensitive to the weights and should be confirmed with the plant's sag statistics, harmonic spectrum, ratings, losses, redundancy and life-cycle cost. For the second plant, a shunt APF or DSTATCOM, or a hybrid filter if a resonance study supports it, is the natural candidate; a DVR is hard to justify on the sag evidence.

9.9 Recent Developments and Research Directions

Recent work should be read as an extension of the physical models developed in this chapter. Higher switching frequency changes filter size and loss; grid-forming control changes the converter's synchronization and voltage-source behaviour; predictive and data-assisted control changes how references are computed; digital twins and hardware-in-the-loop change how claims are validated. None of these removes the semiconductor current, DC-energy, transformer, thermal or protection constraints. A recent scoping review of UPQCs in

microgrids found that most studies still rely on PI and SRF control and on offline simulation, with comparatively few experimental or long-term stability assessments [20].

9.9.1 Wide-bandgap devices and higher switching frequency

SiC and GaN devices allow higher switching frequency and lower switching loss than silicon devices in suitable voltage classes. For custom-power equipment this shrinks the ripple filter (Equation (9.25) scales C_f with $1/f_s$) and the interfacing inductor (Equation (9.24) scales L_r with $1/f_s$), and it widens the current-loop bandwidth available for harmonic tracking. The gain is not free: dv/dt , common-mode current, insulation stress, electromagnetic compatibility, gate-drive design and thermal cycling become more demanding, and at medium voltage series connection or multilevel structures remain necessary. A meaningful comparison uses total converter loss, filter volume, junction-temperature swing and conducted emissions, not switching frequency alone.

9.9.2 Grid-forming and weak-grid operation

Conventional custom-power controllers assume a grid stiff enough for a PLL to extract the angle. In converter-dominated or island-capable microgrids this can fail, and the converter can no longer be treated as an ideal current or voltage source behind a fixed impedance. Its closed-loop output impedance $Z_{out}(s)$ interacts with the network impedance $Z_g(s)$, and a screening condition is based on the minor-loop ratio

$$L(s) = \frac{Z_g(s)}{Z_{out}(s)} \quad (9.71)$$

Nyquist assessment must include open-loop unstable poles and use coupled dq or sequence matrices where needed; a scalar impedance ratio is only a screening model. Selected grid-code developments and proposed tests address grid-forming voltage-source behaviour, fault response and harmonic/unbalance support [30]; they do not automatically apply to every custom-power device. RTDS work explores coexistence of grid-forming and grid-following converters [31]. A DSTATCOM or UPQC study must show how voltage-source behaviour interacts with current limiting and harmonic control. IEEE Std 1547-2018 [32] applies to relevant distributed-energy-resource interconnections; applicability to a conditioning-only installation must be assessed separately.

9.9.3 Predictive, repetitive, adaptive and data-assisted control

Model-predictive control can handle multivariable converter constraints explicitly. A generic constrained formulation for a UPQC is

$$J = \sum_{k=0}^{N-1} \left(w_v \|v_L - v_L^{\dot{}}\|^2 + w_i \|i_s - i_s^{\dot{}}\|^2 + w_{dc} (V_{dc} - V_{dc}^{\dot{}})^2 + w_u \|\Delta u\|^2 \right) \quad (9.72)$$

subject to converter voltage, current, DC-energy and switching constraints. Repetitive and resonant controllers target periodic distortion, and adaptive observers or data-driven classifiers can improve disturbance identification or tune references. These methods should be evaluated against conventional SRF or PR control using the same converter limits, sampling delay, sensor noise and grid-impedance uncertainty, and the report should include execution time and robustness to parameter error. A lower simulation THD alone is not evidence of a better practical controller.

In Equation (9.72), voltage, current, DC voltage and control increments have different dimensions. Use per-unit errors on declared bases, or give each weight reciprocal squared units so that the terms form a consistent objective. Include actual computation and PWM delay in the prediction horizon. Gains optimized without those constraints can appear excellent in an ideal model but fail when executed.

9.9.4 Digital twins, HIL and protection-aware validation

A digital twin of a UPQC or DVR is more than a simulation model. At minimum it synchronizes measured converter states with an electrical model, estimates slowly varying parameters such as filter inductance or DC-capacitor condition, compares predicted and measured residuals, and supports control and protection decisions without bypassing hard safety limits. Physics-constrained estimation is preferable to an unconstrained black-box predictor whenever extrapolation could command unsafe converter states. The twin should include saturation, current limiting, bypass logic and protection events, because these govern performance during the disturbances that matter most.

9.9.5 Renewable energy, EV charging and storage integration

PV inverters, battery systems and EV chargers are both sources of power-quality stress and controllable converter resources. A UPQC sharing a DC link with PV or a battery can use spare converter headroom for reactive and harmonic compensation while also managing real power. The scheduling problem is multi-timescale: energy management acts over seconds to hours, harmonic and sag compensation over cycles or milliseconds. The converter current limit couples these services and requires explicit priority logic.

9.9.6 What constitutes research-level validation

A research contribution becomes more convincing as it moves up the validation ladder of Figure 9.57. An analytical model should state network strength and R/X , load type, converter limits and sign conventions. EMT simulation should include switching or a justified averaged model, saturation, delay, DC-link dynamics and disturbance sequences. Sensitivity studies should vary grid impedance, load level and parameter error. Controller-HIL tests validate execution and timing; power-HIL adds power-interface dynamics; a prototype or field benchmark finally exposes sensing, dead time, thermal limits, protection and interference that ideal simulations hide. Comparisons should use identical hardware ratings and report the exact standard metric (for example TDD rather than THD), the detection delay and the transient peak current (Table 9.16).

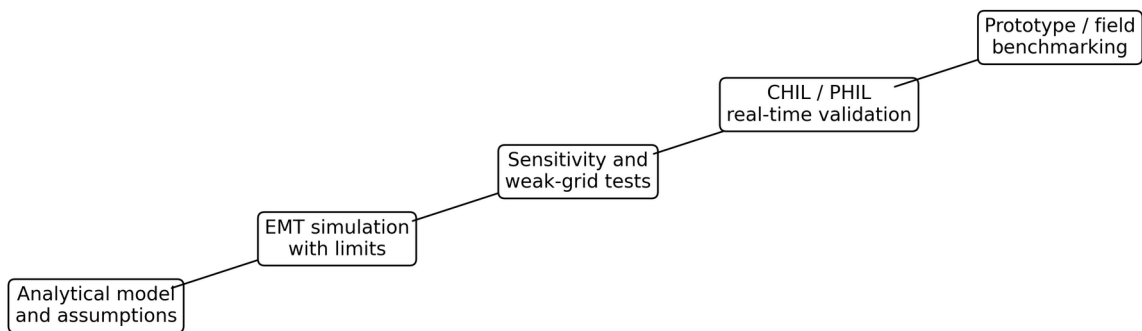


Figure 9.57. Research-validation ladder for advanced custom-power controllers.

Table 9.16. Validation levels and the evidence each provides.

Validation level	Question answered	Minimum evidence
Analytical / phasor	Is the compensation principle feasible?	Sign conventions, power balance, operating region

Validation level	Question answered	Minimum evidence
Averaged dynamic	Do control loops interact acceptably?	Bandwidths, saturation, DC-energy dynamics
Switching EMT	Are ripple and fast transients acceptable?	PWM, filters, delays, device limits
Controller HIL	Does the real controller execute correctly?	Sampling, computation delay, I/O scaling, protection logic
Power HIL / laboratory	Does hardware behave under realistic stress?	Interface stability, thermal and current limits, fault-safe operation

9.9.7 Industry perspective

Industrial selection rests on a measured disturbance profile and the equipment's actual tolerance. Field DVR experience has been reported since the late 1990s [13]. A DVR or UPQC can address sags within its specified envelope; outages require adequate energy, isolation and a suitable voltage-forming path. The analytical and reduced-model illustrations in this chapter explain physical relationships and deliberately stated control assumptions. They do not constitute hardware, protection, standard-compliance or process ride-through validation.

9.9.8 Quantitative bridges from device design to advanced control

Switching frequency and voltage headroom. Figure 9.58 plots Equation (9.24) for Example 9.1: $V_{dc} = 900$ V, $a = 1.2$, $m = 1$ and $\Delta i = 3.913$ A. At the 828 V minimum DC level, fundamental capacitive operation imposes $L_r \leq [828 / (2\sqrt{2}) - 239.600] / (\omega 26.085) = 6.485$ mH. Combining this bound with the ripple equation requires $f_s \geq 4.266$ kHz before harmonic, overload and control margin. Thus increasing switching frequency reduces ripple inductance, while a low-frequency ripple design may require too much fundamental terminal voltage. Loss, EMI and hardware limits still apply; this is an analytical design relationship, not a measured SiC/GaN claim.

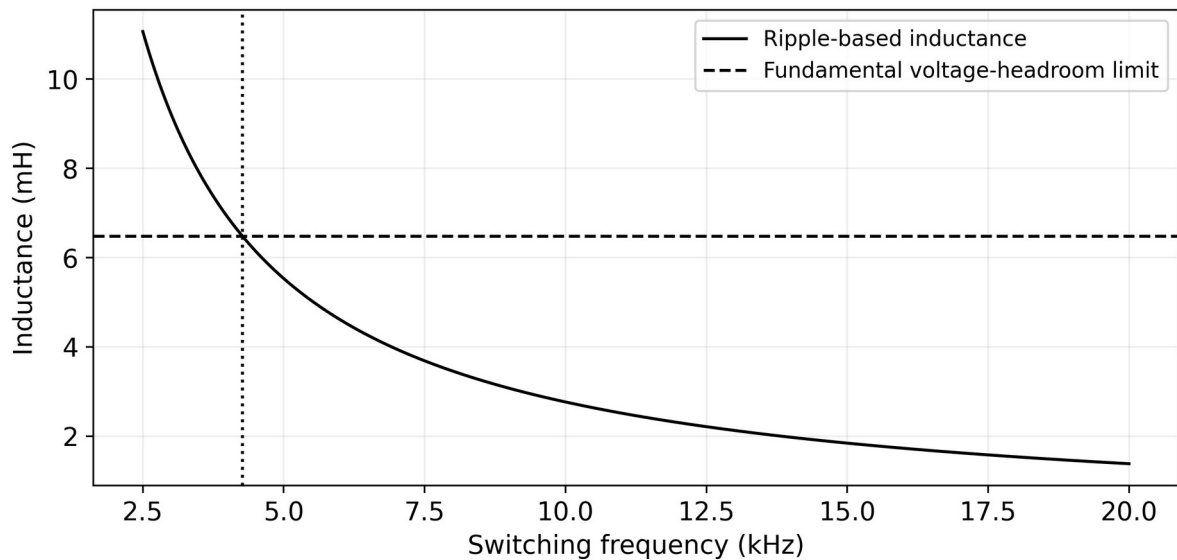


Figure 9.58. Ripple-based inductance and fundamental voltage-headroom bound for Example 9.1. The intersection is 4.266 kHz; lower frequencies fail this specified normal-current DC-window check. Harmonic, overload, thermal and electromagnetic constraints require further checks.

Delay and harmonic tracking. Figure 9.59 uses the declared illustrative transfer function $H(s) = e^{-sT_d} / (1 + s\tau)$, with $\tau = 0.5$ ms and $T_d = 0.1$ ms. Its magnitude is $[1 + (2\pi f\tau)^2]^{-1/2}$ and phase is $-\tan^{-1}(2\pi f\tau) - 2\pi fT_d$. At 50 Hz, fifth and seventh harmonics lie at 250 and 350 Hz. High reference-generation accuracy cannot remove the amplitude and phase errors of a finite-bandwidth tracking loop; selective resonant control, delay compensation and stability margins must be considered together.

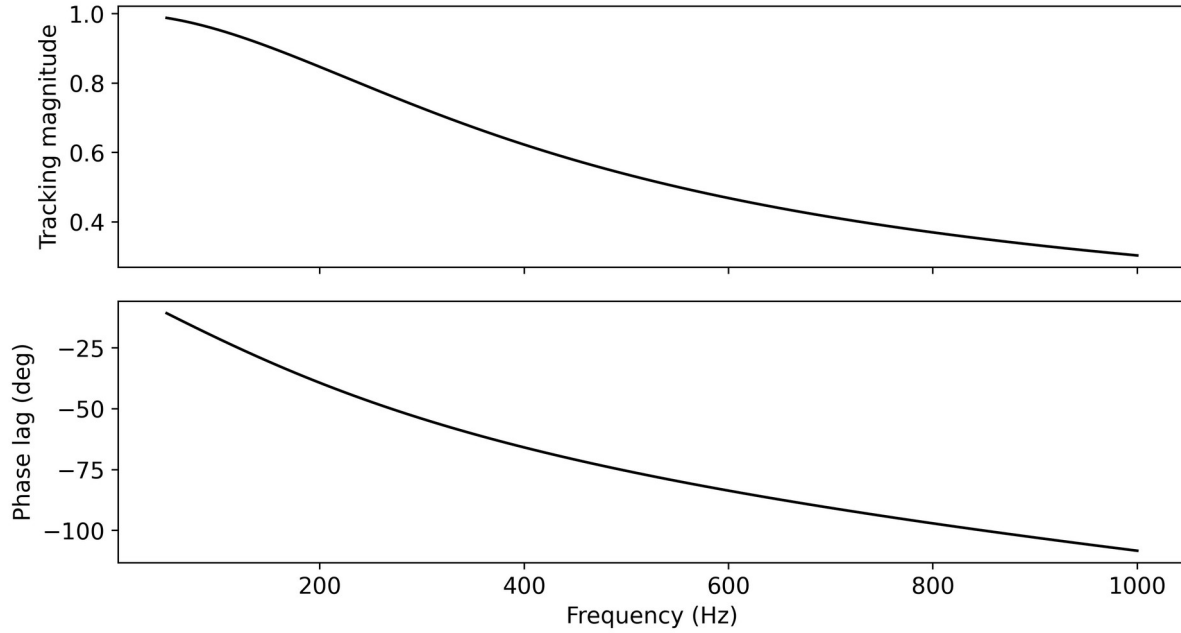


Figure 9.59. Magnitude and phase of a first-order tracking loop with explicitly stated transport delay. This is an analytical frequency response, not a tuned controller for a particular converter.

Sharing current headroom. For balanced orthogonal components, $I_{rms}^2 = I_d^2 + I_q^2 + I_H^2 \leq I_{max}^2$. Figure 9.60 fixes $I_d = 0.6 I_{max}$ and shows the remaining feasible combinations of reactive and harmonic RMS current. A renewable-energy or storage real-power command therefore uses headroom that would otherwise support power quality. With unbalance, zero sequence or large waveform crest factor, phase-by-phase peak and neutral limits must be checked in addition to this RMS screening circle. A supervisor must declare which service is reduced when the boundary is reached.

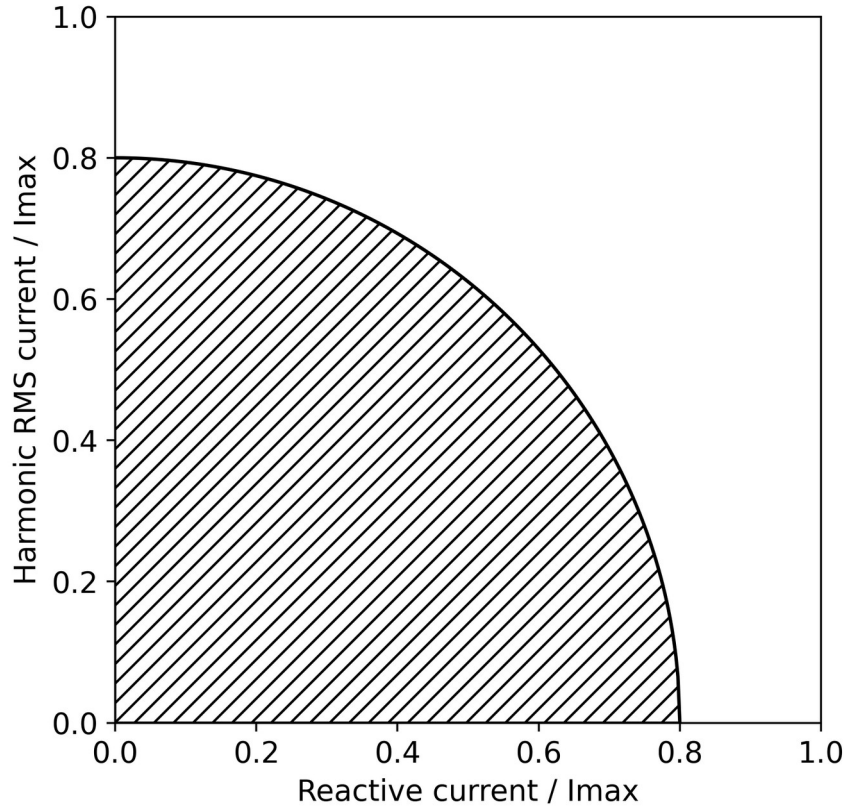


Figure 9.60. RMS current-allocation region for balanced orthogonal components with $I_d=0.6I_{max}$. The boundary is $I_q^2 + I_H^2 = 0.64I_{max}^2$; it excludes phase-peak, unbalance and neutral-leg constraints.

Key Equations

Table 9.17. Key equations and their assumptions.

Equation	Relationship	Meaning and scope
(9.2)	$P F = \cos \phi_1 / \sqrt{1 + T H D_I^2}$	True power factor with sinusoidal voltage
(9.4)	$i_s = i_L - i_c$	DSTATCOM KCL, i_c positive into the bus
(9.7)	Quadratic in I_q	Reactive source current for ZVR
(9.12)	$\dot{i}_d = \dot{i}_{Ld} + i_{loss}$	SRF reference; $\dot{i}_q = 0$ for UPF
(9.14)	$p = v_\alpha i_\alpha + v_\beta i_\beta, q = v_\alpha i_\beta - v_\beta i_\alpha$	Amplitude-invariant; $P = 3 p / 2$
(9.21)	$V_{dc} = 2\sqrt{2} V_{LL} / (\sqrt{3} m)$	Zero-drop SPWM minimum; use terminal voltage and lowest DC window for design
(9.24)	$L_r = \sqrt{3} m V_{dc} / (12 a f_s \Delta i)$	Interfacing inductor
(9.27)	$V_L = V_s + V_{inj}$	Series KVL, voltage-rise convention

Equation	Relationship	Meaning and scope
(9.30)	$P_{DVR} = (V_{Lc} - V_s) I_L \cos \phi$	Standalone in-phase DVR
(9.33)	$X_{max} = 1 - \cos \phi$	Zero-real-power limit, standalone DVR
(9.35)	$\cos \psi = (V_{Lc}^2 - V_s^2 - U^2) / (2 V_s U)$	Feasible angle for specified U
(9.42)	$C = 2 E / (V_{max}^2 - V_{min}^2)$	Energy-based capacitance
(9.44)	$\dot{W}_{dc} = \sum P_{into} - P_{loss}$	Shared DC-link balance
(9.45)	$I_s = I_L \cos \phi / (1 - X)$	Right-shunt UPQC source current
(9.47)	$S_{se} = V_{Lc} I_L \cos \phi \tan \delta$	UPQC-Q series duty
(9.49)	$P_{se} = V_{Lc} X I_L \cos \phi / (1 - X), S_{se} = P_{se} $	UPQC-P series duty
(9.55)	$h_r \approx \sqrt{S_{sc} / Q_c}$	Capacitor-bank parallel resonance
(9.57)	$I_{sh} = \frac{Z_F I_{Lh} + V_{sh}}{Z_s + Z_F + K}$	Series APF with shunt passive filter
(9.59)	$TDD = 100 \sqrt{\sum I_h^2 / I_L}$	Maximum-demand denominator
(9.62)	$V_h / V_1 = h I_h / I_{sc}$	Voltage distortion from a harmonic current
(9.64)	$I_h = (\sum I_{h,i}^\alpha)^{1/\alpha}$	IEC planning diversity estimate; not a guaranteed coherent-sum bound
(9.68)	Normalized UPQC-P rating	Fundamental S_1 base; balanced RMS aggregate duty, not unequal-leg nameplate
(9.70)	$C_{ann} = C R F C_{cap} + C_{OM} + C_{loss}$	Life-cycle cost

Common Misconceptions

- “A capacitor-supported DVR can correct any sag by injecting in quadrature.” For a standalone DVR, quadrature means perpendicular to actual load current. Sustained lossless restoration needs $X \leq 1 - \cos \phi$; a unity-pf load needs positive real power for a sag, although a charged capacitor may supply it briefly.
- “Quadrature to the supply voltage and quadrature to the current are the same thing.” They coincide only when the series current is in phase with the supply, as in a right-shunt UPQC with unity source pf.
- “A zero-power series converter needs no DC energy.” It still has conduction, switching and magnetic losses.
- “UPQC-Q cannot correct any swell.” This restriction applies to the right-shunt unity-source-pf geometry. The left-shunt zero-series-power arrangement uses the load-current reference and can restore suitable swells, as Example 9.26 shows.

- “The UPQC-P series rating is $V_{Lc} X I_L \cos \phi$.” The series converter carries the increased source current, so the factor $1/(1 - X)$ is required.
- “The smallest series VA means the smallest installation.” Check the shunt duty, the nominal-voltage duty and each converter’s own maxima.
- “A three-leg DSTATCOM can remove neutral current.” It needs a fourth leg, a split DC link or a zero-sequence transformer.
- “A neutral leg needs no rating because its AC terminal is at neutral potential.” Neutral VA is not obtained by multiplying neutral current by phase voltage, but the leg still requires RMS and peak-current, thermal and full DC-bus switch-voltage checks. Summed unequal phase duties do not establish an equal-leg converter nameplate.
- “A hybrid converter carries only residual source current.” In series with a passive branch it carries the branch current.
- “A shunt filter is always the right answer for harmonics.” For voltage-source type loads a series impedance is more effective, and a passive shunt filter can resonate with the source.
- “Passing current THD proves IEEE 519 compliance.” TDD, individual limits, voltage limits and the statistical tests are all required, and the low-voltage voltage-THD limit is 8 %, not 5 %.

Chapter Summary

Custom power devices apply the converter principles of transmission FACTS to distribution power quality. Problems divide into voltage-quality problems, which a series device corrects, and current-quality problems, which a shunt device corrects.

The capacitor-supported DSTATCOM injects assigned reactive, unbalanced, zero-sequence and harmonic current so the source supplies balanced active current, or retains controlled reactive current for voltage regulation. Zero-sequence compensation needs a rated neutral path. Source references can use unit templates, SRF, pq or adaptive methods, with explicit synchronization and current limits. Sizing must combine modulation headroom, real-energy imbalance, ripple, peak/RMS current and thermal/protection duty.

The DVR injects series voltage through a suitable converter/transformer or transformerless arrangement. Storage, rectifier support or self-supported operation determines its energy capability. In-phase injection minimizes fundamental voltage magnitude; pre-sag compensation preserves waveform trajectory; phase rotation trades voltage and energy. Sustained lossless standalone quadrature restoration requires $X \leq 1 - \cos \phi$. Finite stored energy can support real power for a limited duration. Voltage/current headroom, phase tolerance, transformer flux and fault bypass remain essential.

The UPQC coordinates series voltage and shunt current on a common DC link. UPQC-P minimizes series injection magnitude; right-shunt UPQC-Q supplies purely reactive series duty and corrects sags but not swells in that geometry; left-shunt zero-power arrangements can correct suitable swells. UPQC-S uses an explicit angle/objective to share duty. Each installed converter must cover its own worst voltage/current condition, including nominal operation and filter drops.

Harmonic filtering depends on the load model, network impedance and feedback law. Shunt filters naturally serve current-source loads; series impedance can suppress voltage-source-load harmonics. Series/hybrid arrangements reduce impedance sensitivity only with sufficient stable harmonic gain and suitable current paths; small converter VA is an outcome to calculate. IEEE Std 519 assessment uses PCC demand-current bases, individual harmonics, voltage criteria and statistics. Device selection combines the measured disturbance profile, equipment tolerance, installed ratings, storage, protection and life-cycle cost.

Solved Numerical Practice Problems

The following problems are chapter-level synthesis exercises. Unless stated otherwise, three-phase quantities use 415 V line-to-line (239.6 V phase), and each solution connects the calculation with converter physics, rating limits and model assumptions.

Example 9.36. Injection transformer rating

A DVR for a 415 V, 20 kVA system must inject up to 171 V per phase on the line side, with a converter-side voltage of 50 V. Find the turns ratio, the transformer kVA and the converter-side current.

Solution. $n=50/171=0.292$. $I_s=27.82$ A, so $S=3 \times 171 \times 27.82=14.27$ kVA. The converter-side current is $27.82 \times 171/50=95.1$ A.

Example 9.37. DVR with source impedance: why zero-power injection fails

A 415 V supply with $Z_s=0.3+j1.8\Omega$ per phase feeds a star load $Z_L=18+j12\Omega$ per phase. The source EMF sags by 25 %. A DVR at the load must keep the load voltage at 239.6 V. (a) Find the load current and voltage without the DVR. (b) Can a capacitor-supported DVR restore the voltage? (c) Find the minimum-magnitude injection and its real power.

Solution. (a) $V_M=0.75 \times 239.6=179.7$ V, $I=179.7/|Z_s+Z_L|=7.84$ A and $|V_L|=7.84 \times 21.63=169.6$ V (293.8 V line). (b) With the load restored, $|I_L|=239.6/21.63=11.08$ A and the circuit resistance absorbs $11.08^2 \times 18.3=2245$ W per phase. The sagged EMF can deliver at most $179.7 \times 11.08=1990$ W at that current, so zero-real-power restoration is impossible. (c) $V_{inj}=V_L(1+Z_s/Z_L)-V_M$ is smallest when $V_L(1+Z_s/Z_L)$ is aligned with V_M : $|V_{inj}|=239.6 \times 1.0595 - 179.7=74.15$ V per phase. The DVR then supplies $P_{DVR}=1.97$ kW (three-phase) and its duty is 2.46 kVA. A quadrature injection of $\sqrt{239.6^2 - 179.7^2}=158.5$ V is not a valid zero-power answer here; the power-balance test in (b) settles feasibility before any injection angle is chosen.

Example 9.38. Unbalanced sag compensation

At the DVR terminals $V_p=0.85 \angle 0^\circ$ pu, $V_n=0.15 \angle 0^\circ$ pu and $V_0=0$. Restore the balanced target $[1, a^2, a]$ pu.

Solution. From Equation (9.36), $V_a=1$, $V_b=-0.5-j0.6062=0.7858 \angle -129.52^\circ$ and $V_c=0.7858 \angle 129.52^\circ$ pu. Equation (9.37) gives $V_{inj,a}=0$, $V_{inj,b}=-j0.2598$ and $V_{inj,c}=+j0.2598$ pu. Phase a already meets its target; b and c need equal magnitudes with opposite angles. Restoring phase magnitudes alone does not ensure balanced phase relationships.

Example 9.39. Real and reactive power of an angled injection

A single-phase DVR injects 60 V at 45° ahead of a 15 A load current. Find its P, Q and VA.

Solution. $P=60 \times 15 \cos 45^\circ=636.4$ W, $Q=636.4$ var and $S=900$ VA.

Example 9.40. IDVR real-power sharing

Three series compensators share a DC bus. Feeder 2 needs 12 kW for 150 ms; feeders 1 and 3 can each absorb up to 8 kW while keeping acceptable voltage. Check sufficiency, neglecting losses.

Solution. 16 kW is available against 12 kW needed. Commanding 6 kW absorption on each supporting converter gives $P_1=-6$, $P_2=12$ and $P_3=-6$ kW, and the mean DC energy is constant. A standalone storage-only DVR would need $12 \times 0.15=1.8$ kJ. With total loss P_{loss} the absorption must sum to $12+P_{loss}$ kW, so up to 4 kW of loss can be covered. This establishes steady power sufficiency, not zero DC capacitance: transient delay and ripple still need buffer energy.

Example 9.41. Preliminary series and DC-link design of a 415 V, 30 kW four-wire UPQC

Design the series part of a 415 V, 30 kW lossless right-shunt UPQC-P for $\pm 30\%$ supply variation. Use $V_{dc,max}=800$ V, 5% droop, $f_s=10$ kHz, $m_{max}=1$, $a=1.2$ and 10% line-current ripple. Choose transformer ratio $n=3$. The shunt rated compensating current is **specified separately** as 69.56 A, giving a declared 600 J transient energy budget. Check the series converter voltage/current and calculate DC capacitance; full shunt harmonic design requires its own load spectrum.

Solution. Maximum line injection is $0.3(239.600)=71.880$ V. Ratio $n=3$ steps down from a 215.641 V converter winding. Deepest-sag line current is $30\,000/[3(0.7)(239.600)]=59.623$ A and converter current is 19.874 A; transformer duty is 12.857 kVA. At the largest swell line current is 32.105 A, giving the smallest allowed ripple 3.210 A. Equation (9.53) gives $L_{converter}=8.992$ mH and $L_{line}=0.9991$ mH. For in-phase series injection, fundamental terminal voltage is $\sqrt{215.641^2 + (\omega L_{converter} 19.874)^2} = 222.83$ V, below $760/(2\sqrt{2})=268.70$ V available at minimum DC voltage. Terminal fundamental duty is 13.286 kVA. With the larger sag ripple, the switch-current screen is $1.25[\sqrt{2}(19.874)+5.9623/3]=37.62$ A; a 50 A class is preliminary. DC overshoot screening gives 880 V and suggests a 1200 V class before detailed switching tests. For the separately specified 600 J energy budget, $C=1200/(800^2-760^2)=19.231$ mF. A four-leg neutral path avoids midpoint-current injection into split capacitors, but unbalanced power ripple remains. The shunt side has zero-drop SPWM headroom at 760 V; its filter and harmonic voltage requirements must also fit that window. This is preliminary sizing, not a complete hardware design.

Example 9.42. Hybrid branch current and residual source harmonic

For the topology of Figure 9.46(b), a load draws 40 A RMS of fifth harmonic. Assume, for this simplified same-phase exercise, that the passive branch diverts 92 % under nominal tuning and 85 % when detuned. Find the residual source harmonic and the fifth-harmonic current through the active stage. Is a 10 A stage enough?

Solution. Nominal: $I_{s,5}=40 \times 0.08=3.2$ A and the branch carries 36.8 A. Detuned: 6.0 A residual and 34.0 A in the branch. The active stage in series with the branch carries 36.8 A or 34.0 A, so a 10 A stage is insufficient; a 45 A stage gives 22 % margin over the nominal case, before the branch's fundamental capacitive current is added (Equation (9.58)). Assigning the 6 A residual to this converter would confuse it with a separate shunt residual-current stage. In a real network the current division is complex-valued and depends on the source and filter impedances (Equation (9.54)).

Example 9.43. PCC TDD and individual-harmonic screening

An installation at a 120 V to 69 kV PCC has $I_L=400$ A and $I_{sc}=14$ kA. Its RMS spectrum is $I_5=14$, $I_7=9$, $I_{11}=5$, $I_{13}=3.5$ and $I_{17}=2$ A, with $I_1=380$ A. Screen it.

Solution. $I_{sc}/I_L=35$, the 20 to 50 band: TDD limit 8 %, individual limits 7 % (5th, 7th), 3.5 % (11th, 13th) and 2.5 % (17th). $I_H=\sqrt{318.25}=17.84$ A, so TDD = 4.46 % and THD = 4.69 %. The individual values are 3.5, 2.25, 1.25, 0.875 and 0.5 %, all within limits. A shunt APF that reduced every harmonic by 80 % would bring TDD to 0.89 %, but the spectrum already passes this screen, so such a filter would need a separate justification. Full compliance also needs the statistical assessment of Section 9.7.4.

Unsolved Practice Problems

Unless stated otherwise, UPQC problems use the lossless right-shunt topology with unity source pf and RMS single-phase quantities; standalone DVR problems use the actual load current.

Problem 9.1 (UPQC-Q) A 230 V load draws 18 A at 0.75 pf; the supply sags by 18 %. Find V'_s , I_s , V_{inj} and S_{se} .

Problem 9.2 For the load of Problem 9.1, find V_{inj} and S_{se} in UPQC-P mode.

Problem 9.3 For Problem 9.1, find δ and the shunt current.

Problem 9.4 For Problems 9.1 and 9.3, find the total UPQC-Q operating VA.

Problem 9.5 (Standalone DVR, in-phase) 400 V (line), 15 kW, 0.85 pf, 25 % sag, 180 ms. Find V_{ph} , V_{inj} , I_{DVR} , P_{DVR} and the ideal E .

Problem 9.6 For Problem 9.5, the DC bus may fall from 650 V to 600 V. Find C_{dc} .

Problem 9.7 (IDVR) Feeder 2 needs 10 kW; feeders 1 and 3 can supply 5 and 4 kW. Is this sufficient?

Problem 9.8 A load has 5th = 7 A, 7th = 3.5 A and fundamental 45 A. Find the THD and the APF compensating current.

Problem 9.9 A facility has $I_L = 350$ A with 5th = 12 A, 7th = 8 A, 11th = 4 A and 13th = 3 A. Find the TDD.

Problem 9.10 (Hybrid filter) A load draws 35 A of harmonic current; the passive branch diverts 88 %. Find the residual source harmonic and the harmonic current in an active stage in series with the branch.

Problem 9.11 (Unbalanced sag) $V_p = 0.80$ pu and $V_n = 0.12$ pu in phase with V_p , $V_0 = 0$. Find the DVR injection magnitude in each phase to restore a balanced 1.0 pu set.

Problem 9.12 (TDD compliance) The facility of Problem 9.9 has $I_{sc}/I_L = 28$. Check the TDD and each individual harmonic.

Problem 9.13 (Interruption bound) A UPQC DC bus stores 10 kJ and must retain 75 % of it. With an island-capable path and a 12 kW load, find the ideal ride-through time.

Problem 9.14 (Strategies) A right-shunt UPQC protects 230 V, 18 A, 0.75 pf against a 15 % sag. Find the series voltage and VA in UPQC-Q and UPQC-P modes.

Problem 9.15 (Loss-inclusive energy) A standalone DVR supplies 6000 W for 180 ms at 95 % efficiency. Find the energy and its excess over the ideal value.

Problem 9.16 (DSTATCOM, UPF) A 415 V, 40 kW, 0.75 pf load is compensated to UPF. Find the source current before and after, the DSTATCOM current and its kVA.

Problem 9.17 (DSTATCOM design) For a 415 V DSTATCOM rated 50 A, find V_{dc} ($m=1$), L_r ($f_s=10$ kHz, $a=1.2$, 5 % ripple) using a 700 V bus, and C_{dc} for a 5 % droop with $\Delta E = 3 V_{ph} a I_c t$, $t=10$ ms.

Problem 9.18 (Load balancing) A 30 kW UPF single-phase load is connected between lines b and c of a 415 V system. Find the load current, the balanced source current, the DSTATCOM currents and its kVA.

Problem 9.19 (Harmonic compensation) A 415 V load draws 60 A fundamental at unity displacement with 5th, 7th and 11th harmonics of 18, 12 and 6 %. Find the current THD, the DSTATCOM current and its kVA.

Problem 9.20 (Capacitor-supported DVR limit) A standalone capacitor-supported DVR protects a 230 V, 0.75 pf load. Find the maximum sag correctable without real power, the injection magnitude at that limit and the load-voltage phase shift.

Problem 9.21 (BESS energy) A 400 V, 50 kVA, 0.9 pf load is protected by an in-phase DVR against a 40 % sag for 300 ms with 95 % efficiency. Find P_{DVR} and the energy in kJ and Wh.

Problem 9.22 (Injection transformer) A 415 V, 30 kVA DVR must inject 0.35 pu with a converter-side voltage of 100 V. Find the line-side voltage, ratio, kVA and converter-side current.

Problem 9.23 (UPQC-S) A right-shunt UPQC with a 110 V series limit protects 230 V, 20 A, 0.85 pf against a 25 % sag. Find ψ , δ , P_{se} , S_{se} , S_{sh} and the total.

Problem 9.24 (Left-shunt UPQC-Q) A left-shunt UPQC-Q protects 230 V, 20 A, 0.9 pf during an 8 % sag. Using the smaller root, find θ_L , $|V_{inj}|$, S_{se} and the shunt operating VA.

Problem 9.25 (Neutral current) A four-wire load draws $I_a = 30 \angle 0^\circ$, $I_b = 20 \angle -150^\circ$ and $I_c = 25 \angle 100^\circ$ A. Find the neutral current the DSTATCOM must supply.

Problem 9.26 (Resonance) A 400 V bus has a short-circuit level of 15 MVA. Find the parallel resonant order with a 250 kvar capacitor bank, and the tuning order with a 5.67 % detuning reactor.

Problem 9.27 (Series hybrid) Repeat Example 9.28 with $K = 2$ pu and find the maximum source 5th-harmonic current.

Problem 9.28 (Economics) For Example 9.34, find the number of exposing events per year below which the DVR is not justified if the trip cost falls to 15 k\$.

Answer Key (Unsolved Problems)

9.1 $V_s' = 188.60$ V, $I_s = 16.463$ A, $V_{inj} = 131.64$ V, $S_{se} = 2167.3$ VA.

9.2 $V_{inj} = 41.40$ V, $S_{se} = 681.6$ VA.

9.3 $\delta = 34.92^\circ$, $I_{sh} = 2.483$ A.

9.4 2738.4 VA.

9.5 $V_{ph} = 230.94$ V, $V_{inj} = 57.74$ V, $I_{DVR} = 25.47$ A, $P_{DVR} = 3750$ W, $E = 675$ J.

9.6 $C_{dc} = 21.6$ mF.

9.7 9 kW available; insufficient by 1 kW.

9.8 THD = 17.39 %, $I_{comp} = 7.826$ A.

9.9 TDD = 4.36 %.

9.10 Residual 4.20 A; active-stage branch harmonic current 30.80 A.

9.11 0.08, 0.28 and 0.28 pu.

9.12 TDD 4.36 % < 8 %; 5th 3.43 % and 7th 2.29 % (< 7 %); 11th 1.14 % and 13th 0.86 % (< 3.5 %). Passes the spectrum screen.

9.13 Usable energy 2500 J; $t_{max} = 0.208$ s.

9.14 UPQC-Q: 121.16 V, 1924.3 VA; UPQC-P: 34.5 V, 547.9 VA.

9.15 $E = 1136.8$ J, 5.3 % above 1080 J.

9.16 74.20 A before, 55.65 A after; $I_{DST} = 49.08$ A; 35.28 kVA.

9.17 Zero-drop SPWM bound is 677.7 V; with a 700 V bus the ripple calculation gives 3.37 mH and the specified apparent-power-derived budget gives 18.05 mF. However, 5% droop reaches 665 V, below 677.7 V even before the inductor drop. Thus the proposed window is not a feasible complete SPWM design. Increase DC voltage, restrict droop and recheck filter/harmonic headroom, or explicitly redesign for another modulation method.

9.18 $I_{Lb} = -I_{Lc} = 72.29 \angle -90^\circ$ A; $I_s = 41.74$ A; $I_{ca} = 41.74 \angle 180^\circ$, $I_{cb} = 41.74 \angle -60^\circ$, $I_{cc} = 41.74 \angle 60^\circ$ A; 30 kVA.

9.19 THD = 22.45 %; 13.47 A; 9.68 kVA.

9.20 25 %; 152.1 V; the load voltage leads by 41.41° .

9.21 18 kW; 5.684 kJ = 1.579 Wh.

9.22 83.86 V; $n = 100/83.86 = 1.192$; 10.5 kVA; 35.0 A.

9.23 $\psi = 73.08^\circ$, $\delta = 27.23^\circ$, $P_{se} = 725.6$ W, $S_{se} = 2493.3$ VA, $S_{sh} = 726.6$ VA, total 3219.9 VA.

9.24 $\theta_L = 13.87^\circ$, $|V_{inj}| = 56.37$ V, $S_{se} = 1127.5$ VA, $I_{sh} = 4.15$ A at 211.6 V, 877.6 VA.

9.25 $16.83 < 60.30^\circ$ A.

9.26 $h_r = \sqrt{15/0.25} = 7.75$; $h_t = 1/\sqrt{0.0567} = 4.20$.

9.27 $0.0052 + 0.0048 = 0.0100$ pu, half the value with $K = 1$ pu.

9.28 $41.0 / (0.55 \times 15) = 4.97$, so about 5 events per year.

References

- [1] N. G. Hingorani, "Introducing custom power," *IEEE Spectrum*, vol. 32, no. 6, pp. 41-48, Jun. 1995.
- [2] B. Singh, A. Chandra and K. Al-Haddad, *Power Quality: Problems and Mitigation Techniques*. Chichester, U.K.: Wiley, 2015.
- [3] A. Ghosh and G. Ledwich, *Power Quality Enhancement Using Custom Power Devices*. Boston, MA: Kluwer, 2002.
- [4] V. Khadkikar, "Enhancing electric power quality using UPQC: a comprehensive overview," *IEEE Trans. Power Electron.*, vol. 27, no. 5, pp. 2284-2297, May 2012.
- [5] *IEEE Recommended Practice for Monitoring Electric Power Quality*, IEEE Std 1159-2019.
- [6] Information Technology Industry Council, *ITI (CBEMA) Curve Application Note*, Washington, DC.
- [7] K. R. Padiyar, *FACTS Controllers in Power Transmission and Distribution*. New Delhi: New Age International, 2007.
- [8] B. Singh, P. Jayaprakash and D. P. Kothari, "A T-connected transformer and three-leg VSC based DSTATCOM for power quality improvement," *IEEE Trans. Power Electron.*, vol. 23, no. 6, pp. 2710-2718, Nov. 2008.
- [9] B. Singh and J. Solanki, "A comparison of control algorithms for DSTATCOM," *IEEE Trans. Ind. Electron.*, vol. 56, no. 7, pp. 2738-2745, Jul. 2009.
- [10] C. Schauder and H. Mehta, "Vector analysis and control of advanced static VAR compensators," *IEE Proc. C*, vol. 140, no. 4, pp. 299-306, Jul. 1993.
- [11] H. Akagi, Y. Kanazawa and A. Nabae, "Instantaneous reactive power compensators comprising switching devices without energy storage components," *IEEE Trans. Ind. Appl.*, vol. IA-20, no. 3, pp. 625-630, May/Jun. 1984.
- [12] H. Akagi, E. H. Watanabe and M. Aredes, *Instantaneous Power Theory and Applications to Power Conditioning*. Hoboken, NJ: IEEE Press/Wiley, 2007.
- [13] N. H. Woodley, L. Morgan and A. Sundaram, "Experience with an inverter-based dynamic voltage restorer," *IEEE Trans. Power Del.*, vol. 14, no. 3, pp. 1181-1186, Jul. 1999.
- [14] J. G. Nielsen and F. Blaabjerg, "A detailed comparison of system topologies for dynamic voltage restorers," *IEEE Trans. Ind. Appl.*, vol. 41, no. 5, pp. 1272-1280, Sep./Oct. 2005.
- [15] A. Ghosh, A. K. Jindal and A. Joshi, "Design of a capacitor-supported dynamic voltage restorer (DVR) for unbalanced and distorted loads," *IEEE Trans. Power Del.*, vol. 19, no. 1, pp. 405-413, Jan. 2004.
- [16] J. G. Nielsen, F. Blaabjerg and N. Mohan, "Control strategies for dynamic voltage restorer compensating voltage sags with phase jump," in *Proc. IEEE APEC*, 2001, pp. 1267-1273.
- [17] S. S. Choi, B. H. Li and D. M. Vilathgamuwa, "Dynamic voltage restoration with minimum energy injection," *IEEE Trans. Power Syst.*, vol. 15, no. 1, pp. 51-57, Feb. 2000.
- [18] H. Fujita and H. Akagi, "The unified power quality conditioner: the integration of series- and shunt-active filters," *IEEE Trans. Power Electron.*, vol. 13, no. 2, pp. 315-322, Mar. 1998, doi: 10.1109/63.662847.
- [19] V. Khadkikar and A. Chandra, "UPQC-S: a novel concept of simultaneous voltage sag/swell and load reactive power compensations utilizing series inverter of UPQC," *IEEE Trans. Power Electron.*, vol. 26, no. 9, pp. 2414-2425, Sep. 2011.
- [20] B. Khan, S. Jadapalli, G. V. Swathi, S. K. Maitra, P. Singh, U. Choudhury and M. L. Biramo, "Power quality improvement of microgrids using unified power quality conditioner (UPQC): a scoping review," *Electric Power Systems Research*, vol. 253, Art. no. 112573, 2026, doi: 10.1016/j.epsr.2025.112573.
- [21] B. Singh, K. Al-Haddad and A. Chandra, "A review of active filters for power quality improvement," *IEEE Trans. Ind. Electron.*, vol. 46, no. 5, pp. 960-971, Oct. 1999.

- [22] F. Z. Peng, H. Akagi and A. Nabae, "A new approach to harmonic compensation in power systems: a combined system of shunt passive and series active filters," *IEEE Trans. Ind. Appl.*, vol. 26, no. 6, pp. 983-990, Nov./Dec. 1990.
- [23] F. Z. Peng, "Harmonic sources and filtering approaches," *IEEE Ind. Appl. Mag.*, vol. 7, no. 4, pp. 18-25, Jul./Aug. 2001.
- [24] H. Fujita and H. Akagi, "A practical approach to harmonic compensation in power systems: series connection of passive and active filters," *IEEE Trans. Ind. Appl.*, vol. 27, no. 6, pp. 1020-1025, Nov./Dec. 1991.
- [25] H. Akagi, "Active harmonic filters," *Proc. IEEE*, vol. 93, no. 12, pp. 2128-2141, Dec. 2005.
- [26] *IEEE Standard for Harmonic Control in Electric Power Systems*, IEEE Std 519-2022.
- [27] IEC 61000-4-30:2025, Electromagnetic compatibility (EMC), Part 4-30: Testing and measurement techniques, Power quality measurement methods, edition 4, corrected July 2026. Geneva: IEC, 2025.
- [28] IEC TR 61000-3-6:2008, *Electromagnetic compatibility (EMC), Part 3-6: Limits, Assessment of emission limits for the connection of distorting installations to MV, HV and EHV power systems*. Geneva: IEC, 2008.
- [29] *IEEE Recommended Practice for Evaluating Electric Power System Compatibility With Electronic Process Equipment*, IEEE Std 1346-1998.
- [30] M. K. Khan, K. Kauhaniemi, H. Laaksonen and M. A. Hassan, "Review of recent developments in grid codes: focus on compliance testing and grid-forming inverter-based resources," *Renewable and Sustainable Energy Reviews*, vol. 227, Art. no. 116509, 2026, doi: 10.1016/j.rser.2025.116509.
- [31] N. Mohammed, S. Ahmed, C. Konstantinou and J. M. Guerrero, "RTDS-based evaluation of coexisting grid-forming and grid-following inverter technologies for renewable-integrated AC, DC, and hybrid microgrids," *International Journal of Electrical Power & Energy Systems*, vol. 181, Art. no. 112137, 2026, doi: 10.1016/j.ijepes.2026.112137.
- [32] *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces*, IEEE Std 1547-2018.
- [33] M. H. J. Bollen, *Understanding Power Quality Problems: Voltage Sags and Interruptions*. New York: IEEE Press, 2000.
- [34] B. Bae, J. Lee, J. Jeong and B. Han, "Line-interactive single-phase dynamic voltage restorer with novel sag detection algorithm," *IEEE Trans. Power Del.*, vol. 25, no. 4, pp. 2702-2709, Oct. 2010.
- [35] Imperix, "Proportional resonant controller," Technical Note TN110, updated May 16, 2024, doi: 10.66800/0110. <https://imperix.com/doc/implementation/proportional-resonant-controller>.